AKADEMIKA

# FOL-A-P

## ADVANCE FIBER OPTIC COMMUNICATION TRAINER

## EXPERIMENTAL MANUAL

**…. A MESSAGE FROM**

AKADEMIKA

Today's system designers are faced with tomorrow's problems. FIBEROPTIC COMMUNICATION is one of the important subject need to teach while learning electronics.

It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

**OUR MOTTO;**

- Light years ahead – refers to leadership.

As leaders in our industry in India, We are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that we have a complete, truly innovative & superior training products we are so committed to quality that we back our products with a complete comprehensive warranty.

**AKADEMIKA LAB SOLUTIONS**

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# SAFETY RULES

Carefully follow the instructions contained in this handbook, as they provide you with important points on safety during the installation, use and maintenance. Keep this handbook at hand for any further help.

Set all accessories in order on unpacking, and check integrity with respect to its checklist. Check the Kit for any visible damage.

Before connecting the power supply to the equipment, be sure patch cords are connected correctly, jumpers are correct as per experimental position.

In **FOL-A-P** signals are to be monitored with an oscilloscope, the scope input channel should be AC coupled, unless otherwise indicated. Ensure that X10 oscilloscope probes are used throughout.

These kits must be employed only for the use for which it has been conceived, i.e. as educational kits, and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of fault or malfunctioning of the trainers, turn off the power supply and do not tamper the kits. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

# WARRANTY

These kits are warranted against defects in workmanship and materials. If any failure resulting from a defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting patch chords, Crystals, etc. and other imported items.

**This warranty is subject to the following conditions and limitations.** The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solutions sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the event warranty service is needed, purchaser should get in touch with service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kit /system has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solutions judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs and maintenance of old and decrepit products, be sure to contact our service center or sales outlet.

# INDEX

# TECHNICAL SPECIFICATION

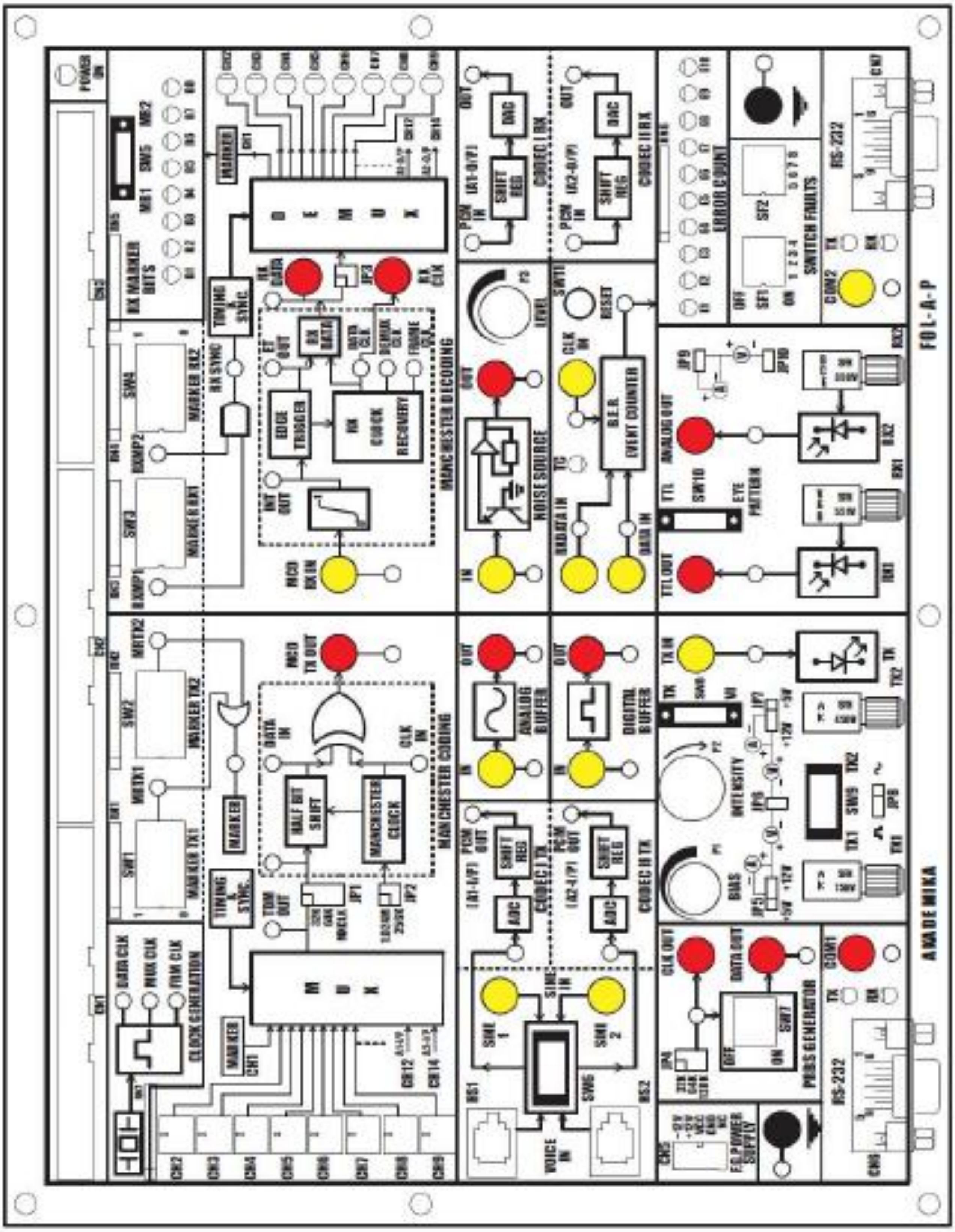
| | |
|---|---|
| **Transmitter:** | Two Nos. Siemens Fiber Optic LED<br>1. Peak wavelength of emission 950nmInfrared (SFH450V)<br>2. Peak wavelength of emission 660nm Visible (SFH756V) |
| **Receiver:** | Two Nos. Siemens Photo Detector<br>1. Photo Transistor Detector (SFH350V)<br>2. Photo Detector with Logic Output (SFH551V) |
| **Modulation Techniques:** | Digital communication with Pulse Code Modulation (PCM) using Motorola MC 145502 CODEC Chip. |
| **Coding / Decoding:** | Manchester Coding / Decoding Techniques. |
| **Noise Generator:** | White noise source output. |
| Amplitude: | 0 to 5Vpp. |
| **PRBS Generator:** | 16 Bit switch selectable. |
| Clock: | 32, 64, 128 KHz. |
| **Bit Error Rate Measurement:** | 10 bit counter with LED indication up to 1024 count. |
| **Multiplexing:** | Time Division Multiplexing 16 Channels (64 Kbits/Sec) |
| **Frame Marker:** | Two 8 bit user selectable markers in alternate frames. |
| **Data rate:** | 1.024 MBits/Sec. |
| **Voice PCM:** | Two Nos. with Telephone Handsets (A- Law) |
| **Analog Bandwidth:** | 3.45 KHz. |
| **FWHM Spectral Width:** | 100 nm. |
| **Fiber Optic Cable Type:** | Plastic Optical, Step Index, Multimode. |
| **Core Refractive Index -n1:** | 1.492. |
| **Clad Refractive Index -n2:** | 1.406. |
| **Numerical Aperture:** | 0.50. |
| **Acceptance Angle:** | 60 degrees. |
| **Fiber Diameter:** | 1000 microns. |
| **Outer Diameter:** | 2.2 mm. |
| **No. of Fibers:** | Two. |
| **Length:** | 01 & 03 Meters. |

# Accessories

## FOL-A-P

| | | Quantity |
|---|---|---|
| 1 | Mini Links PCS2 Patch cord Male To Male | 8 |
| 2 | Short Links PCS2 Patch cord Male To Male | 2 |
| 3 | FOL-A-P EXPT Manual | 1 |
| 4 | Power Supply | 1 |
| 5 | Power cord | 1 |
| 6 | Jumper to Crocodile 30 cm with double wire | 4 |
| 7 | Jumper Caps Small ( 5mm) | 5 |
| 8 | Plastic Fiber Step Index multimde_1000um, 1 meter | 1 |
| 9 | Plastic Fiber Step Index multimde_1000um, 3 meter | 1 |
| 10 | Numerical Aperture Jig ACR- 14/0 | 1 |
| 11 | Steel Rule ( 15 cm /6") | 1 |
| 12 | Telephone Handset With RG-11 & GE 200 Capsule | 2 |
| 13 | Serial To USB converter cables | 2 |

# FUNCTIONAL BLOCK



# INTRODUCTION

Fiber Optics technology is rapidly becoming a familiar & indispensable part of human life. You have undoubtedly heard of Fiber Optics. All giant telephone companies like AT & T; sprints have saturated the airways & print media with advertisements heralding this bright & new technology. Futurists, science writers & entrepreneurs forecast fantastic growth for Fiber Optics applications.

And their enthusiasm is valid. Today medical fiber optic systems allow physicians to peer inside the human body without surgery. Military commanders demand portable battlefield communication systems, which use superior fiber optic transmissions. The massive bundles of copper wiring which once carried telephone conversations between continents now are being replaced with the optical fibers from coast to coast.

Very few technologists ever come close to the fantastic growth rate that market analyst's love for them to predict. Fiber optics has exceeded those rates. The next decade promises even more wonders. The influence of fiber optics will pervade our homes, workplaces & recreational facilities. Your television reception will take on startling clarity. Newsprint may largely become a thing of the past as environmentally friendly electronics newspapers become feasible. Two-way fiber optic will allow you to "attend college" in the comfort of your home.

Anyone who takes an interest in fiber optics & pursues a career in the field today is on the ground floor of opportunity. The technology is sufficiently new that few " experts" exist. Many science colleges, universities & engineering colleges have included fiber optics in their syllabus. The future beckons exciting with the prospect of making new discoveries finding new applications for the technology.

This kit is an introduction to fiber optics communication technology for instructors, students & hobbyists. Welcome to the fascinating & expanding world of fiber optics. We hope that you find the kit challenging, stimulating & - yes – even fun.

## CIRCUIT DESCRIPTION

### 1.1 Clock And Timing

The 2.048MHz crystal oscillators generate a 2.048MHz clock. U1 (74HCT14) is used around crystal for generation of clock. This clock is divided by 2 using U5 (74HCT74) to generate 1.024MHz. This clock is further divided by two ripple counters U2 & U3 (74HC193) to generate, 256 KHZ, 128 KHz, 64 KHz, 32 KHz and 8 KHz. These are generated for the complete time division multiplexed transmission of the data from 16 channels. The data bits are transmitted at the rate of 1.024 MHz Each channel is selected for on period of 1/128 KHz. Thus per channel 8 bits of data is transmitted. The channel selection frequency is 8 KHz. Thus a single frame of 16 channels exists for a period of 125 µs. Marker selection clock is also generated here. U6 (74LS193) is used to generate the clock signals QA, QB, QC, QD for multiplexing.

### 1.2 TX Marker Generation

The marker setting is achieved using switches SW1 & SW2. The two eight bit markers are alternately transmitted at the start of each frame. U7 & U8 (IC 74LS165) are used respectively for Marker 1 and Marker 2 to convert them from parallel to serial form. These signals are ANDed with Marker selection clock and final outputs are ORed to get one channel data.

### 1.3 Audio To PCM Conversion (Voice And Codec Section)

The voice signals from telephone handsets are processed and applied to CODEC U12 & U16 (MC 145502). The analog to digital conversion process inside the CODEC samples the signal at a frequency of 8 KHz and this sampled digitized data is applied to the multiplexer input.
Present techniques of voice communication use standards such as A-law / μ-law companded PCM voice coding at 64 Kbits/sec. When the analog speech signal is converted to pulse code modulation, it is first filtered using a low pass filter with a cut-off at about 3.4 KHz. The analog signal is sampled at 8 KHz as per the Nyquist Criteria. Each sample is quantized and coded into eight bits per sample.
Actually the voice signal features something undesirable and that is its amplitude range varies greatly from one conversion to another. If the quantization levels are uniformly spaced then it certainly creates problems. If the amplitude of the signal is small, quantization levels have to be closely spaced. This gives proper resolution. But if the signal amplitude is large than this fine resolution will result in increasing the number of code bits.
Hence, normally a technique of unequal spacing of quantization levels is used. In the Fiber trainer kit, the audio signal from the telephone handset is processed using telephone interface chip U10 & U15 (TEA1062A) and amplifier U11 (LF353) circuit. It is then applied to voice coder. This chip actually performs analog to digital conversion and vice-versa.
The digital data output is in pulse code modulated form. The CODEC chip used exhibits both A-law and u-law companding techniques. Here, we have selected A-law of companding. The digital output of CODEC is connected as i/p to the multiplexer. The received digital data is again converter into analog form by the chip. The clock considerations and other features of the CODEC chip used in this kit can be studied in details from the datasheets provided with the manual.

### 1.4 Time Division Multiplexing

In case of communication systems, signals which are transmitted usually carry voice or video information with them & are interpreted by human eye or an ear, which has slow response. Persistence of vision as well as of hearing has given rise to the concept of time division multiplexing. In time division multiplexing various signals are sampled & transmitted for a fixed duration of time one after the other. At the receiving end these signals are extracted in the same order & form of transmission.
To implement this scheme we have used 16 channel digital multiplexer U9 (74LS150) at transmission end with clock generator for timing of signals. One channel is reserved for marker transmission, two channels for voice data transmission, eight channels take their inputs from eight data switches & five channels are available as expansion channels for user. Each channel has a data rate of 64 Kbits / Sec.
Now the repetition rate of frame depends on channel sampling frequency. Since we are transmitting audio signals on these channels, we should sample channel at least twice the highest frequency component in audio signal which is 4 KHz. This determines frame frequency of 8 KHz. Within this period of 125 micro sec. we transmit 16 channels hence, each channel ON period comes about 125/16 micro sec. This corresponds to the frequency of 128 KHz. Lastly, since we transmit 8 bits of data per channel, data rate can be derived as 7.8125 /8 =0.9765625 & 1 / 09765625 = 1.024 MHz.
The different frequencies required for the generation of a frame are obtained from the clock generation circuit. The synchronization between these three frequencies is achieved using a stable crystal clock and the divider circuitry of a counter. The rising edge of frame frequency (8 KHz) Is synchronized with the Marker generation block. The marker is generated by a parallel to serial shift register and it always appears in the first channel of a

frame. Channels 2 to 9 in frame receive data from 8 keys (K1-K8). The digitized data from CODEC I and from CODEC II appears in 12th and 14th slots respectively. Remaining are expansion channels.

### 1.5 Manchester Coding

The multiplexed data is Manchester coded and transmitted through the fiber. The data to be coded is applied to a flip flop U18 (74HCT74). An inverted data clock of 1.024 MHz is given as clock input to this flip-flop. The data output from the flip-flop is thus half bit delayed. This delayed data is then EX-ORed U19 (IC 74HC86) with the data clock. The output is the Manchester coded data. It is then transmitted through BUFFER U17 (74HCT04) and then through fiber link.

### 1.6 Analog Buffer

Audio Pre-Amplifier is built around Q3 (BC548) transistor in common emitter mode.

### 1.7 Digital Buffer

This is a TTL BUFFER built around op-amp U1 (74HCT14).

### 1.8 Fiber Optics Transmitter

We have used driver circuitry based on a standard TTL gate to drive a NPN transistor (Q2), which modulates the LED SFH450V source (Turns it ON and OFF).
The transmitter module takes the input signal in electrical form & then transforms it into optical (light) energy containing the same information. The optical fiber is the medium, which carries this energy to the receiver. Fiber optic transmitters are typically composed of a buffer, driver & an optical source. The buffer section provides both electrical connection & isolation between the optical transmitter & analog information source. The driver section provides electrical power to the optical source in a fashion that duplicates the pattern of data being fed to the transmitter. Finally the optical source (LED) converts the electrical current to light energy with the same pattern. The LED SFH450V operates outside the visible light spectrum. Its optical output is centered at near infrared wavelength of 950nm. This LED is coupled to the transistor driver in common emitter mode. In the absence of input signal there is no output light. When signal is present transistor works in saturation state and it gives bias to transmitter LED, which emits full light. Optical signal is then carried over by the optical fiber. Transmitter- LED, digital, DC coupled transmitters are one of the most popular variety due to their ease of fabrication. We have used two driver circuitries. One is base on a standard TTL gate to drive a NPN transistor (Q2), which modulates the LED SFH756V source (Turns it ON and OFF).

### 1.9 Fiber Optics Receiver

There are various methods to configure detectors to extract digital data. Usually detectors are of linear nature. We have used a photo detector TTL receiver having TTL type output (SFH 551/V). Usually it consists of integrated photodiode, transimpedance amplifier and level shifter. Another detector used on kit is photo detector analog receiver (SFH 350V) with output proportional to light intensity.  This one is analog output type detector. For analog output detector gain control potentiometer P5 is provided.

## 1.10 Manchester Decoding

The received data is Manchester decoded to separate clock and data and is applied to a demultiplexer.The received data is integrated using RC integrator. The output of this integrator is EX-ORed U19 (IC 74HC86) with the received data. The output of this EX-OR gate gives a low going narrow pulse for each transition of the received data. This signal is applied to a monostable multivibrator U20 (IC 74LS121), which is configured to give a high pulse for each high to low transition at its input. The on period of this output waveform is adjusted such that the multivibrator output is of 1.024 MHz. This is the received clock, which is then applied as clock input to Flip-flop U18 (IC74HCT74). The output of this Flip-flop is the decoded data. Note that the output is taken at the inverted output pin of Flip-flop. The received data is one clock period shifted. This is due to the particular coding-decoding circuitry used.

## 1.11 Marker Detection (RX Marker)

Marker used in Time Division Multiplexing is a unique bit pattern placed at some fixed position in the frame & is used to determine the start of the frame at the receiver. Sometimes a different marker is used in alternate frames to counter the possibility of data bits containing the marker sequence generating a false marker. Double Marker- The marker is supposed to be a unique bit pattern in the data stream, which is identified at the receiver to identify the beginning of frame. Question arises, what happens if the marker pattern is contained elsewhere as data?

Using the two 8-bit markers hundreds of combinations can be formed for TDM communication. Few settings in the marker pattern may match the data and the receiver comparator may generate a false detection pulse, disturbing the TDM communication.

To avoid this situation the markers are set in a way such that the detection pulses are generated at proper time and proper TDM communication is achieved. It is highly impossible that the data will have this pattern repeating at the same position in every frame. However, to avoid even this situation, the marker is sometimes chosen to be different in alternate frames. Same received data is applied to the marker detection circuitry where low going pulses are generated corresponding to the each marker received and are used as synchronizing signal for timing circuitry. There by two markers are separated from the Multiplexed data. U21, U22 & U23 are used here.

## 1.12 Time Division Demultiplexing

The demultiplexer forms the data distribution system from where the samples are separated and are applied to corresponding channels.U28 (IC 74HC193) & U24 (74HC154) are used to generate channel selection signals. U25, U26, U27 separates out individual channel signals from the TDM received data.The switch channels directly drive the LEDs and the decoded voice data is again applied to the CODEC.

## 1.13 Marker Indication

From the received markers marker-1 is separated out and given to serial to parallel converter U34 (IC 74HC164). Parallel converted marker is passed through buffer latch U35 (IC 74HC374) and output of latch drive LED’s to indicate received marker pattern. The marker MR1 and MRCLK is generated using U30 (PAL CE16V8).

## 1.14 PRBS Generator

From the received markers marker-1 is separated out and given to serial to parallel converter U34 (IC 74HC164). Parallel converted marker is passed through buffer latch U35 (IC 74HC374) and output of latch drive LED's to indicate received marker pattern. The marker MR1 and MRCLK is generated using U30 (PAL CE16V8).

## 1.15 RS 232 Interface

U50 (IC MAX232C) is standard RS232 interface IC is used. The two D type 9 pin connectors are used for Interfacing PC's. Here data Transmitted/Received by the PCs is fed to CN 6 and CN 7 connectors. COM 1and COM 2 posts are used to connect the electrical signal to the Fiber Optic Transmitter and Receiver Circuit.

## 1.16 Bit Error Rate Counter

Transmitted data and received data bits are passed through buffer U43 (74HCT04) to D-flipflop U41 (IC 74HCT74). Output of D-flipflop is then EXORed using U19 (IC 74HC86) and then ANDed with inverted clock using U42 (IC 74HC08). This bit pattern is given as clock input to decade counters U44 (IC 74HC393). Output of counters drive LED's to indicate the error in the received bits in the form of decimal number. Counter is reset using switch SW11.

## 1.17 Noise Source

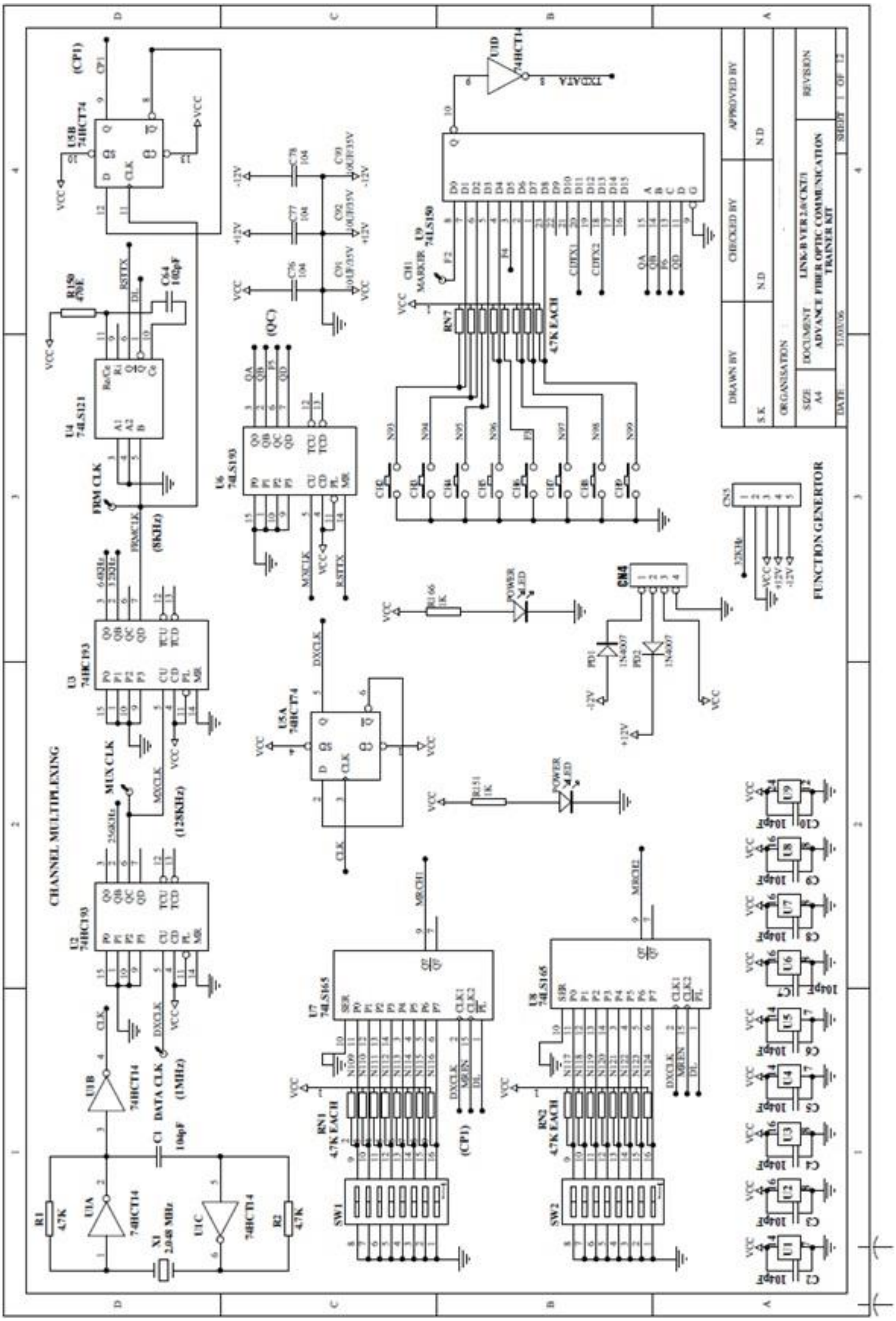
The noise source consists of a Transistor T1 (BC547) whose base is grounded, collector is left open and emitter is given supply voltage. In this biasing condition it produces a white noise in the order of some tens of mV. The noise is amplified by two amplifier stages U51 (IC TL084) with a potentiometer P3 for level regulation.

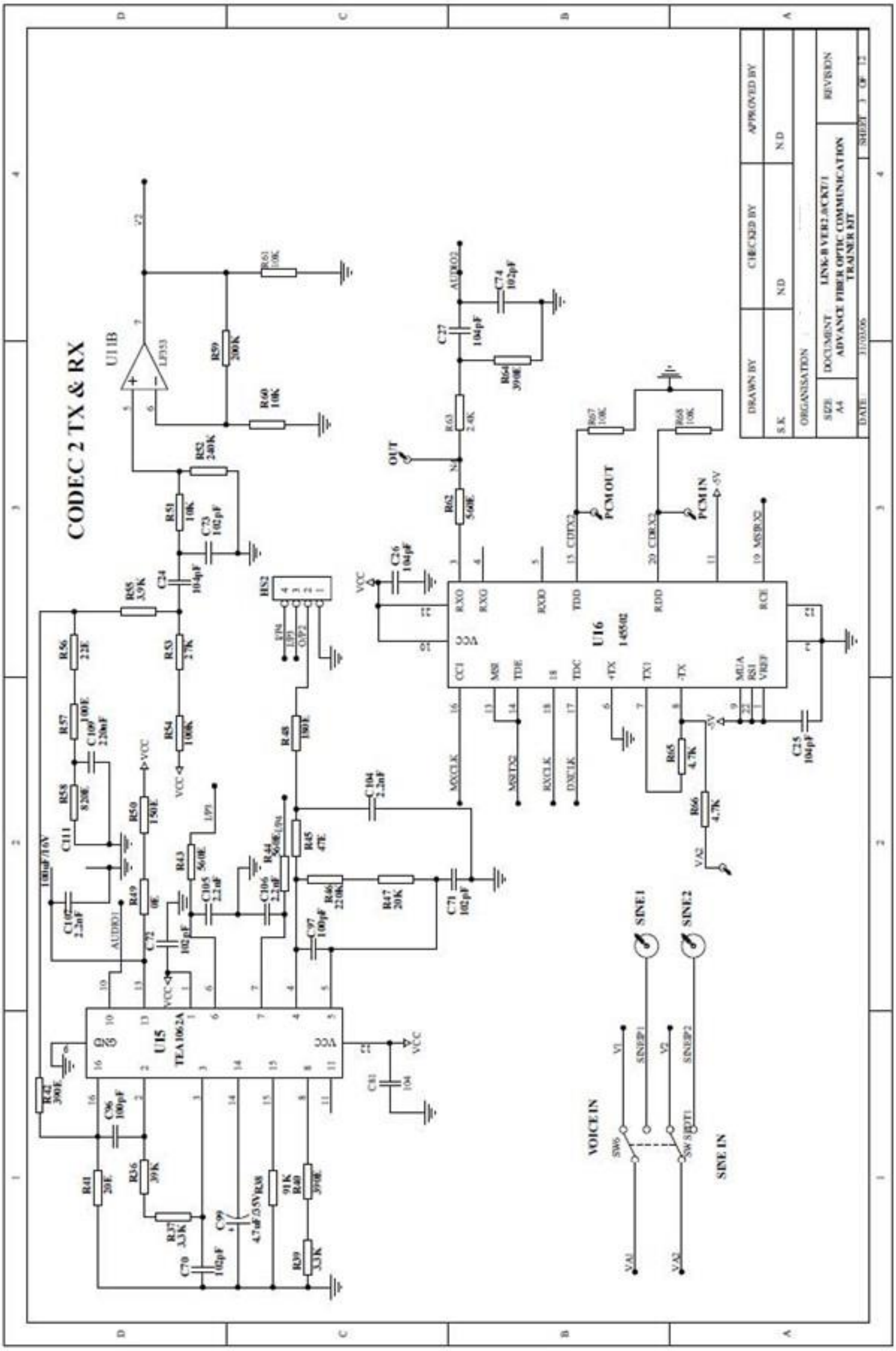
# SWITCH FAULTS

Switch fault section on is provided with 8 switches. These switches can be used to simulate fault conditions in various parts of the circuit. In order to keep FO-A-Pkit fully operational, all switches should be set to OFF position before use. Switch faults are introduced one at a time.

- Put switch **1 (SF1)** in Switch Fault section to **ON** position. This will open TX marker signal at i/p of U9. This will open pin 8 of U9 (74LS150) i.e. channel one is removed at MUX section. No marker is transmitted and hence the receiver synchronization is disturbed.
- Put switch **2 (SF1)** in Switch Fault section to **ON** position. This will open pin 4 of U9 (74LS150). The channel 6 signal goes in undefined state & o/p response is irrespective of the CH6 switch position.
- Put switch **3 (SF1)** in Switch Fault section to **ON** position. This will open pin 13 of U9 (74LS150). This will remove the MUX clock QC. Due to which few channels at the MUX o/p will be lost. Since they will not be selected.
- Put switch **4 (SF1)** in Switch Fault section to **ON** position. This will open pin 8of U12 (145502). The MSIRX1 signal is removed. This will disable the codec selection during demux and codec response cannot be observed.
- Put switch **5 (SF2)** in Switch Fault section to **ON** position. This will open pin 2 MCDCLK of U19 (74HC86). This will affect the Manchester coding and the Manchester coded o/p will be in a wrong format.
- Put switch **6 (SF2)** in Switch Fault section to **ON** position. This will disturb the Manchester decoding o/p and data will be disturbed. R71 comes in parallel toPR1 & hence the RC time constant changes & RX CLK gets disturbed.
- Put switch **7 (SF2)** in Switch Fault section to **ON** position. This will open pin 20 of U24 (74HCT154). Demux clock Q4 will be removed from Demux IC and hence few channels at i/p will not be selected.
- Put switch **8 (SF2)** in Switch Fault section to **ON** position. This will open pin 11of U35 (74LS374). This will open the latching signal of U35 and marker settings hence the U35 will not accept new marker settings & will display the previous settings till the latch signal is connected.

# CIRCUIT DIAGRAM




CODEC 2 TX & RX
U11B
LF353
U16
145502
U15
TEA1062A
HS2
VOICE IN
SINE IN
SINE1
SINE2
PCM OUT
PCM IN
OUT
DRAWN BY
CHECKED BY
APPROVED BY
S.K
N.D
ORGANISATION
DOCUMENT
LINK-B VER2.0/CKT/1
ADVANCE FIBER OPTIC COMMUNICATION TRAINER KIT
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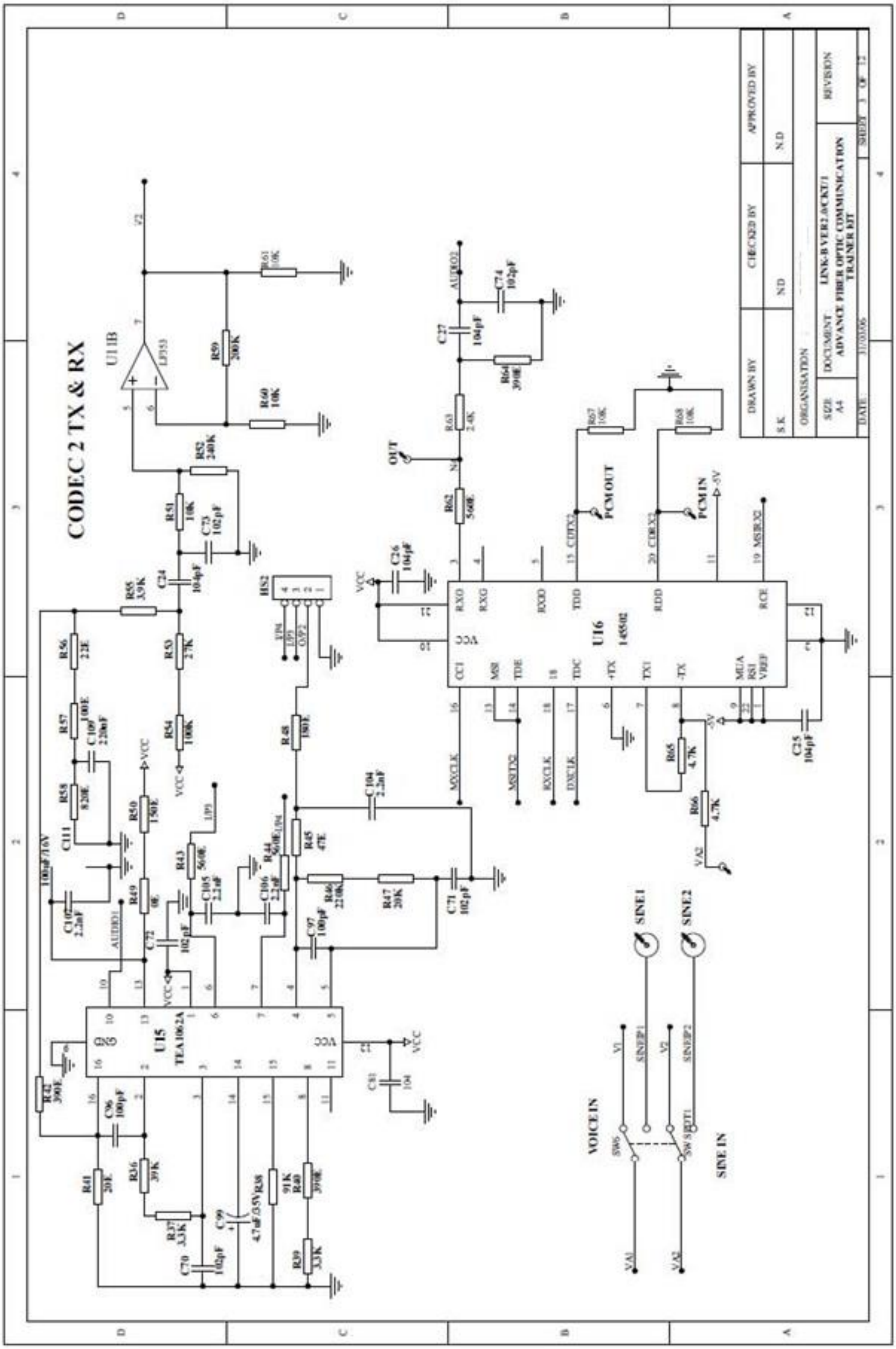
SIZE
A4
DATE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U16
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PCM IN
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SINE2
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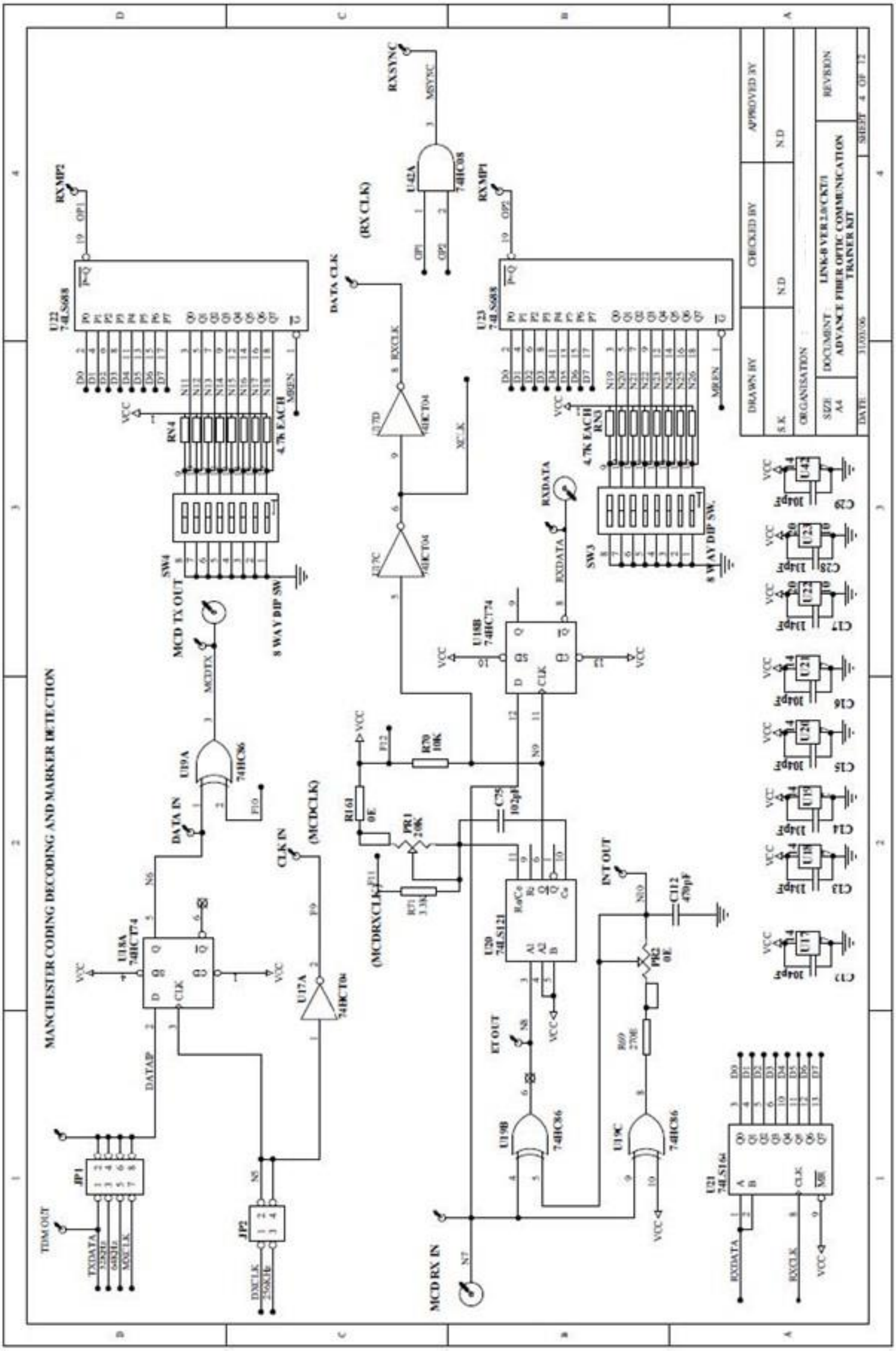
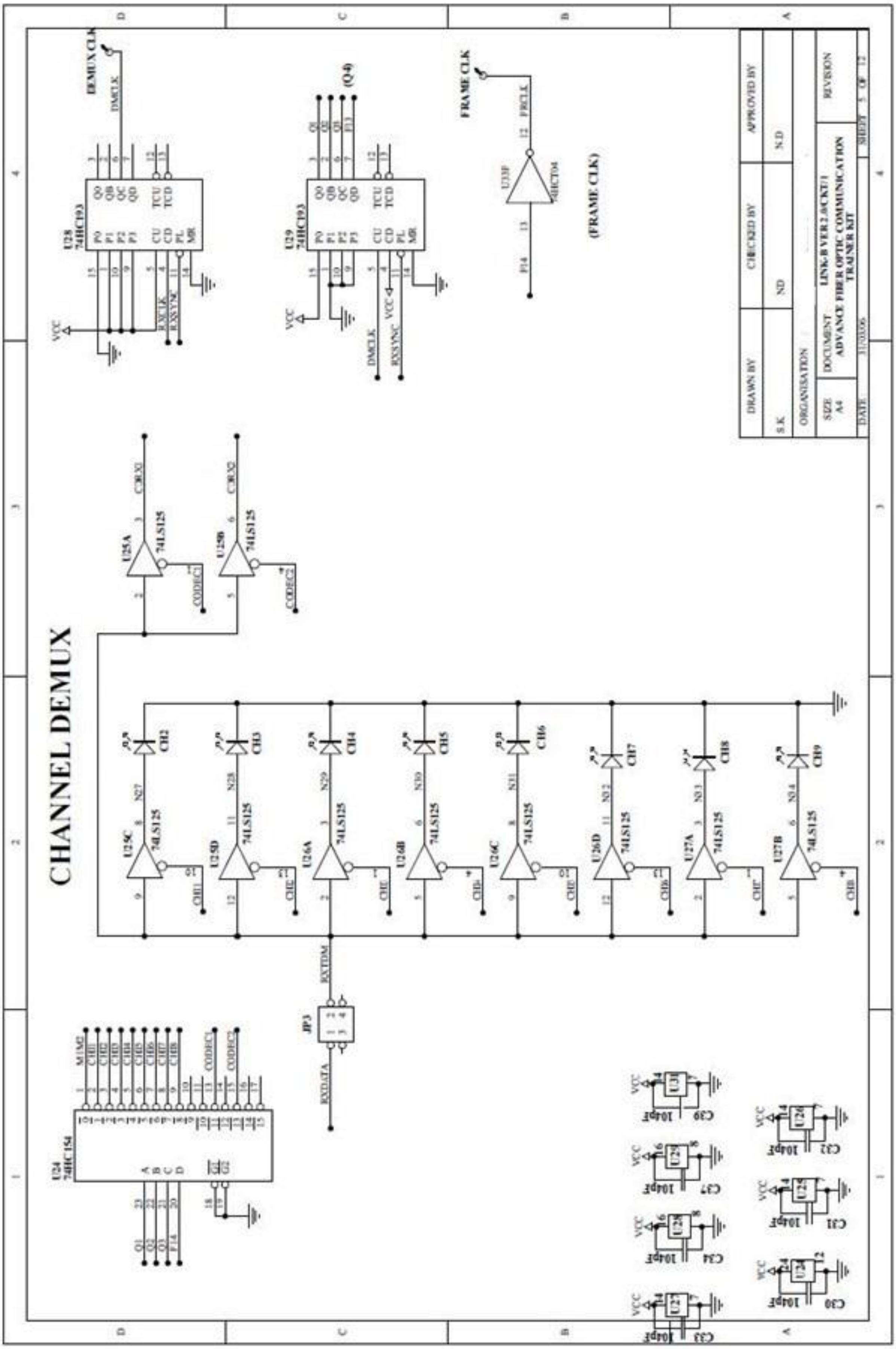
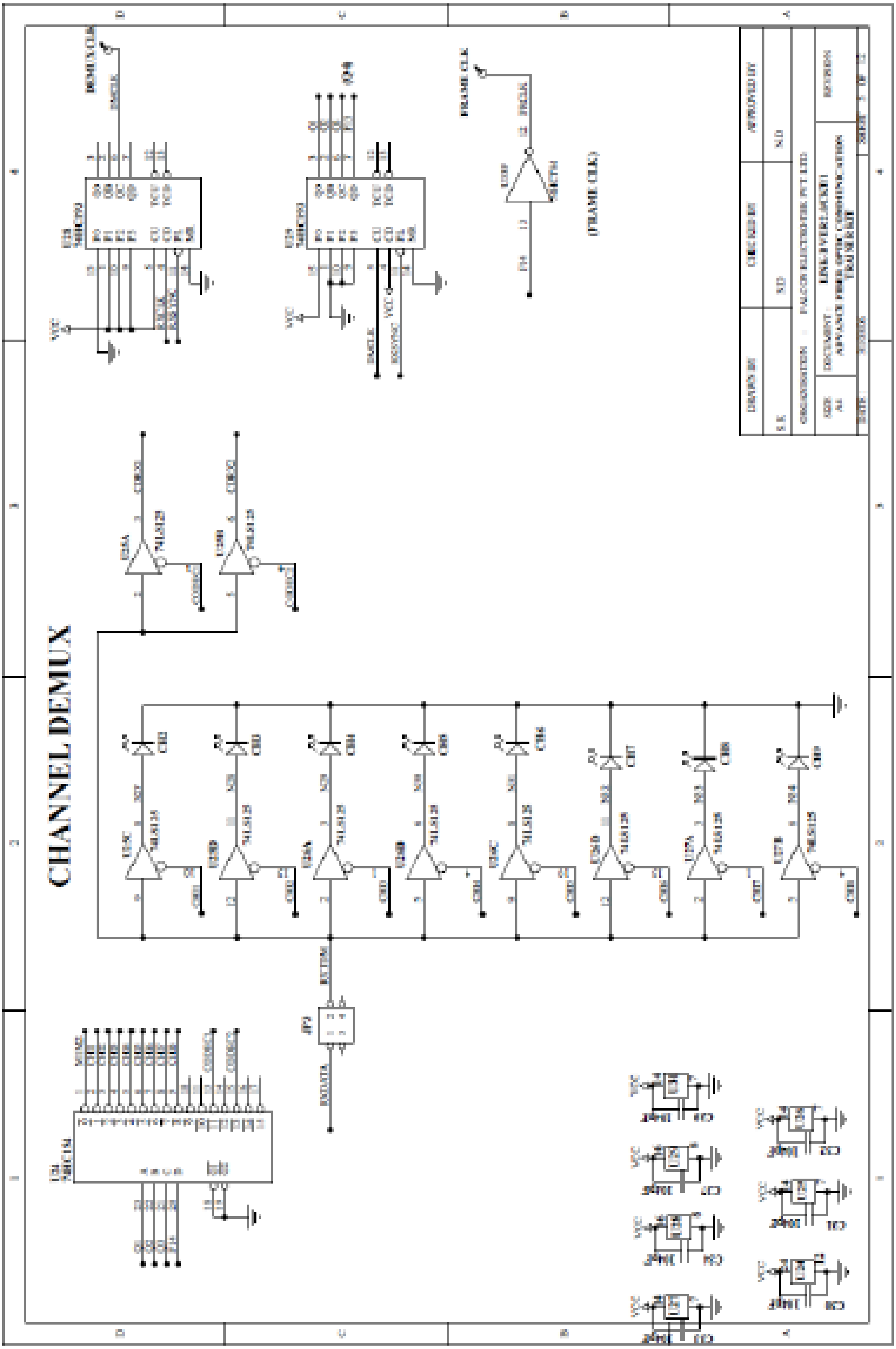
ORGANISATION
REVISION
ADVANCE FIBER OPTIC COMMUNICATION TRAINER KIT


MANCHESTER CODING DECODING AND MARKER DETECTION
LINK-B VER 2.0/CKT/I
ADVANCE FIBER OPTIC COMMUNICATION TRAINER KIT
RXSYNC
RXMP1
RXMP2
DATA CLK
(RX CLK)
RXDATA
MCD TX OUT
8 WAY DIP SW
DATA IN
CLK IN
(MCDCLK)
(MCDRXCLK)
INT OUT
ET OUT
MCD RX IN
TDM OUT


CHANNEL DEMUX
DEMUX CLK
FRAME CLK
(FRAME CLK)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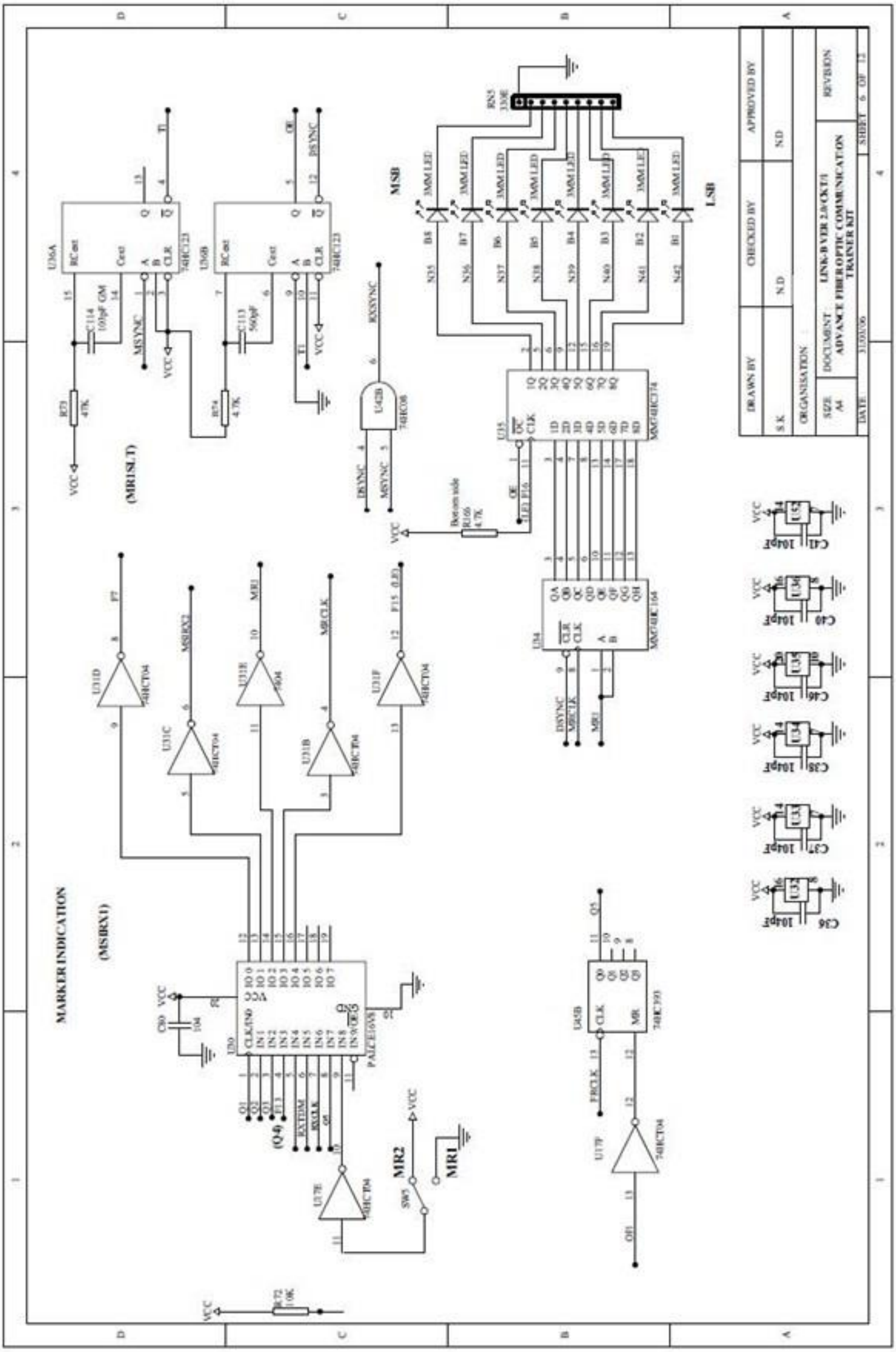
MARKER INDICATION
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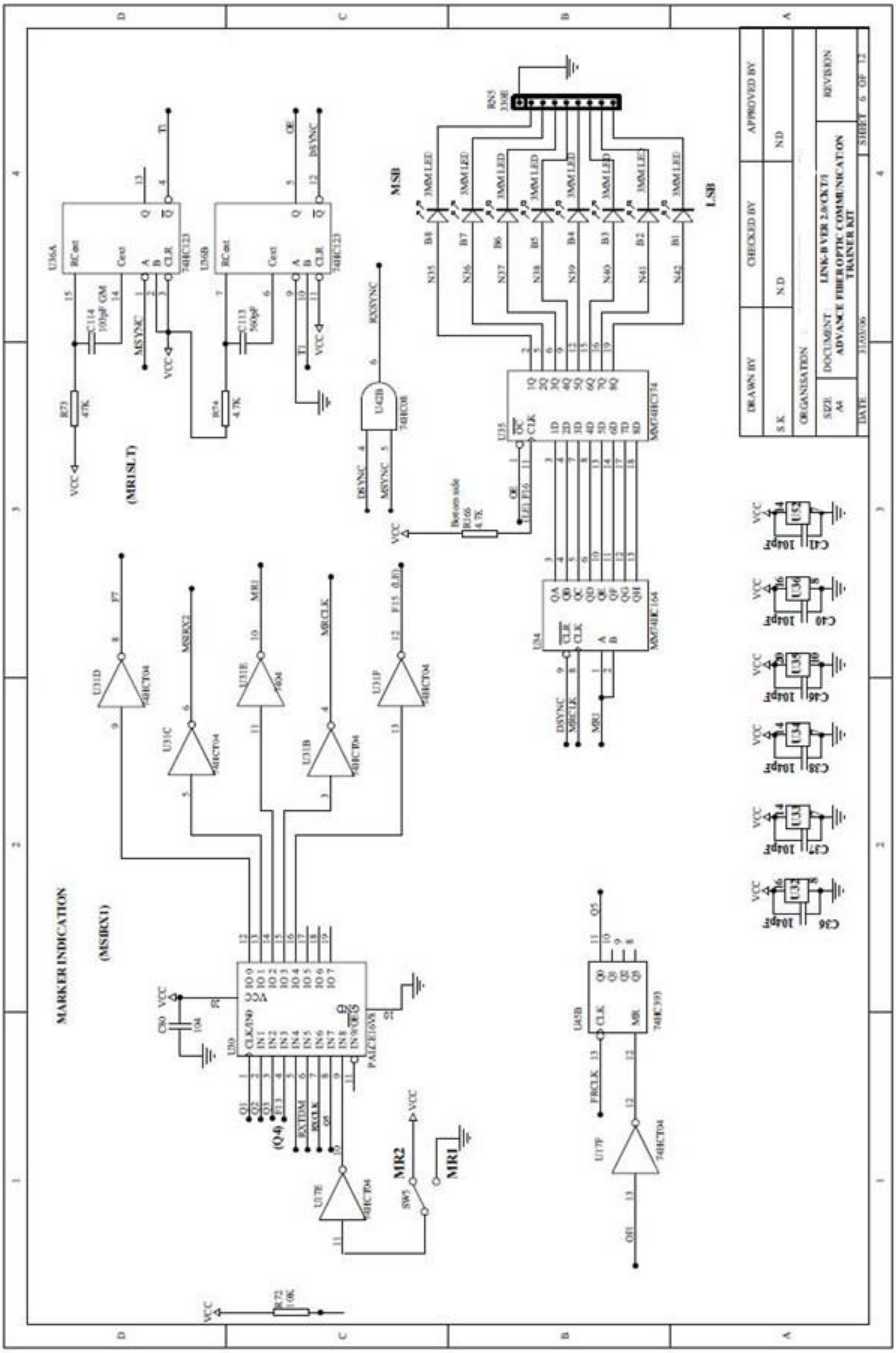
MSB
LSB
MR2
MR1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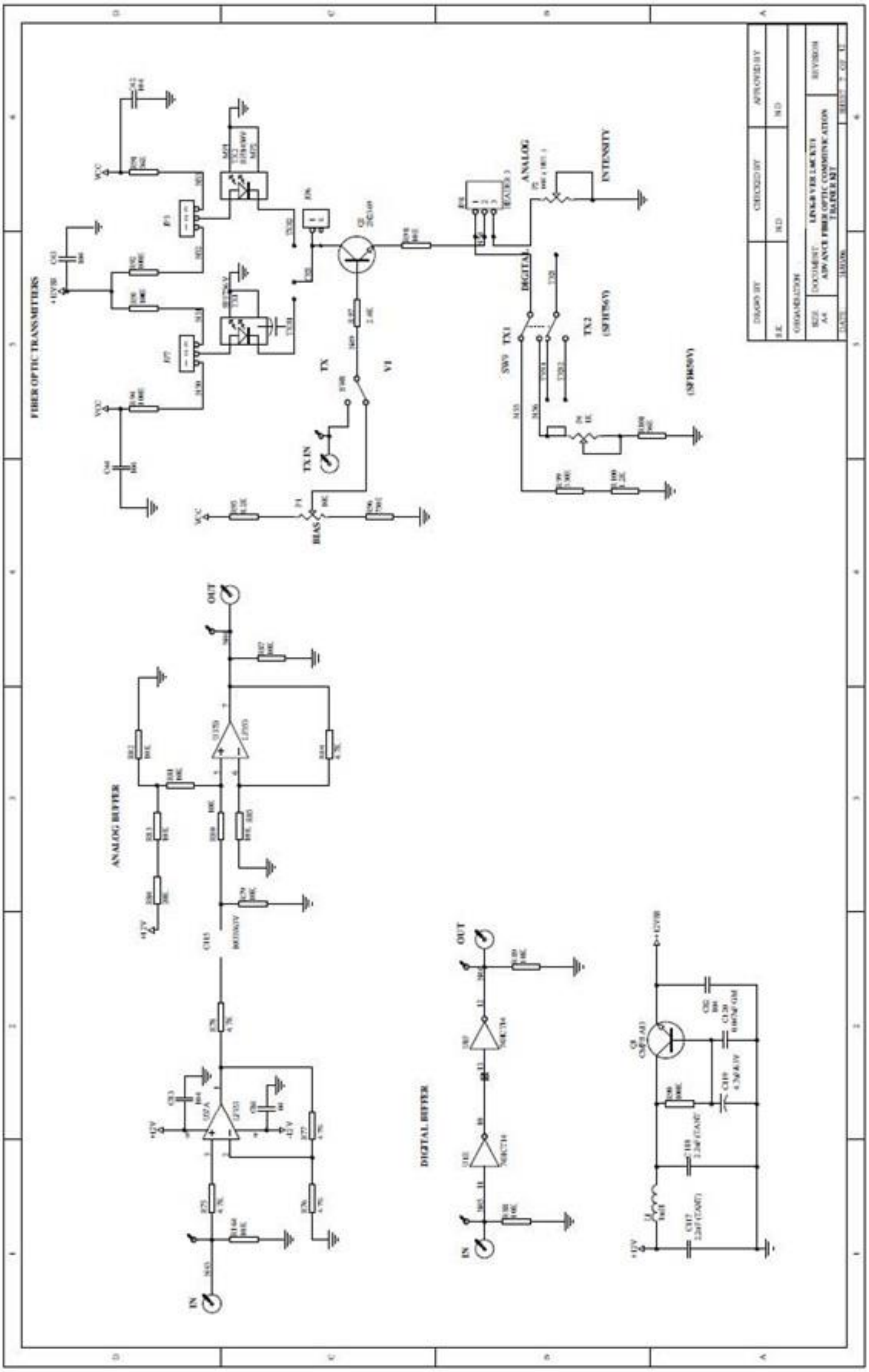
MSB
LSB
MR1
MR2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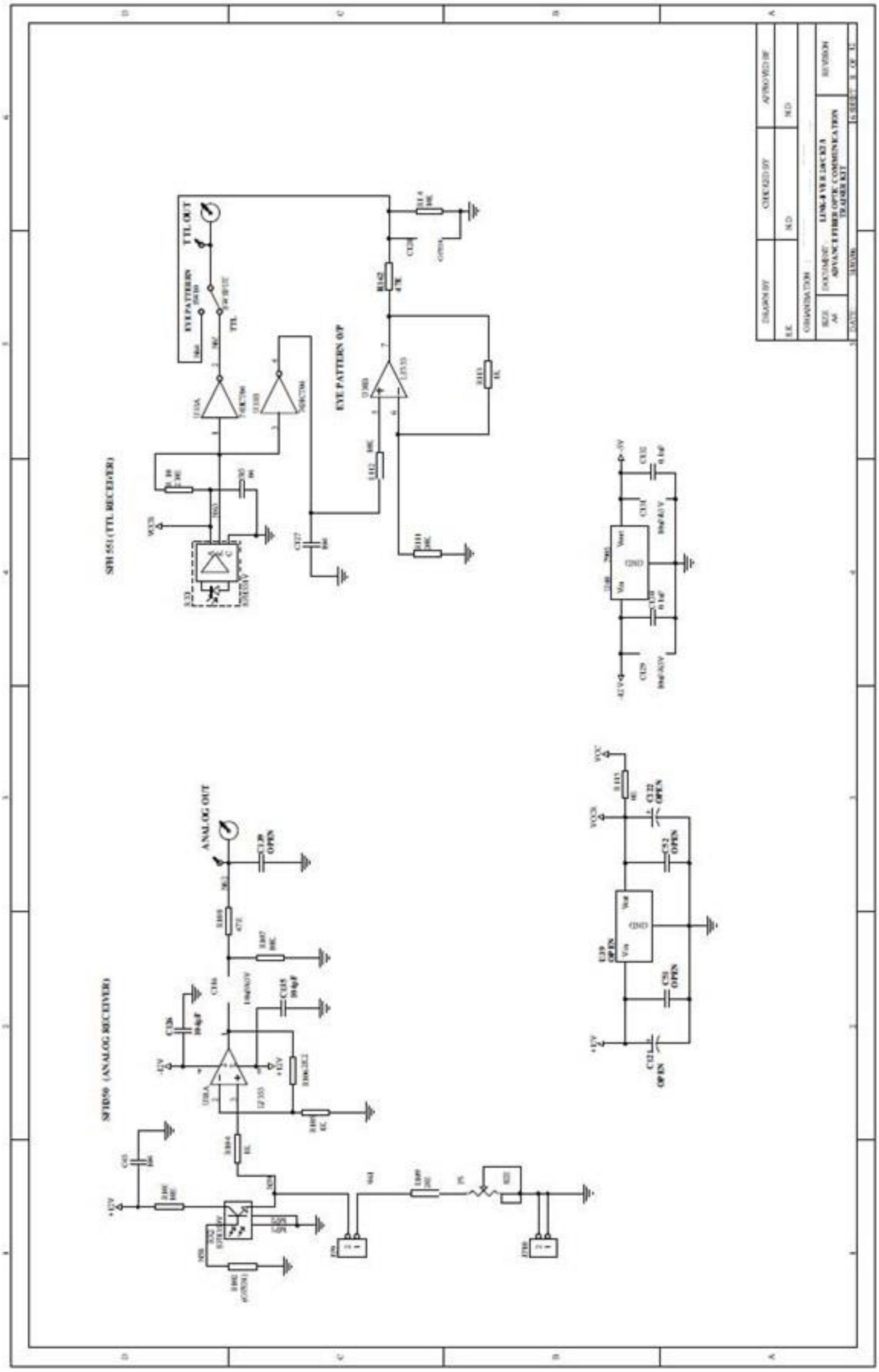
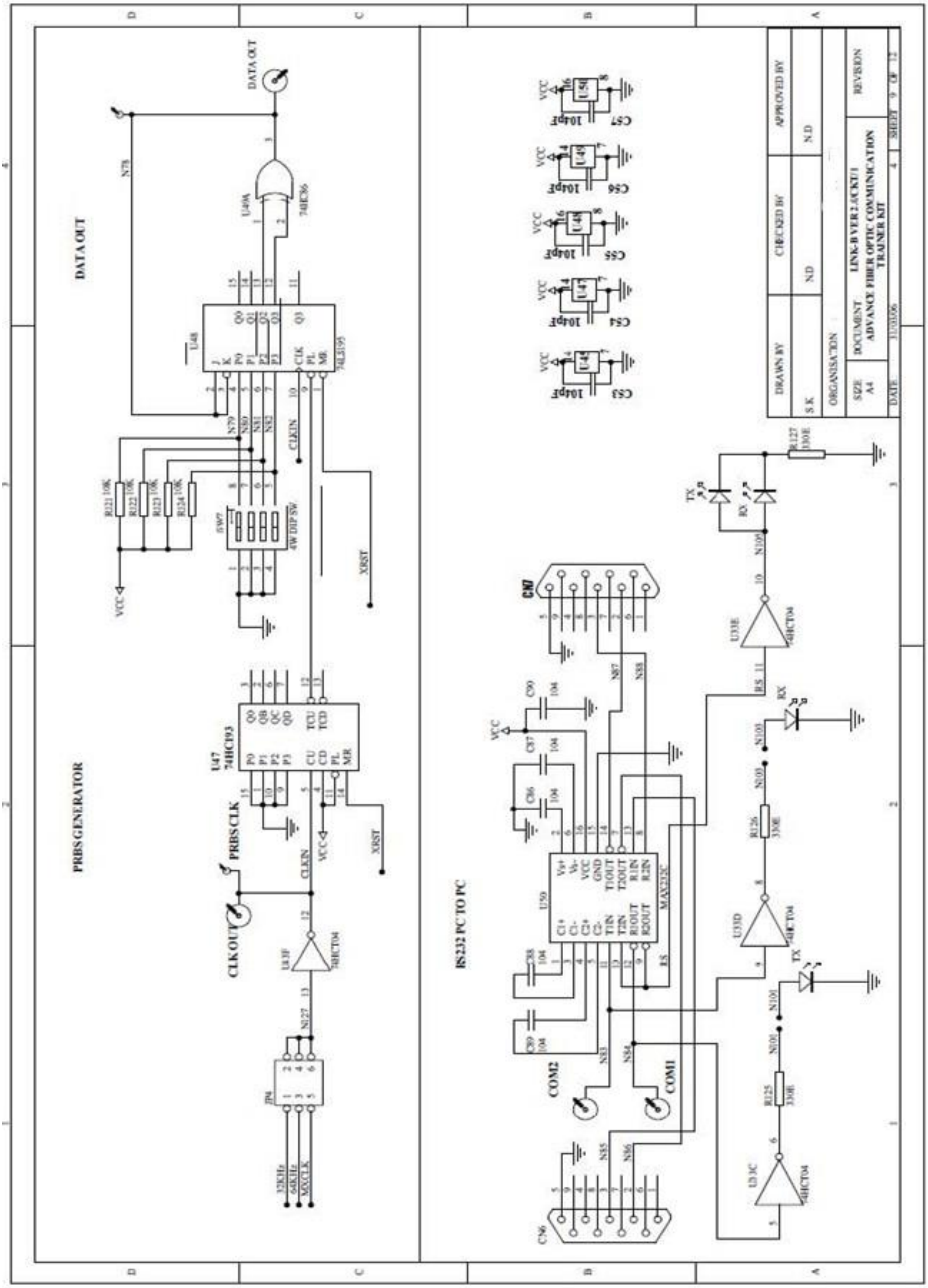
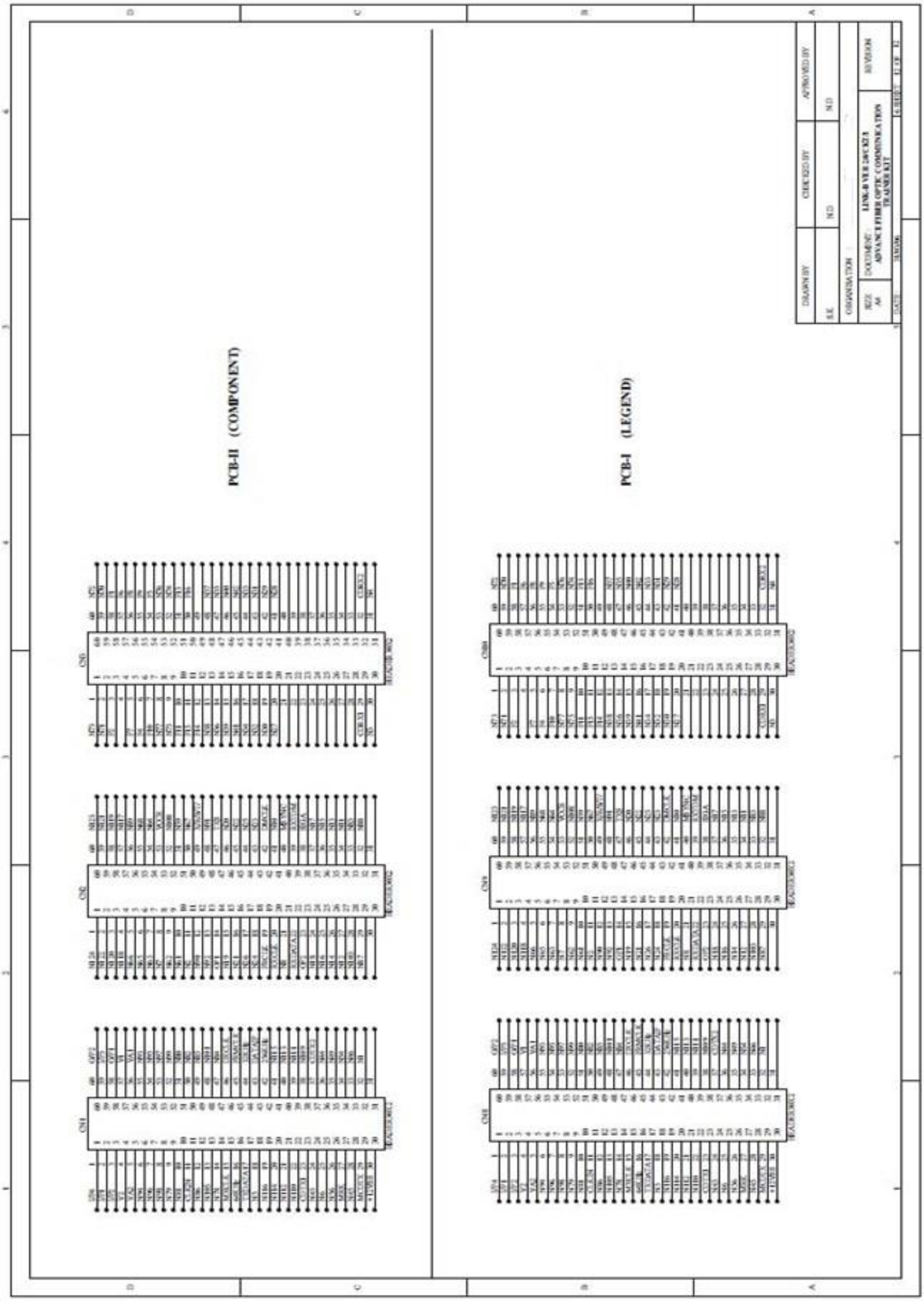
DIGITAL BUFFER
INTENSITY
ANALOG
DIGITAL
BIAS
TX IN
IN
OUT


TTL OUT
EYE PATTERN O/P
ANALOG OUT
SFH 551/1 (TTL RECEIVER)
SFH250 (ANALOG RECEIVER)


DATA OUT
PRBS GENERATOR
RS232 PC TO PC
PRBS CLK
CLK OUT
COM1
COM2
MAX232C
74LS195
74HC193
74HC86
4W DIP SW
DRAWN BY
CHECKED BY
APPROVED BY
ADVANCE FIBER OPTIC COMMUNICATION TRAINER KIT


PCB-II (COMPONENT)
PCB-I (LEGEND)

# ABOUT FIBER OPTICS

Before fiber optics came along, the primary means of real time data communication was electrical in nature. It was accomplished using copper wire or by transmitting electromagnetic (radio) waves through free space. Fiber optics changed that by providing an alternative means of sending information over significant distances – using light energy. Although initially a very controversial technology, today fiber optics has been shown to be very reliable & cost effective.

Light as utilized for communication has major advantages because it can be manipulated (modulated) at significant higher frequencies than electrical signals can. For example, a fiber optic cable can carry up to 100 million times more information than a telephone line! The fiber optic cable has lower energy loss & wider bandwidth capabilities than copper wire.

As you will learn, fiber optic communication is a quite simple technology, closely related to electronics. In fact, it was research in electronics that set the stage for fiber optics to develop into the communication giant that it is today. Fiber optics became reality when several technologies came together at once. It was not an immediate process, nor was it easy, but it was most impressive when it occurred. An example of one critical product, which emerged from that technological merger, was the semiconductor LED that we used in our kit.

**How Optical Fiber Works?**

The principle of operation of optical fiber lies in the behavior of light. Light travels in straight line through most optical materials, but that's not necessarily the case at the junction (interface) of two materials of different refractive indices. Air & water are a case in point as shown in FIG. A. The light ray traveling through air actually is bent as it enters the water. The amount of bending depends on the refractive indices of the two materials involved & also on the angle of the incoming (incident) ray of light as it strikes the interface. The angle of the incident ray is measured from a line drawn perpendicular to the surface. The same is true for the angle of the outgoing (transmitted) ray of light after it has been bent.

Snell's Law explains the mathematical relationship between the incident ray & the refracted ray

$$n_1 \bullet \sin \theta_1 = n_2 \bullet \sin \theta_2$$

In which $n_1$ & $n_2$ are the refractive indices of the initial and secondary materials respectively, and $\Theta_1$ and $\Theta_2$ are the incident and transmitted angles.

Snell's Law says that refraction (bending) of light cannot take place when the angle of incidence grows too large (as when light travels from a material with a high refractive index to one with a low refractive index). If the angle of incidence exceeds a certain critical value (in which the product on nl& the sine of the angle equals or exceeds one.) light cannot exit.

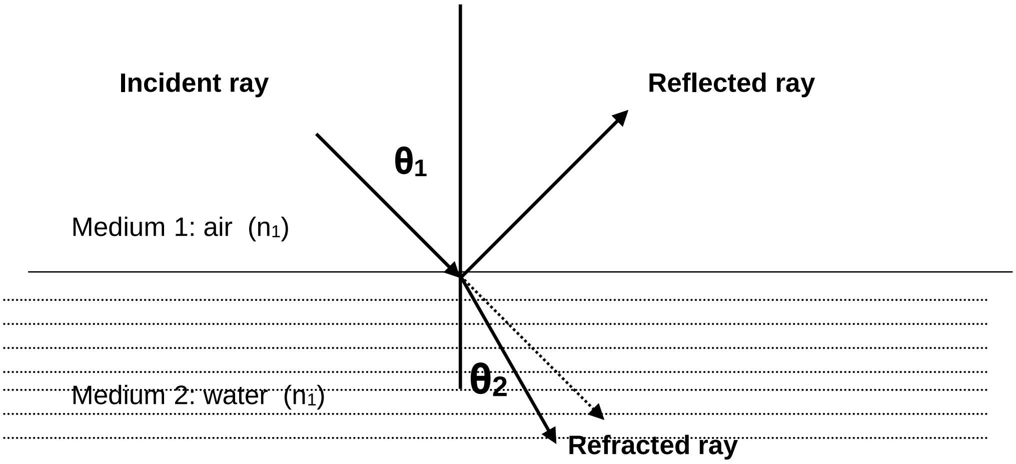



**Fig. A: The Different Portions Of A Light Ray At A Material Interface**

If light cannot exit the material, it is reflected. The angle that is reflected is equal to the angle of incidence. The phenomenon just described is called Total Internal Reflection and it is what keeps light inside an optical fiber.

## Types of Optical Fiber

The simplest fiber optic cable consists of two concentric layers of transparent materials. The inner portion (the core) transports the light. The outer covering (the cladding) must have a lower refractive index than the core, so the two are made of different materials. The cable used in this kit also has a jacket to protect the optical properties of the core & cladding.

Optical fiber is generally made from either plastic or glass. This kit uses plastic fiber that is very easy to terminate & does not require special tools. Plastic fiber is generally limited to uses involving distance of less than 100 meters because of high attenuation (higher loss).

Glass fiber on other hand has very low attenuation (low loss) & hard to cut, requires special end connections & is more expensive.

The core of fiber is typically made of silica doped with impurities, which increases the refractive index relative to pure silica. The cladding, which surrounds the fiber core & creates an optical interface, is typically made from pure silica. The outer buffer coating is a plastic cover.

The first type of fiber optic cable put to use was called Step Index. In this design the cladding has a different index of refraction from the core. The light bounces off the sides & is reflected back into the fiber core. The problem with this design is that the reflected light must travel a slightly longer distances than that which travels down the center of the fiber, thus limiting the maximum transmission rate. This design was improved with the use of Graded Index Fiber. In this design, the index of refraction decreases in proportion to the distance away from the center of the fiber core. The light moves more quickly in the outer portion, thus compensating for the additional distance. The change in index also has the effect of ‘bending’. The light reflects back towards the core. This change increases the transmission capacity. In the newest Single Mode design, the diameter of the fiber core is made so small that all the light travels in a straight line. Even the latest fiber optic facility in use today uses less than 5% of the maximum theoretical capacity of a Single mode Fiber.

## Single Mode V/S Multimode

There are essentially two different types of fiber optic transmission schemes in use. Multimode fiber optic systems are typically used in short haul applications such as LANS & FDDI. Recent growth in multimode installations has been driven by LAN/FDDI applications. The term Multimode means that the diameter of the fiber optic core is large enough to propagate more than one mode (electromagnetic wave). Because of the multiple modes the pulse that is transmitted down the fiber tends to become stretched over distance. This is referred to as Modal Dispersion & has the effect of reducing available bandwidth.

**Fig. B: Modal Dispersion of Light**

Typical wavelengths used in multimode applications are 850 nm & 1300 nm. There are other wavelengths used (e.g. 660 nm for plastic fiber, 820/870 nm) in specialized applications such as the military. The vast majority of multimode applications are 850 or 1300 nm wavelengths. Most multimode cables in use today are graded index in type with a cladding diameter of 125 microns & fiber core diameter of 62.5 or 50 microns (Typically referred to as 62.5/125 or 50/125 fiber)

Single Mode Systems are generally used in long haul applications (10’s or 100’s of kilometers) & local loop applications in both the carrier & CATV markets. Some large utilities such as electrical power companies also utilize single mode fiber optic backbones. As the name implies single mode fiber is designed to propagate only one mode of light. It is therefore not affected by modal dispersion & has higher bandwidth capacity. It is considered to be higher quality transmission system. Wavelengths used in single mode applications are 1300 & 1550 nm. The majority of today’s installed systems uses 1300 nm wavelength technology & is designed for higher speed applications. These systems are more sensitive to back reflections from connectors & sharp cable bends. They often require additional tests, such as Optical Return Loss, to be run. Most single mode cable in used today are step index in type with cladding diameter of 125 micron & fiber core diameter of 9 micron (Typically referred to as 9/125 fiber).

## Advantages of Fiber Optics Over Conventional Copper Cables

- Much greater bandwidth (Several orders of magnitude are theoretically possible).
- Immunity to electrical disturbances, ground loops, cross talks etc. in addition, no EMI radiation is generated.
- Much lighter. A 3-inch bundle of 900 twisted copper pairs can carry about 21,000 channels of traffic & weighs about 25,000 pounds/mile. A1/2 inch fiber link with 12 fiber strands can carry about 3,00,000 channels & weighs about 200 pounds/mile. In addition less duct space is required to make the cable within building.
- Better in hostile environment, not affected as much by temperature, water etc.
- Lower transmission loss. Typical loss on a fiber link is 0.2 dB/Km. On a copper based facility, one can usually expect at least 5 db/km. For longer links, fewer repeaters are required.
- Better security as it is not possible to simply bridge onto the facility & monitor the traffic.

With all these properties of optical fiber, it is becoming very important in sensing field also. Most of the properties of optical fiber are useful for sensors. Further optical fiber sensors offer extreme sensitivity & can be used to sense almost any physical parameters like temperature, velocity, current, voltage etc.

# ABOUT FOL-A-P
# ADVANCE FIBER OPTIC COMMUNICATION TRAINER KIT

The purpose of FOL-A-P Trainer Kit is to provide you hands on experience on the various Fiber Optics & digital Communication Techniques. In preparing this manual, we assumed you have a basic grasp of digital & transistor circuits. An experiment starts with basic principles of Fiber Optics & then go on to more sophisticated experiments. Every advanced experiment uses concept, procedures and results of the previous experiment. So you are advised not to skip in between experiments. In this manual before starting the procedure of every experiment objective of the experiment & basic theory related to that is given. You are advised to read it before starting the experiment because it will help you to understand the experiment very well. Also wherever necessary refer the datasheets of various components provided in the backside of the circuit description manual.

Before starting the experiments check following things in the kit.
• Trainer Kit FOL-A-P.
• Plastic Fiber of 1 & 3 Meter lengths.
• NA JIG.
• Steel Rule.
• 2 Nos. of telephones handsets.
• Power cord and connecting Cables.
• Power supply.
• Experimental Manual.

**Optical Fiber Preparation Instructions**
The Fibers provided with this kit are multimode, step index, 1000 micron Plastic Fiber & are factory prepared for use. They are not require to prepared again even after several operations, if handled carefully. In case the Fiber is broken or damaged you have to prepare each end of the Optical Fiber cable very carefully so it transmits light effectively.

• Cut off the ends of the cable with a single-edge razor blade or sharp knife. Try to obtain a precise 90-degree angle.

• Wet the polishing paper with water or light oil & place it on a flat, firm surface. Hold the Optical Fiber upright, at angle to the paper & polish the Fiber tip with a gentle ”Figure 8” motion. You may get the best result by supporting the upright Fiber against some flat object such as a portion of the printed circuit board.

# EXPERIMENT NO. 1

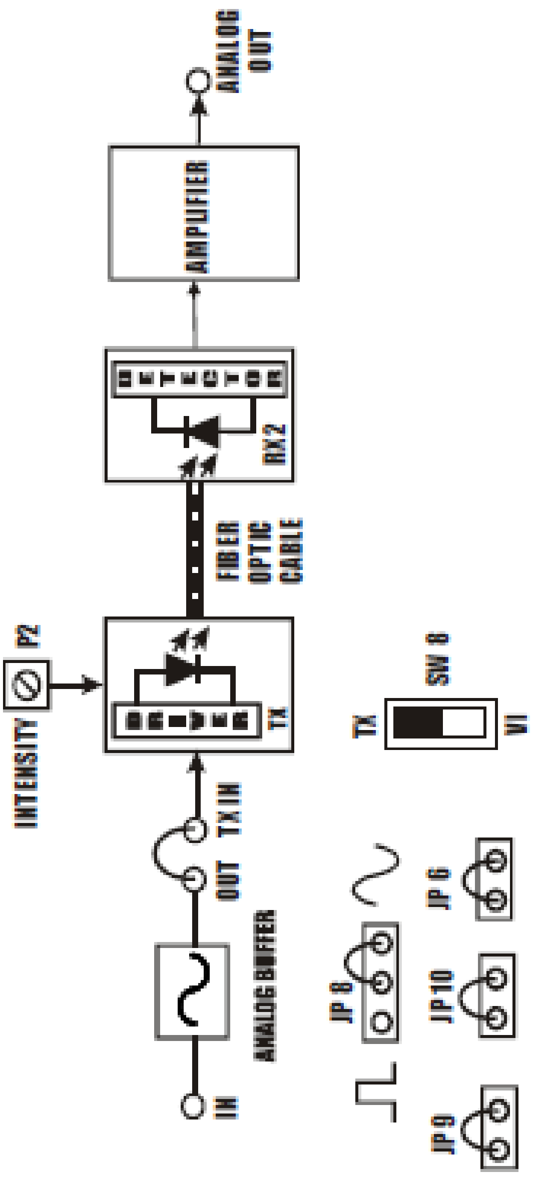
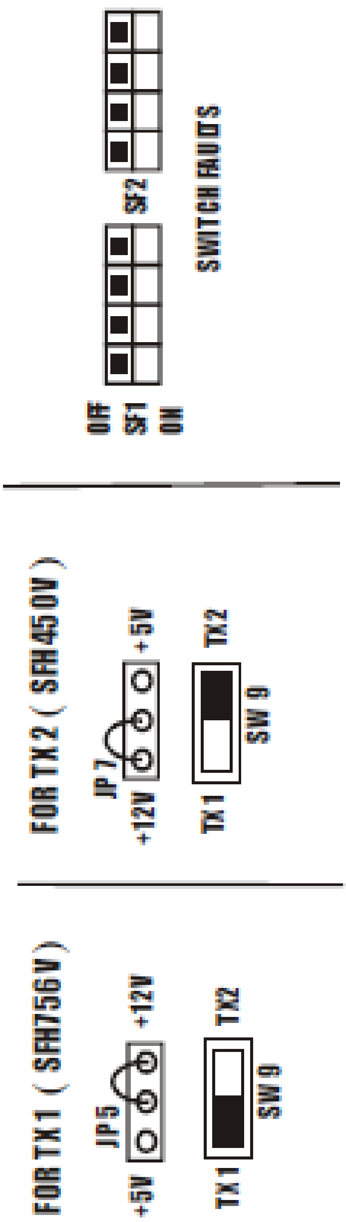



FIG. 1.1 BLOCK DIAGRAM FOR SETTING UP FIBER OPTIC ANALOG LINK

JUMPER SETTING DIAGRAM FOR ANALOG LINK



# EXPERIMENT NO.1

## NAME

Setting Up a Fiber Optic Analog Link & Bandwidth Measurement.

## OBJECTIVE

The objective of this experiment is to study a 660 nm/ 950 nm Fiber Optic Analog Link. In this experiment you will study the relationship between the input signal & received signal.

## THEORY

Fiber Optic Links can be used for transmission of digital as well as analog signals. Basically a fiber optic link contains three main elements, a transmitter, an optical fiber & a receiver. The transmitter module takes the input signal in electrical form & then transforms it into optical (light) energy containing the same information. The optical fiber is the medium, which carries this energy to the receiver. At the receiver, light is converted back into electrical form with the same pattern as originally fed to the transmitter.

**Transmitter**

Fiber optic transmitters are typically composed of a buffer, driver & optical source. The buffer electronics provides both an electrical connection & isolation between the transmitter & the electrical system supplying the data. The driver electronics provides electrical power to the optical source in a fashion that duplicates the pattern of data being fed to the transmitter. Finally the optical source (LED) converts the electrical current to light energy with the same pattern. The LED SFH756V supplied with the kit operates inside the visible light spectrum. Its optical output is centered at near visible wavelength of 660 nm. The emission spectrum is broad, so a dark red glow can usually be seen when the LED is on. The LED SFH450V supplied with the kit operates outside the visible light spectrum. Its optical output is centered at near infrared wavelength of 950 nm.

**Receiver**

The function of the receiver is to convert the optical energy into electrical form, which is then conditioned to reproduce the transmitted electrical signal in its original form. The detector SFH250V used in the kit has a diode type output. The parameters usually considered in the case of detector are its responsivity at peak wavelength & response time. SFH250V has responsivity of about 4 $\mu$A per 10 $\mu$W of incident optical energy at 950 nm and it has rise& fall time of 0.01 $\mu$ sec.

PIN photodiode is normally reverse biased. When optical signal falls on the diode, reverse current starts to flow, thus diode acts as closed switch and in the absence of light intensity, it acts as an open switch. Since PIN diode usually has low responsivity, a trans impedance amplifier is used to convert this reverse current into voltage. This voltage is then amplified with the help of another amplifier circuit. This voltage is the duplication of the transmitted electrical signal, which can be amplified.

# EQUIPMENTS

- FOL-A-P Kit with power supply
- Patch chords
- 20MHz Dual Channel Oscilloscope
- Function Generator
- 1-Meter Fiber Cable

**NOTE**: Keep Switch Faults In Off Position

# PROCEDURE

- Make connections as shown in **FIG.1.1.** Connect the power supply cables with proper polarity to FOL-A-P Kit. While connecting this, ensure that the power supply is OFF.
- Keep switch **SW8** towards **TX** position.
- Keep switch **SW9** towards **TX1** position.
- Keep Jumper **JP5** towards **+12V** position.
- Keep Jumpers **JP6, JP9, JP10** shorted.
- Keep Jumper **JP8** towards **sine** position.
- Keep Intensity control pot **P2** towards **minimum** position.
- Switch ON the power supply.
- Feed about **2Vpp** sinusoidal signal of **1KHz**from the function generator to the **IN** post of Analog Buffer.
- Connect the output post **OUT** of Analog Buffer to the post **TX IN** of Transmitter.
- Slightly unscrew the cap of SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the one-meter fiber into the cap. Now tighten the cap by screwing it back.
- Connect the other end of the Fiber to detector SFH350V (Photo Transistor Detector) very carefully as per the instructions in above step.
- Observe the detected signal at post **ANALOG OUT** on oscilloscope as shown in **FIG 1.2.** Adjust Intensity control pot **P2** Optical Power control potentiometer so that you receive signal of 2Vpp amplitude.
- To measure the analog bandwidth of the phototransistor, vary the input signal frequency and observe the detected signal at various frequencies.
- Plot the detected signal against applied signal frequency and from the plot determine the 3dB down frequency.
- Repeat the same procedure as above for second transmitter SFH450V by making the following changes. Analog bandwidth of SFH350 for TX1 SFH756 is about 300 KHz while for TX2 SFH450 is below 300 KHz.
- Keep switch **SW9**towards **TX2** position.
- Keep Jumper **JP7** towards **+12V** position.

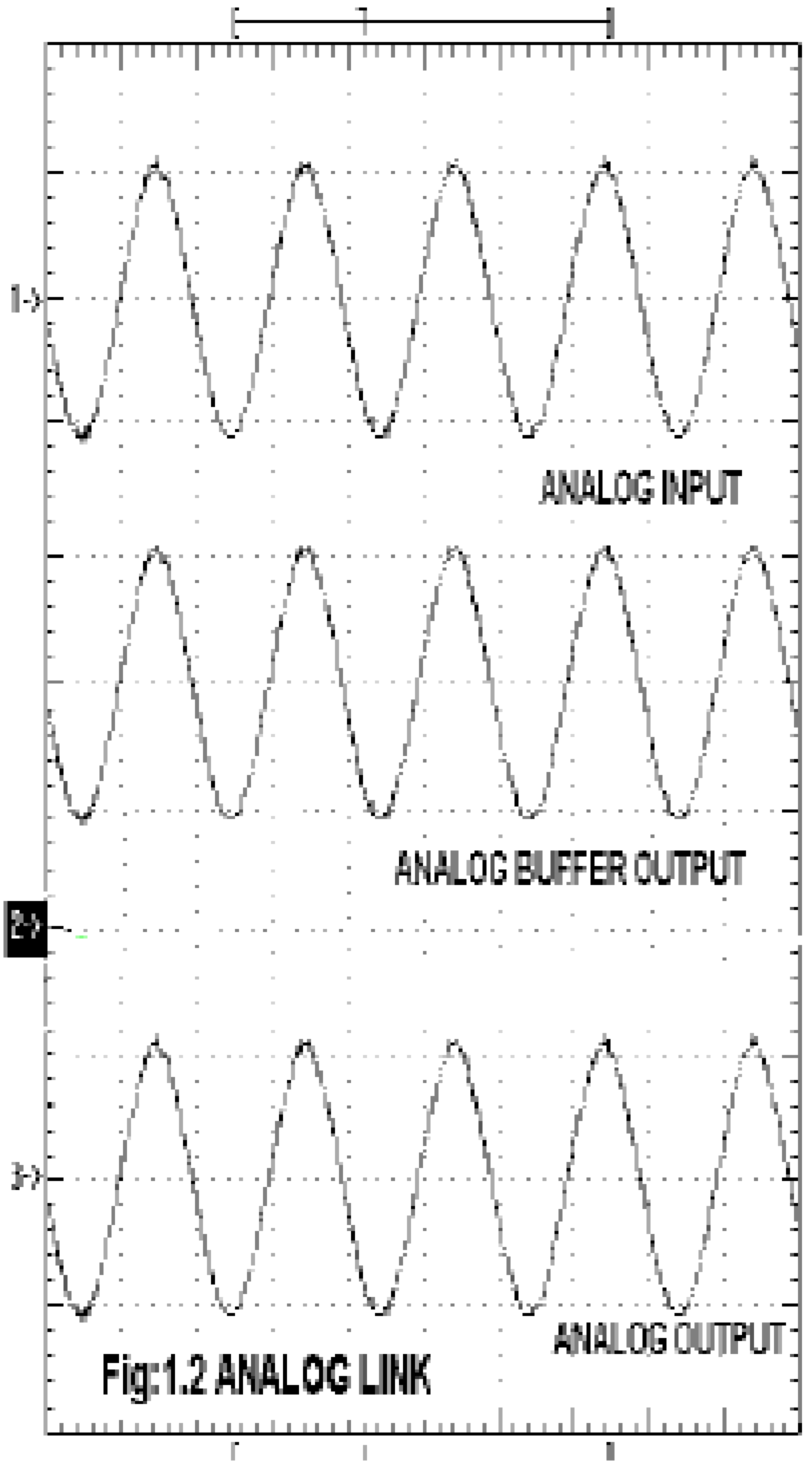



Fig:1.2 ANALOG LINK

Analog Bandwidth Measurement

# PROCEDURE

- Make connections as shown in **FIG.1.1.** Connect the power supply cables with proper polarity to FOL-A-P Kit. While connecting this, ensure that the power supply is OFF.
- Keep switch **SW8** towards **TX** position.
- Keep switch **SW9** towards **TX1** position.
- Keep Jumper **JP5** towards **+12V** position.
- Keep Jumpers **JP6, JP9, JP10** shorted.
- Keep Jumper **JP8** towards **sine** position.
- Keep Intensity control pot **P2** towards **minimum** position.
- Switch ON the power supply.
- Feed about **2Vpp** sinusoidal signal of **1KHz** from the function generator to the **IN** post of Analog Buffer.
- Connect the output post **OUT** of Analog Buffer to the post **TX IN** of Transmitter.
- Slightly unscrew the cap of SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the one-meter fiber into the cap. Now tighten the cap by screwing it back.
- Connect the other end of the Fiber to detector SFH350V (Photo Transistor Detector) very carefully as per the instructions in above step.
- Observe the detected signal at post **ANALOG OUT** on oscilloscope as shown in **FIG 1.2.** Adjust Intensity control pot **P2** Optical Power control potentiometer so that you receive signal of 2Vpp amplitude.
- To measure the analog bandwidth of the phototransistor, vary the input signal frequency and observe the detected signal at various frequencies.
- Plot the detected signal against applied signal frequency and from the plot determine the 3dB down frequency.
- Repeat the same procedure as above for second transmitter SFH450V by making the following changes. Analog bandwidth of SFH350 for TX1 SFH756 is about 300 KHz while for TX2 SFH450 is below 300 KHz.
- Keep switch **SW9**towards **TX2** position.
- Keep Jumper **JP7** towards **+12V** position.

| Input Voltage (Vin) | Input Frequency (KHz) | Output Voltage_V0 (V) | 20Log(V0/Vin) |
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Bandwidth =

# EXPERIMENT NO. 2

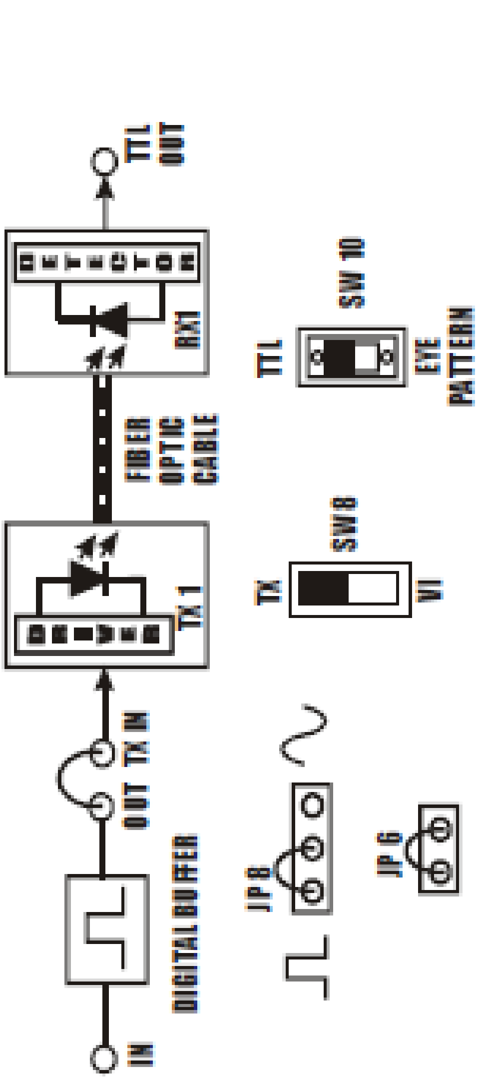
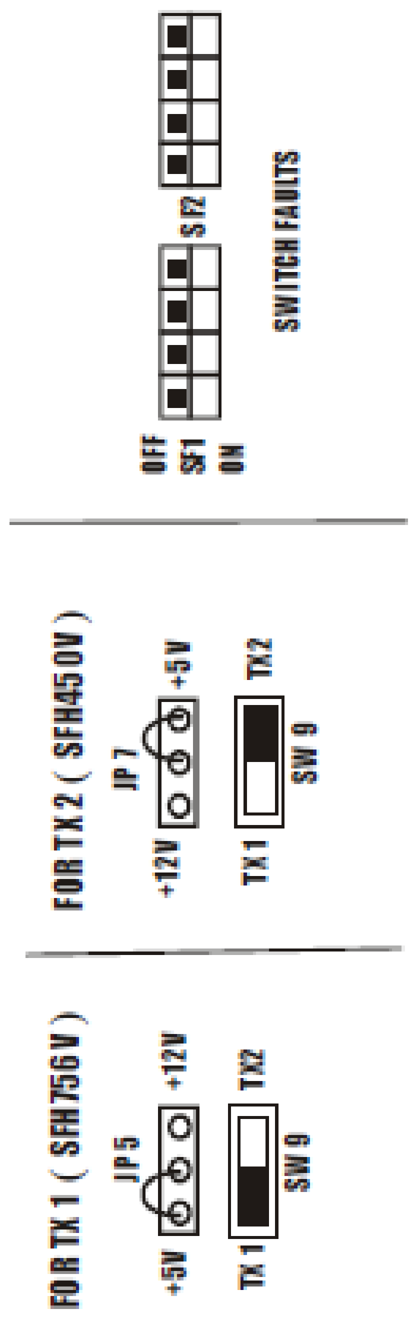



FIG. 2.1 BLOCK DIAGRAM FOR SETTING UP FIBER OPTIC DIGITAL LINK

JUMPER SETTING DIAGRAM FOR DIGITAL LINK



# EXPERIMENT NO. 2

## NAME

Setting Up a Fiber Optic Digital Link

## OBJECTIVE

The objective of this experiment is to study a 660nm & 950nm Fiber Optic Digital Link. Here you will study how digital Signal can be transmitted over Fiber Cable & reproduced at the receiver end.

## THEORY

In the Experiment No.1 we have seen how analog signals can be transmitted &received using LED, Fiber & Detector can be configured for the digital applications to transmit binary data over Fiber. Thus basic elements of the link remain same even for digital application.

**Transmitter**
LED, digital DC coupled transmitters are one of the most popular varieties due to their ease of fabrication. We have used a standard TTL gate to drive a NPN transistor, which modulates the LED SFH450V or SFH 756V source (Turns it on & off).

**Receiver**
SFH-551V is a digital opto-detector. It delivers a digital output, which can be processed directly with little additional external circuitry. The integrated circuit inside the SFH551V opto-detector comprises the photodiode device, a trans-impedance amplifier, a comparator and a level shifter. The photodiode converts the detected light into a photocurrent. With the aid of an integrated lens the light emanating from the plastic Fiber is almost entirely focused on the surface of the diode. At the next stage the trans-impedance amplifier converts the photocurrent into a voltage. In the comparator, the voltage is compared to a reference voltage. In over to ensure good synchronism between the reference and the trans-impedance output voltage, the former is derived from a second circuit of a similar kind, which incorporates a “blind” photodiode. The comparator derives a level shifter with an open collector output stages. Here a catch diode (similar to Schottky-TTL) prevents the saturation of the output transistor, thus limiting the output voltage to the supply voltage.

## EQUIPMENTS

- FO-A-P Kit with power supply
- Patch chords
- 20MHz Dual Channel Oscilloscope
- Function Generator
- 1-Meter Fiber Cable

**NOTE:** Keep All Switch Faults In Off Position.

## PROCEDURE

- Make connections as shown in **FIG.2.1.** Connect the power supply cables with proper polarity to FO-A-P Kit. While connecting this, ensure that the power supply is OFF.
- Switch ON the power supply.
- Feed **TTL** Square wave signal of **1KHz** from the function generator to the **IN** post of Digital Buffer.
- Connect the output post **OUT** of Digital Buffer to the post **TX IN** of Transmitter.
- Slightly unscrew the cap of SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the One Meter Fiber into the cap. Now tighten the cap by screwing it back.
- Connect the other end of the Fiber to detector SFH551V very carefully as per the instructions in above step.
- Observe the detected signal at post **TTL OUT** on oscilloscope as shown in **FIG 2.2.**
- To measure the digital bandwidth of the phototransistor vary the input signal frequency and observe the detected signal at various frequencies.
- Determine the frequency at which the detector stops recovering the signal.
- This determines the maximum bit rate on the digital link.
- Keep switch **SW9** towards **TX2** position.
- Keep Jumper **JP7** towards **+5V** position.
- Repeat the same procedure above for second transmitter SFH450V by making the following changes.
- The digital bandwidth of SFH551 for TX1 SFH756 is 3MHz & for SFH450 it is 1MHz.

## WAVEFORM

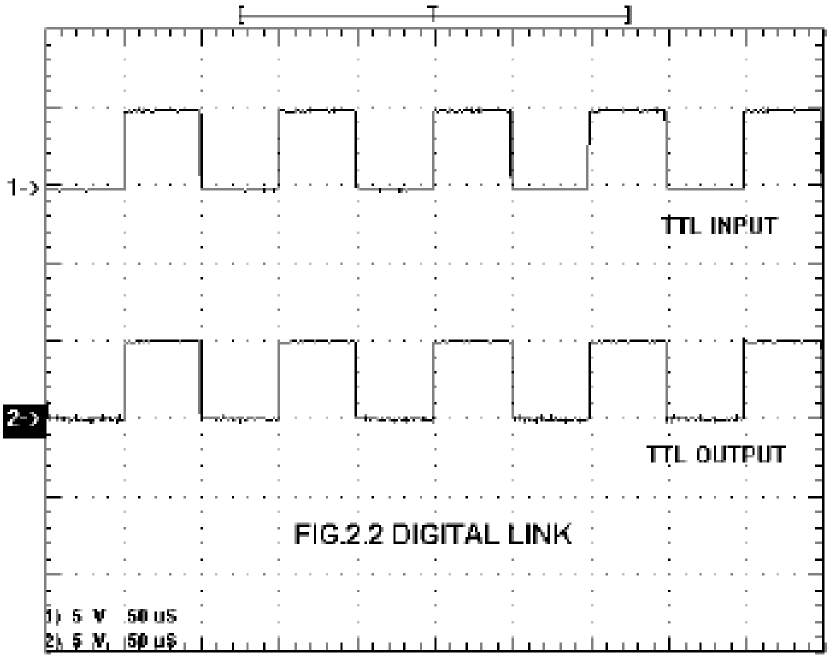


FIG.2.2 DIGITAL LINK

## Measurement of Digital Bandwidth

## PROCEDURE

- Make connections as shown in **FIG.2.1.** Connect the power supply cables with proper polarity to FO-A-P Kit. While connecting this, ensure that the power supply is OFF.
- Switch ON the power supply.
- Feed **TTL** Square wave signal of **1KHz** from the function generator to the **IN** post of Digital Buffer.
- Connect the output post **OUT** of Digital Buffer to the post **TX IN** of Transmitter.
- Slightly unscrew the cap of SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the One Meter Fiber into the cap. Now tighten the cap by screwing it back.
- Connect the other end of the Fiber to detector SFH551V very carefully as per the instructions in above step.
- Observe the detected signal at post **TTL OUT** on oscilloscope as shown in **FIG 2.2.**
- To measure the digital bandwidth of the phototransistor vary the input signal frequency and observe the detected signal at various frequencies.
- Determine the frequency at which the detector stops recovering the signal.
- This determines the maximum bit rate on the digital link.
- Keep switch **SW9** towards **TX2** position.
- Keep Jumper **JP7** towards **+5V** position.
- Repeat the same procedure above for second transmitter SFH450V by making the following changes.
- The digital bandwidth of SFH551 for TX1 SFH756 is 3MHz & for SFH450 it is 1MHz.

## Calculation for Digital bandwidth

Cut off frequency = MHz

Bandwidth = 2 x cut-off frequency

= Mbps

# EXPERIMENT NO. 3

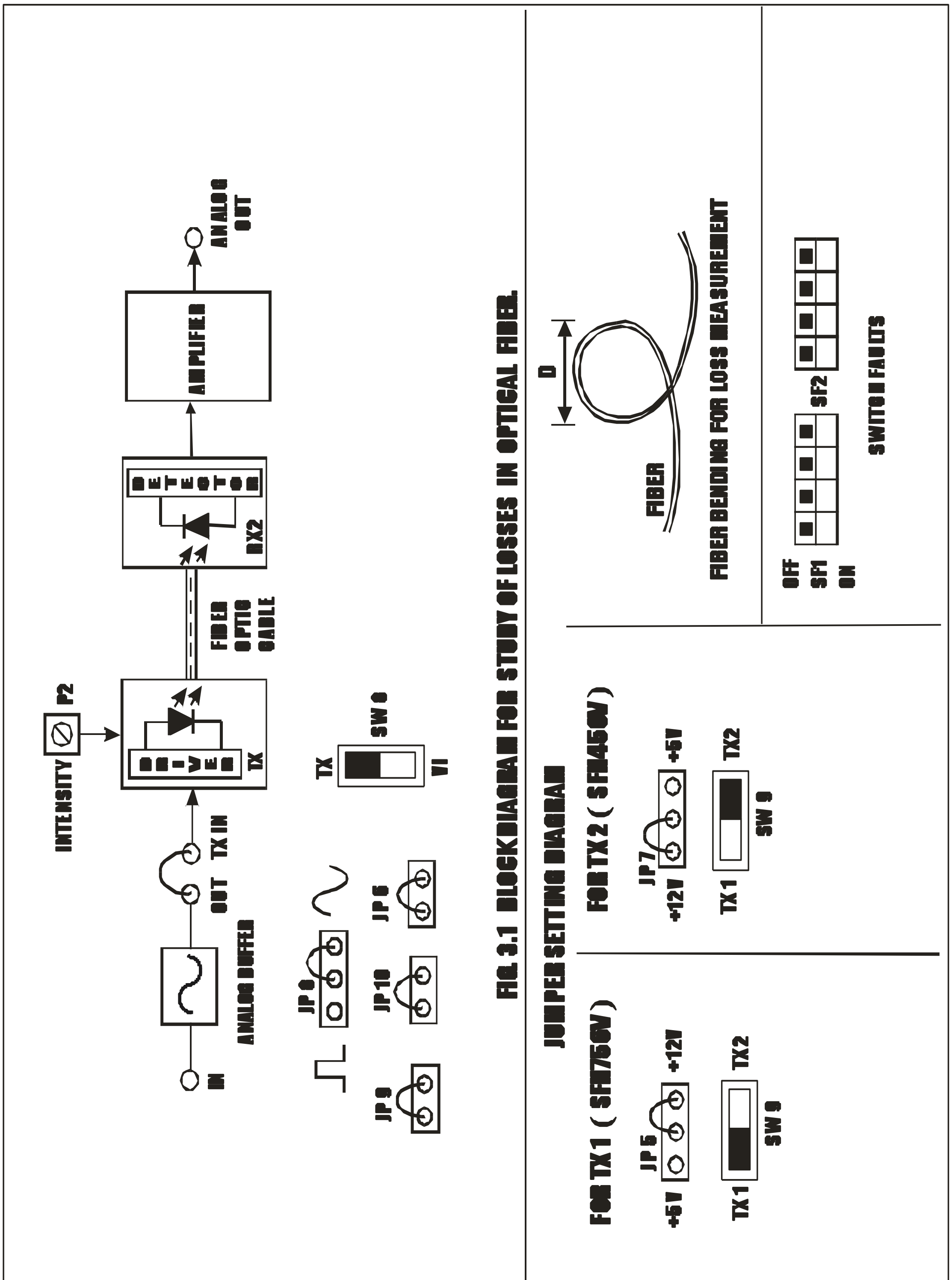



FIG. 3.1 BLOCKDIAGRAM FOR STUDY OF LOSSES IN OPTICAL FIBER.

# EXPERIMENT NO.3

## NAME

Study of Losses in Optical Fiber

## OBJECTIVE

The objective of this experiment is to measure propagation loss & bending losses for two different wavelengths in plastic Fiber provided with the kit.

## THEORY

Optical Fibers are available in different variety of materials. These materials are usually selected by taking into account their absorption characteristics for different wavelengths of light. In case of Optical Fiber, since the signal is transmitted in the form of light, which is completely different in nature as that of electrons, one has to consider the interaction of matter the radiation to study the losses in fiber. Losses are introduced in fiber due to various reasons. As light propagates from one end of Fiber to another end, part of it is absorbed in the material exhibiting absorption loss. Also part of the light is reflected back or in some other directions from the impurity particles present in the material contributing to the loss of the signal at the other end of the Fiber. In general terms it is known as propagation loss. Plastic Fibers have higher loss of the order of 180 dB/Km Whenever the condition for angle of incidence of the incident lights is violated the losses are introduced due to refraction of light. This occurs when fiber is subjected to bending. Lower the radius of curvature more is the loss. Other losses are due to the coupling of Fiber at LED & photo detector ends.

## EQUIPMENTS

- FO-A-P Kit with power supply
- Patch chords
- 20MHz Dual Channel Oscilloscope
- Function Generator
- 1 & 3-Meter Fiber Cable

**NOTE:** Keep All Switch Faults In Off Position.

## PROCEDURE

- Make connections as shown in **FIG.3.1.** Connect the power supply cable with proper polarity to FO-A-P Kit. While connecting this, ensure that the power supply is OFF.
- Keep **SW9** towards **TX1** position for SFH756
- Keep jumpers &**SW8** positions as shown in **FIG.3.1.**
- Keep Intensity control pot **P2** towards **minimum** position.
- Switch ON the power supply.

- Feed about 2Vpp sinusoidal signal of 1KHz from the function generator to the **IN** post of Analog Buffer.
- Connect the output post **OUT** of Analog Buffer to the post **TX IN** of Transmitter.
- Slightly unscrew the cap of SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the 1 Meter Fiber into the cap. Now tighten the cap by screwing it back.
- Connect the other end of the Fiber to detector SFH350V (Photo Transistor Detector) very carefully as per the instructions in above step.
- Observe the detected signal at post **ANALOG OUT** on oscilloscope.
- Adjust Intensity control pot **P2** Optical Power control potentiometer so that you receive signal of 2Vpp amplitude.
- Measure the peak value of the received signal at ANALOG OUT terminal.
- Let this value be V1.
- Now replace 1 meter Fiber by 3 Meter Fiber between same LED and Detector. Do not disturb any settings. Again take the peak voltage reading and let it be V2.
- If ∝ is the attenuation of the Fiber then we have.
  $\propto_{dB}$ = (10/L1-L2) log10 (V2/V1)
  Where α = dB / Km,
  L1 = Fiber Length for V1
  L2 = Fiber Length for V2
  This ∝is for peak wavelength of 660nm
- Now switch off the power supply.
- Keep **SW9** towards **TX1** position for SFH756
- Set the jumpers to form simple analog link using LED SFH450V at 950nm and phototransistor SFH350V (Photo Transistor Detector) with 1 meter Fiber Cable.
- Switch on the power supply.
- Repeat the same procedure as above again for this link to get ∝at 950nm.
- Compare the two ∝values.

## Measurement of Bending Losses

- Set up the 660 nm analog link using 1-meter fiber as per procedure above.
- Bend the Fiber in a loop. (As shown in **FIG. 3.1**) measure the amplitude of the received signal
- Keep reducing the diameter of bend to about 2 cm & take corresponding out voltage readings. (Do not reduce loop diameter less than 1 cm).
- Plot a graph of the received signal amplitude versus the loop diameter.
- Repeat the procedure again for second transmitter.

L1 = 1meter fiber     L2 = 3meter fiber

| V1_o/p voltage at L1 | V2_o/p voltage at L2 | $\propto_{dB}$ |
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# EXPERIMENT NO. 4

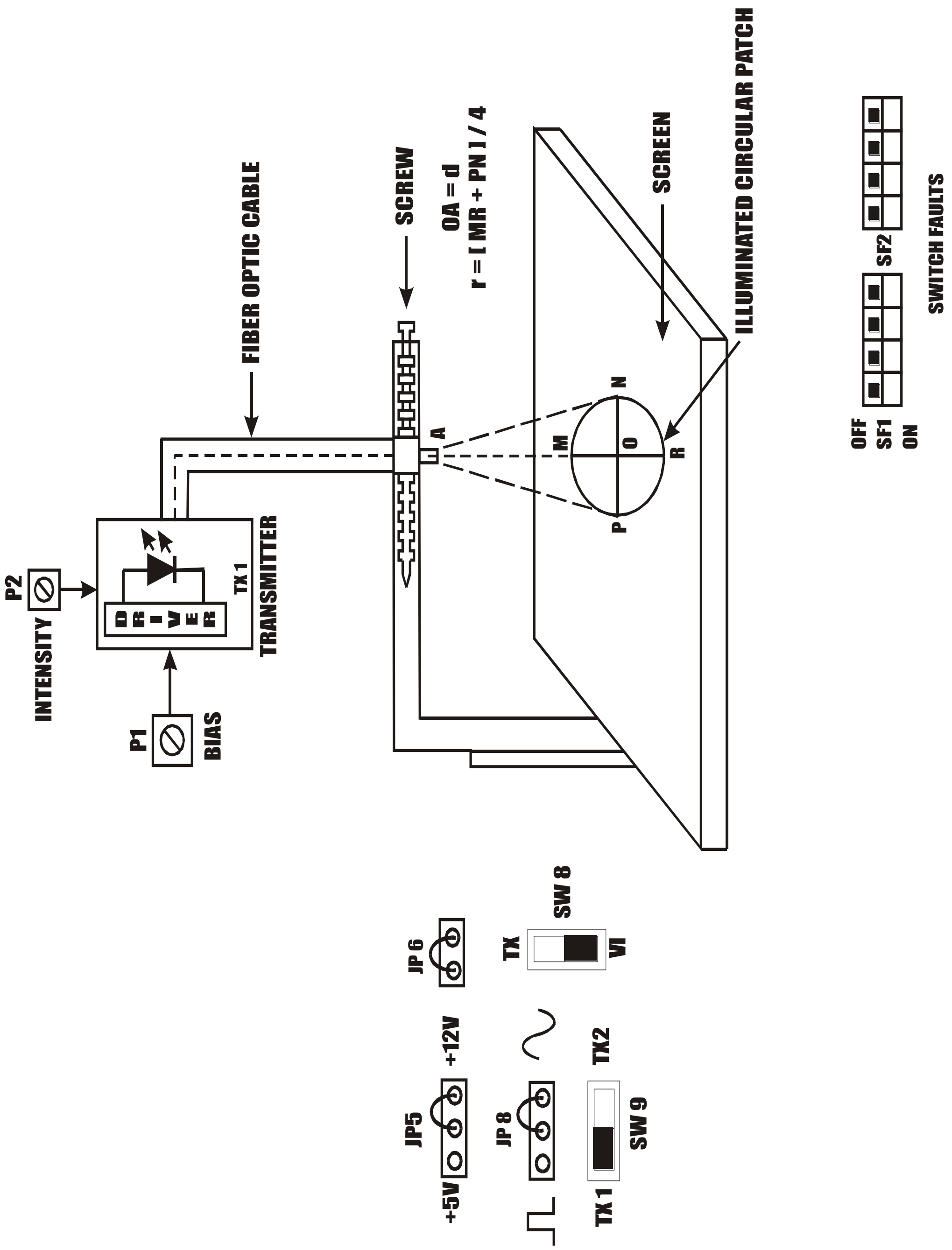


FIG. 4.1 BLOCK DIAGRAM FOR NUMERICAL APERTURE SETUP.

# EXPERIMENT NO. 4

## NAME

Study of Numerical Aperture Of Optical Fiber

## OBJECTIVE

The objective of this experiment is to measure the numerical aperture of the plastic Fiber provided with the kit using 660nm wavelength LED.

## THEORY

Numerical aperture refers to the maximum angle at the light incident on the fiber end is totally internally reflected and is transmitted properly along the Fiber. The cone formed by the rotations of this angle along the axis of the Fiber is the cone of acceptance of the Fiber. The light ray should strike the fiber end within its cone of acceptance; else it is refracted out of the fiber core.

### Consideration In A Measurement

- It is very important that the source should be properly aligned with the cable& the distance from the launched point & the cable be properly selected to ensure that the maximum amount of Optical Power is transferred to the cable.
- This experiment is best performed in a less illuminated room.

## EQUIPMENTS

FO-A-P Kit with power supply
Patch chords
1-Meter Fiber Cable
Numerical aperture measurement Jig
Steel Ruler

**NOTE:** Keep All Switch Faults In Off Position**.**

## PROCEDURE

- Make connections as shown in **FIG.4.1.** Connect the power supply cables with proper polarity to FO-A-P Kit. While connecting this, ensure that the power supply is OFF.
- Keep Intensity control pot **P2** towards **minimum** position.
- Keep Bias control pot **P1** fully clockwise position.
- Switch ON the power supply.
- Slightly unscrew the cap of SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the 1 Meter Fiber into the cap. Now tighten the cap by screwing it back.
- Insert the other end of the Fiber into the numerical aperture measurement fig. Adjust the fiber such that its cut face is perpendicular to the axis of the Fiber.
- Keep the distance of about 5mm between the fiber tip and the screen.
- Gently tighten the screw and thus fix the fiber in the place.
- Increase the intensity pot **P2** to get bright red light circular patch.
- Now observe the illuminated circular patch of light on the screen.
- Measure exactly the distance **d** and also the vertical and horizontal diameters **MR** and **PN** as indicated in the **FIG. 4.1**.
- Mean radius is calculated using the following formula
- **r = (MR+PN)/4**
- Find the numerical aperture of the Fiber using the formula

**NA = sin$\theta_{max}$ = r / d2 + r2**

Where θ max is the maximum angle at which the light incident is properly transmitted through the fiber.

d=OA r= (MR+PH) / 4

| MR (mm) | PH (mm) | r = (MR + PH)/4 | d=OA | NA |
|---|---|---|---|---|
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# EXPERIMENT NO. 5

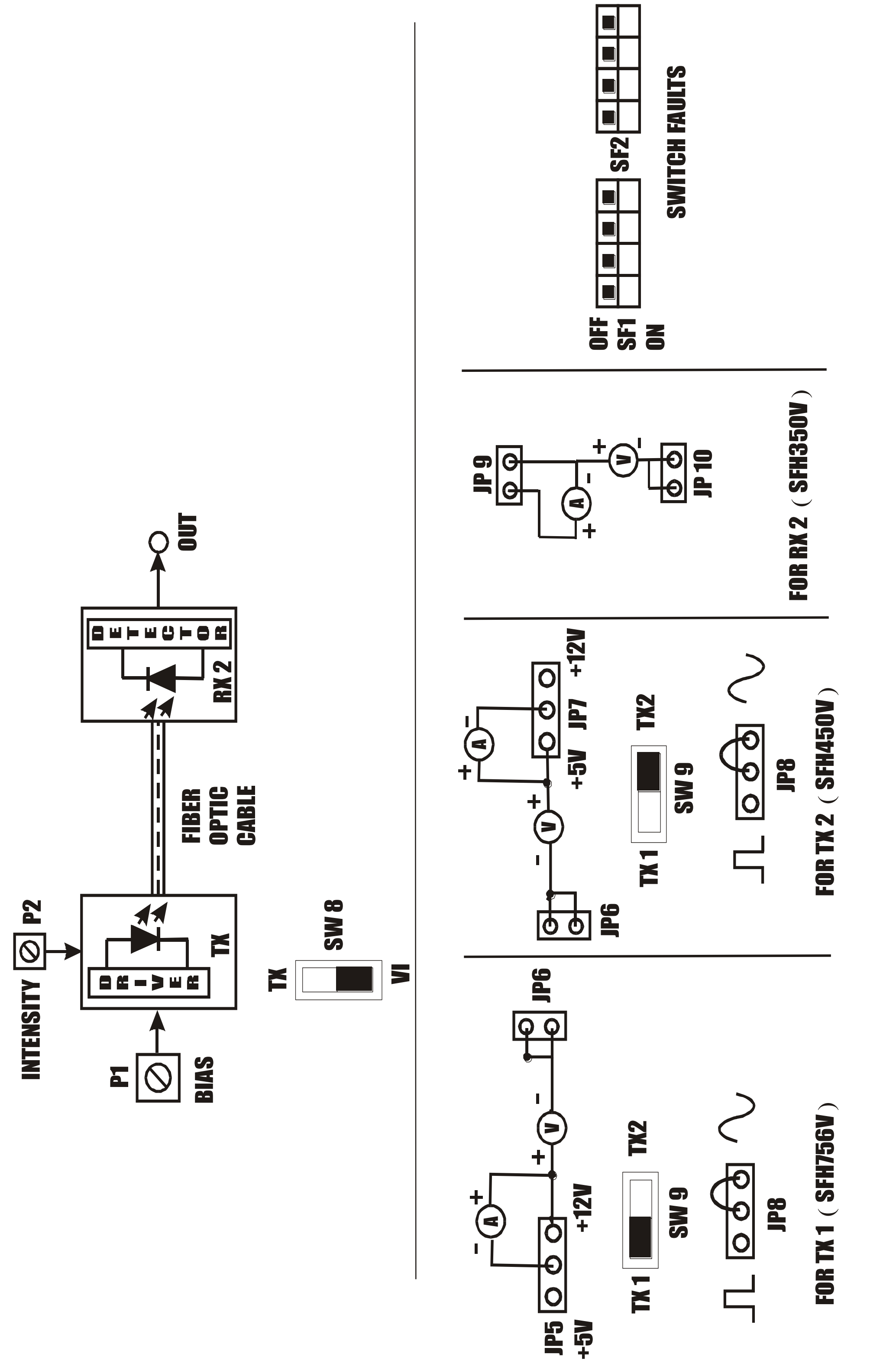


FIG.5.1 BLOCK DIAGRAM FOR CHARACTERISTICS OF FIBER OPTIC LED'S AND DETECTORS.

# EXPERIMENT NO. 5

## NAME

Study of Characteristics of Fiber Optic LED and Photo detector

## OBJECTIVE

To study the characteristics of Fiber Optic LEDs and plot the graph of forward current v/s output optical energy and also to study the photodetector response

## THEORY

In Optical Fiber communication system, Electrical signal is first converted into optical signal with the help of E/O conversion device such as LED. After this optical signal is transmitted through Optical Fiber, it is retrieved in its original electrical form with the help O/E conversion device such as photodetector. Different technologies employed in chip fabrication lead to significant variation in parameters for the various emitter diodes. All the emitters distinguish themselves in offering high output power coupled into the important peak wavelength of
emission, conversion efficiency (usually specified in terms of power launched in optical Fiber peak wavelength of emission, conversion efficiency (usually specified in terms of pore launched in optical Fiber for specified forward current) optical raise and fall times which put the limitation on operating frequency, maximum forward current through LED and typical forward voltage across LED. Photodetectors usually comes in variety of forms photoconductive, photo voltaic, transistor type output and diode type output. Here also characteristics to be taken
into account are response time of the detector which puts the limitation on the operating frequency, wavelength sensitivity and responsivity.

## EQUIPMENTS

FO-A-P Kit with power supply
Patch chords
Jumper to Crocodile connectors
1-Meter Fiber Cable
Voltmeter and Current meter 2Nos. each

**NOTE:** Keep All Switch Faults In Off Position.

# PROCEDURE

- Make connections as shown in **FIG.5.1.** Connect the power supply cables with proper polarity to FO-A-P Kit. While connecting this, ensure that the power supply is OFF.
- Keep switch **SW8** towards **VI** position.
- Keep switch **SW9** towards **TX1** position.
- Keep Jumper **JP8** towards **sine** position.
- Keep Bias control pot **P1** towards **maximum** position &**P2** towards **minimum** position.
- Insert the jumper to crocodile connecting wires (provided along with the kit) in jumper **JP5, JP6, JP9** and **JP10** at positions shown in the fig 5.1.
- Connect the voltmeter and current meter with proper polarities to abovementioned jumpers.
- Switch ON the power supply.
- Slightly unscrew the cap of SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the 1 Meter Fiber into the cap. Now tighten the cap by screwing it back.
- Connect the other end of the Fiber to detector SFH350V (Photo Transistor
- Detector) very carefully as per the instructions in above step.
- Vary intensity control pot **P2** to control current flowing through the LED.
- To get the V-I characteristics of LED, rotate **P2** slowly and measure forward current and corresponding forward voltage. Take number of such readings for various current values and plot V-I characteristics graph for the LED.
- For each reading taken above, find out the power, which is product of V and I. This is the electrical power supplied to the LED. Data sheets for the LED specify optical power coupled into plastic fiber when forward current was 10mA as 200uW. This means that the electrical power at 10mA current is converted into 200uW of optical energy. Hence the efficiency of the LED comes out to be approximately 1.15%.
- With this efficiency assumed, find out optical power coupled into plastic Optical Fiber for each of the reading. Plot the graph of forward current v/s output optical power of the LED.
- Data sheets for the phototransistor detector specified responsivity as 0.8mA for 10uW of incident optical energy.
- Find out the current flowing through phototransistor and voltage across it.
- Plot the graph for the responsivity of phototransistor. Find out the portion where detector response is linear.
- Insert the jumper to crocodile connecting wires (provided along with the kit) in jumper **JP5, JP7, JP8, JP9, JP10** at positions shown in the fig 5.1 and
- Repeat the same procedure & plot the graph for SFH450V.

Table For VI Characteristics Of Fiber Optic Led Sfh756v & Detector SFH350v

| Vf Forward Voltage Of LED SFH756 (V) | If Forward Current of LED SFH756V (mA) | Electrical Power Pi = V * I (mW) | Optical Power of LED 756V Po = Pi * 1.15% (μW) | O/P Current Of SFH350 (mA) | Responsivity R= 0.8mA*Po / 10uW (mA) |
|---|---|---|---|---|---|
| 1.565 | 0.223 | 0.349 | 4 | 0.06 | 0.32 |
| 1.600 | 0.397 | 0.635 | 7.3 | 0.143 | 0.584 |
| 1.640 | 0.728 | 1.19 | 13.6 | 0.335 | 1.088 |
| 1.680 | 1.165 | 1.2 | 13.8 | 0.64 | 1.1 |
| 1.720 | 1.67 | 2.87 | 33 | 1 | 2.64 |
| 1.760 | 2.23 | 3.92 | 45 | 1.45 | 3.6 |
| 1.800 | 2.8 | 5 | 57.5 | 1.96 | 4.6 |
| 1.86 | 3.8 | 7 | 80.5 | 2.76 | 6.4 |
| 1.9 | 4.5 | 8.5 | 97.7 | 3.36 | 7.86 |
| 1.960 | 5.4 | 10.58 | 122 | 4.21 | 9.76 |
| 2 | 6.2 | 12.4 | 142.6 | 4.85 | 11.40 |
| 2.2 | 10 | 22 | 253 | 8.6 | 20.24 |
| 2.3 | 15 | 34.5 | 396.75 | 13.3 | 31.7 |
| 2.38 | 22.4 | 53.3 | 613 | 19.8 | 49.9 |
| 2.42 | 27 | 65 | 751 | 23.5 | 60 |

(Above readings are for reference)

Table For VI Characteristics Of Fiber Optic Led Sfh 450v & Detector Sfh350v

| Vf Forward Voltage Of LED SFH450V (V) | If Forward Current of LED SFH450V (mA) | Electrical Power Pi = V * I (mW) | Optical Power of LED 450V P0 = Pi * 1.15% (μW) | O/P Current Of SFH350 (mA) | Responsivity R=0.8mA*PO -------------- 10uW (mA) |
|---|---|---|---|---|---|
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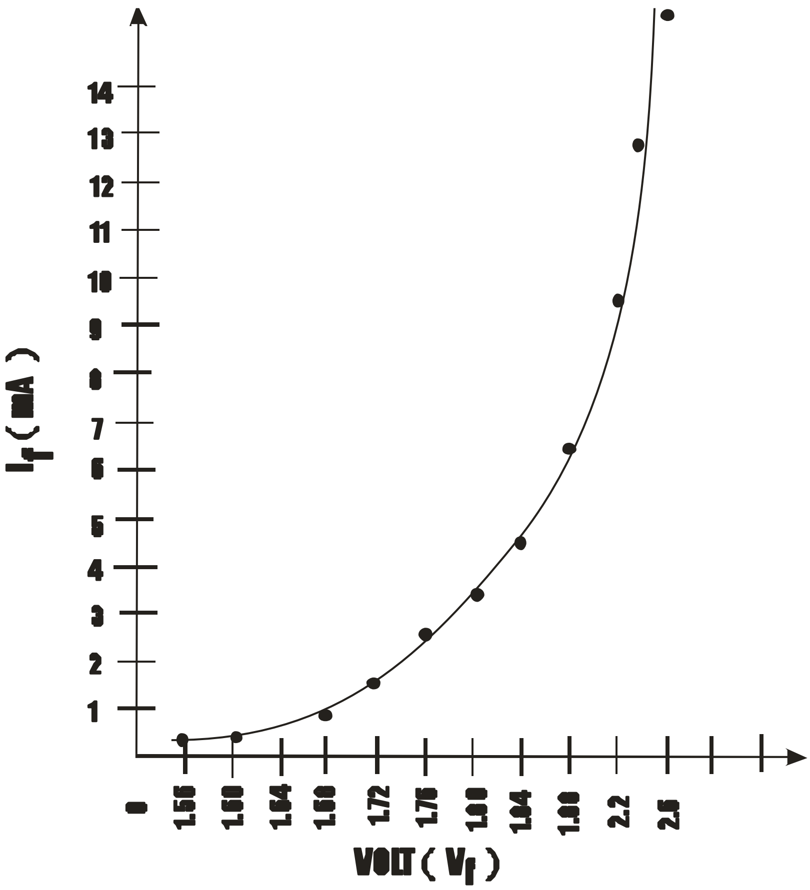
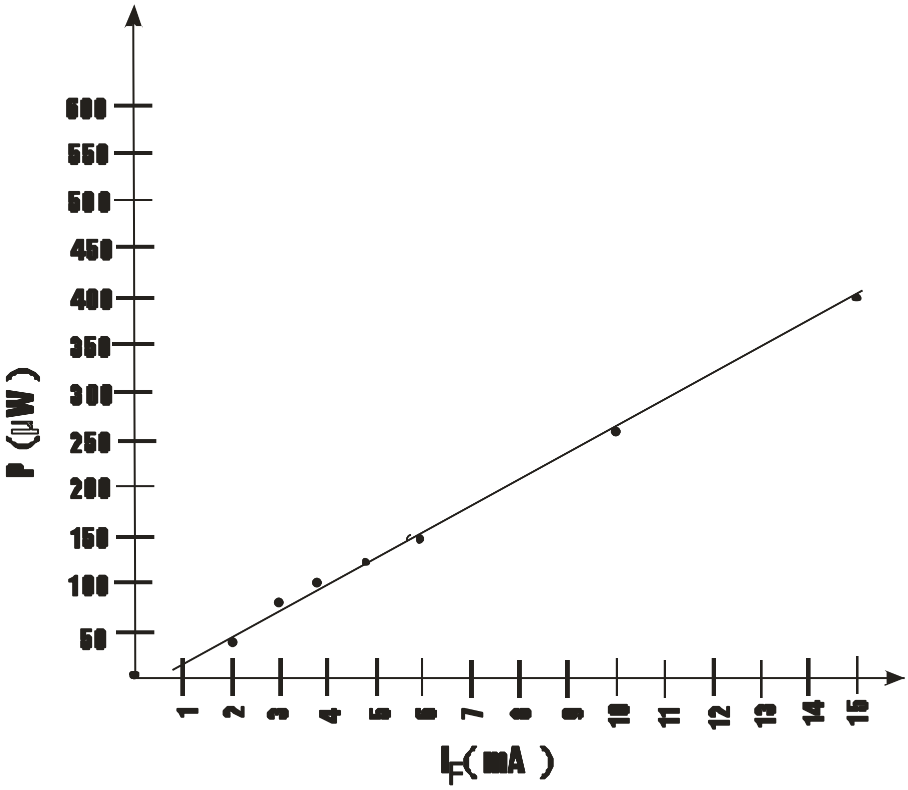
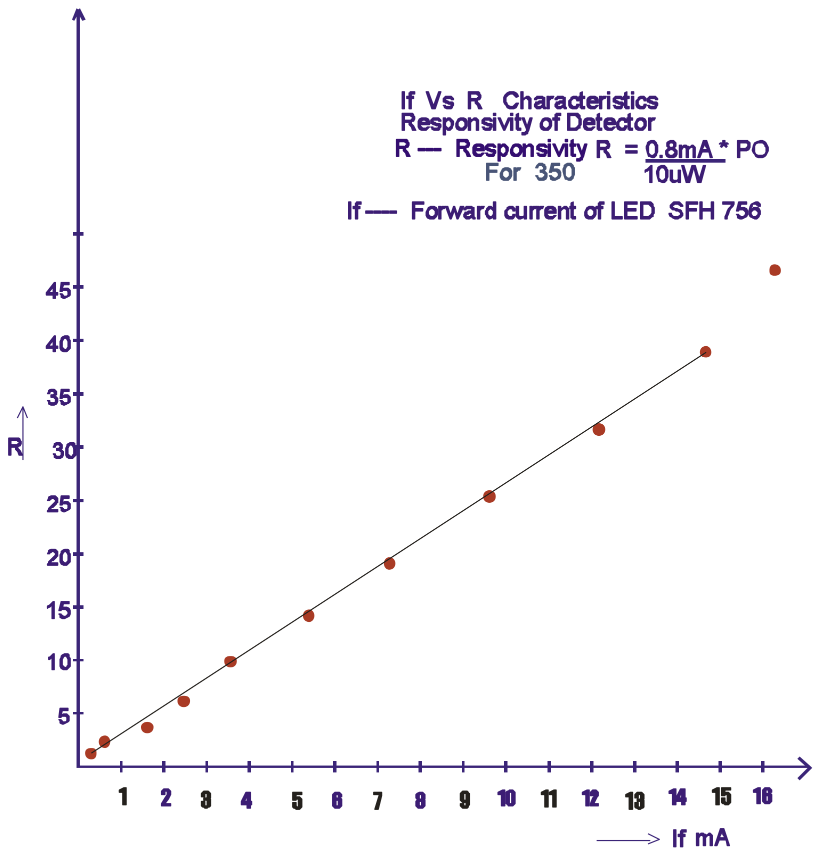
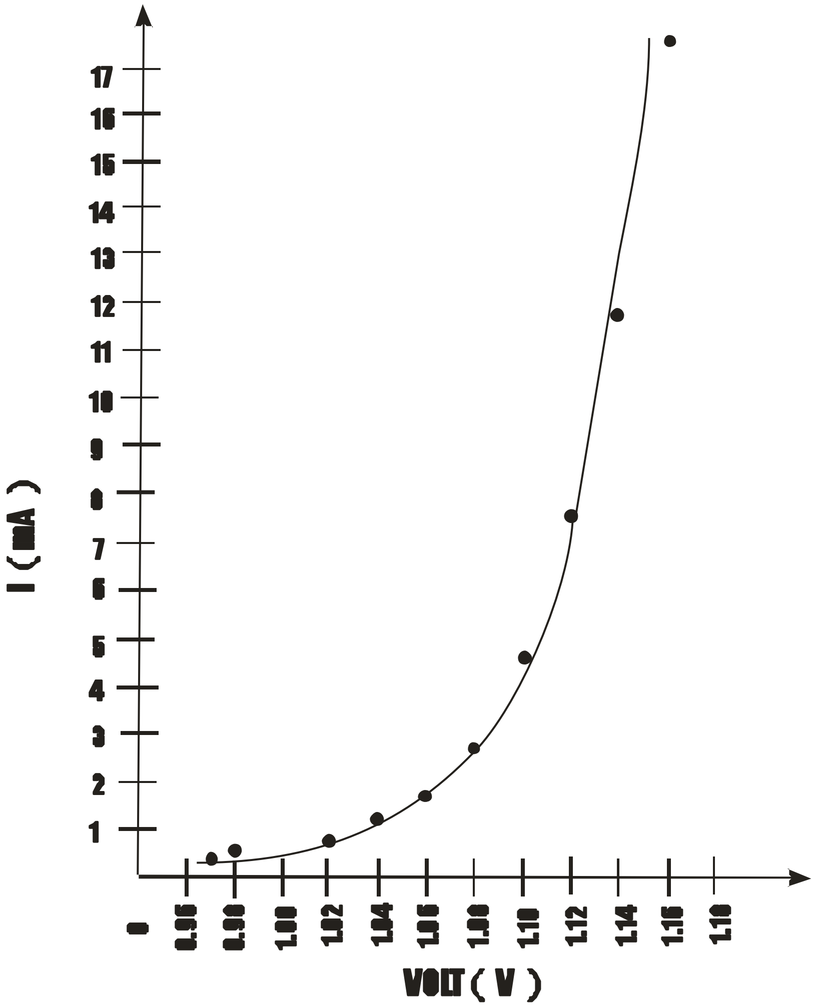



V-I CHARACTERISTICS GRAPH OF LED SFH756V

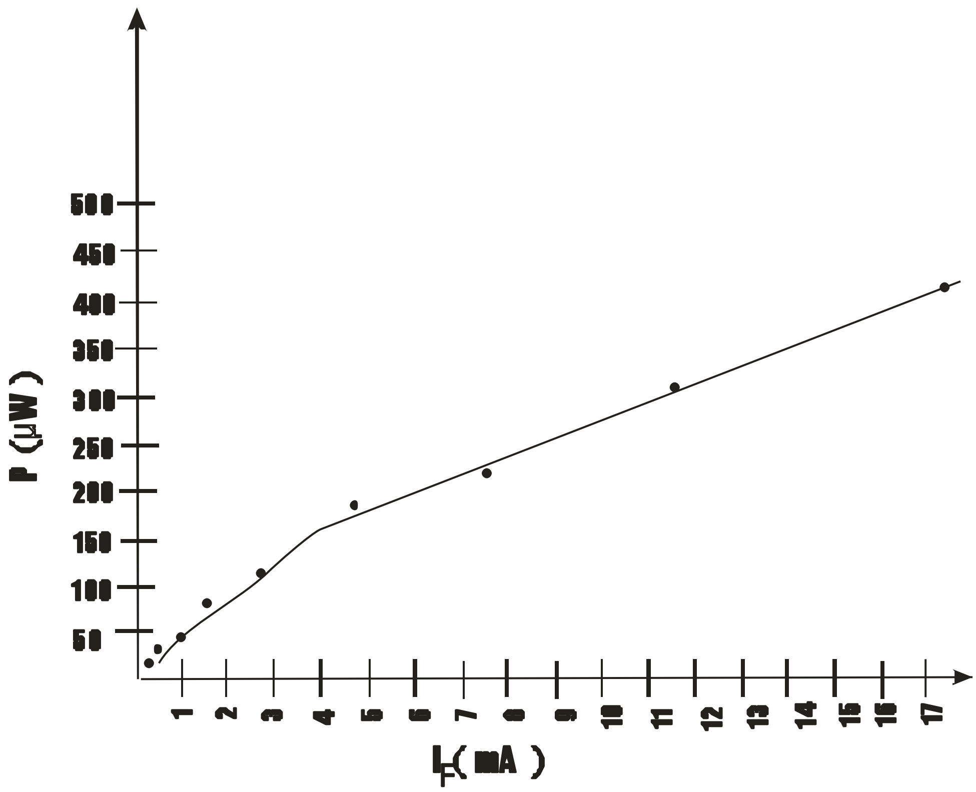



OPTICAL POWER VERSUS CURRENT GRAPH OF LED SFH756V

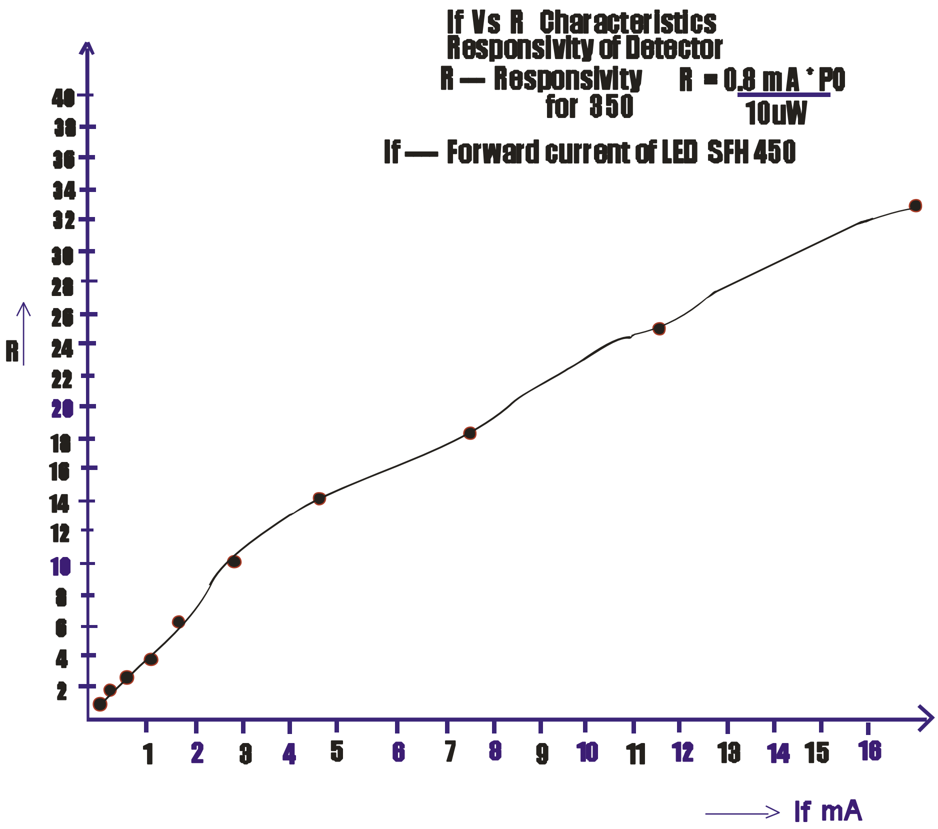





**VI CHARACTERISTICS GRAPH OF LED SFH 450V**




OPTICAL POWER VERSUS CURRENT GRAPH OF LED SFH 450V



# EXPERIMENT
# NO. 6

# EXPERIMENT NO. 6

## NAME

Study of Time Division Multiplexing

## OBJECTIVE

The objective of this experiment is to study simultaneous transmission of several signals using time division multiplexing.

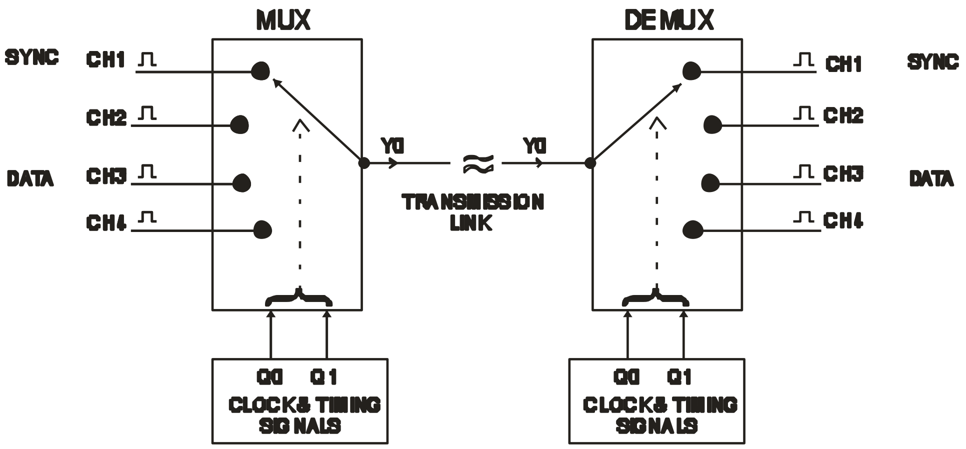
## THEORY

In TDM various signals are sampled and transmitted for a fixed duration of time one after the other. At the receiving end, these signals are extracted in the same order & form of transmission.

**TDM MUX**

Using this technique, more number of data channels can be transmitted on a single transmission link. Hence multiplexing is also termed as Many I/P One O/P logic. The time slot for each channel repeats after regular intervals.

The above definition is explained with the help of an example given below:



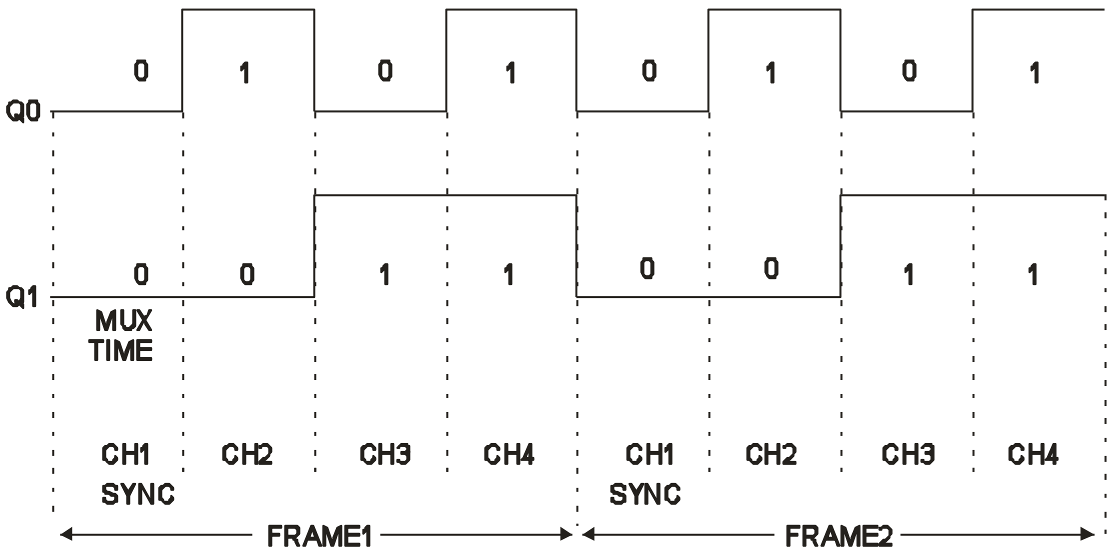

TDM COMMUNICATION

Suppose we need to transmit four channels on a single link, the TDM process is used as shown. CH1, CH2, CH3, CH4 are the four channels, Q0 & Q1 are the channel selection signals (Clock & Timing section). Depending on the condition of Q0, Q1 the channels are selected one by one. The 1st channel of the four is used to carry sync information. Then the data channels are selected sequentially.

The channel selection takes place as shown in table below:

| Q1 | Q0 | CH1 | CH2 | CH3 | CH4 | Y0 (MUX OUTPUT) |
|---|---|---|---|---|---|---|
| 0 | 0 | √ | | | | CH1 |
| 0 | 1 | | √ | | | CH2 |
| 1 | 0 | | | √ | | CH3 |
| 1 | 1 | | | | √ | CH4 |
| 0 | 0 | √ | | | | CH1 |

Now all the channels are multiplexed on a single link & are transmitted over the communication link.



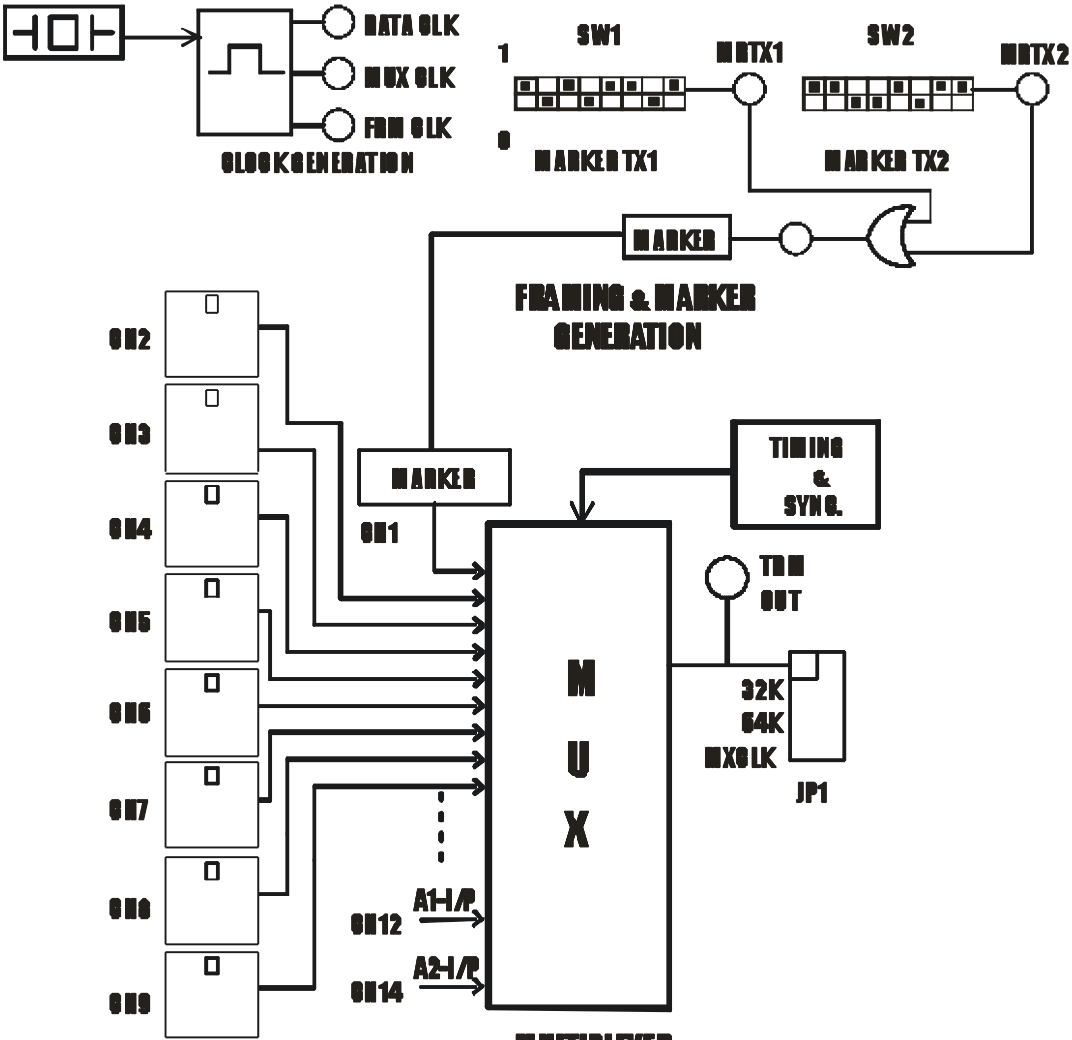

**TDM DEMUX**

At the receiver the same process is carried out to direct the information on corresponding receiving channels. Now as explained in previous section Q0, Q1 are selection clocks and Y0 is MUX data.

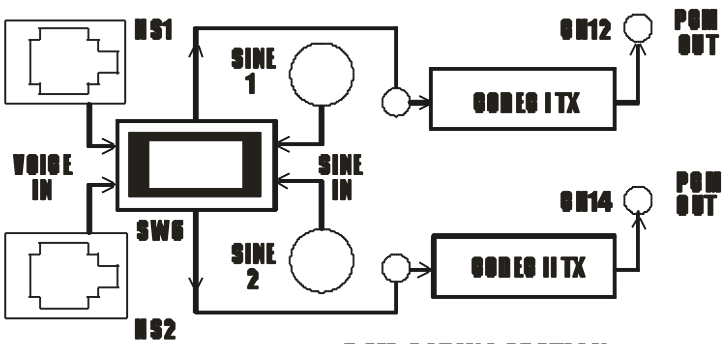
| Data | Q1 | Q0 | CH1 | CH2 | CH3 | CH4 |
|---|---|---|---|---|---|---|
| Y0 | 0 | 0 | Y0 | | | |
| | 0 | 1 | | Y0 | | |
| | 1 | 0 | | | Y0 | |
| | 1 | 1 | | | | Y0 |
| | 0 | 0 | Y0 | | | |

The Q0 & Q1 here in demux section are used to select the channel where the mux output Y0 will be directed. Depending upon the condition of Q0 & Q1 the mux signal Y0 will be directed to CH1, CH2, and CH3 & CH4. The channel one carries the sync information & the rest channels carry the actual information.

# EXPERIMENT
# NO. 7




MULTIPLEXER




PCM CODING SECTION

# EXPERIMENT NO. 7

## NAME

Study of 16-Channel Digital TDM Generation

## OBJECTIVE

The objective of this experiment is to study the technique of generation of TDM data.

## THEORY

The basic TDM data includes

A. Framing & Marker in TDM

B. Digital data & PCM coded data in TDM

### A. Study of framing & Marker in TDM:

**Framing**

The TDM transmission consists of these basic clock & timing signals on the basis of TX channel & information.
**1.** Data Clock : Determines maximum data rate of channels.
**2.** Mux Clock : Determines time slot for each channel.
**3.** Frame Clock : Determines no. of channels to be multiplexed.
In the FO-A-P trainer we have considered Data Rate = 1.024MHz.
A single channel will contain 8 bit data; hence the channel time slot or channel frequency will be 1MHz/8 i.e. 128 KHz.
i.e Mux Clock : 128KHz.
We have provided 16 channels for communication. So the frame clock becomes 128/16 i.e. Frame CLOCK = 8KHz. For selection of 16 channels we need at least 4SIGNALS(2n=16 hence n=4), they are QA, QB, QC & QD with clock frequencies of 64KHz, 32KHz, 16KHz & 8KHz respectively.
The waveforms & table shows how the framing is achieved in TDM & table below shows channel selection in TDM.
The FIG.7.2 shows how frame is formed in TDM.
The FIG.7.3 shows how markers are placed in a frame.
The table shows how the 16 channels are selected depending on the clock signals QA, QB, QC & QD.

| QD | QC | QB | QA | CH 1 | CH 2 | CH 3 | CH 4 | CH 5 | CH 6 | CH 7 | CH 8 | CH 9 | CH 10 | CH 11 | CH 12 | CH 13 | CH 14 | CH 15 | CH 16 | O/P Y0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | √ | | | | | | | | | | | | | | | | CH 1 |
| 0 | 0 | 0 | 1 | | √ | | | | | | | | | | | | | | | CH 2 |
| 0 | 0 | 1 | 0 | | | √ | | | | | | | | | | | | | | CH 3 |
| 0 | 0 | 1 | 1 | | | | √ | | | | | | | | | | | | | CH 4 |
| 0 | 1 | 0 | 0 | | | | | √ | | | | | | | | | | | | CH 5 |

| | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 0 | 1 | | | | | | √ | | | | | | | | | | | CH 6 |
| 0 | 1 | 1 | 0 | | | | | | | √ | | | | | | | | | | CH 7 |
| 0 | 1 | 1 | 1 | | | | | | | | √ | | | | | | | | | CH 8 |
| 1 | 0 | 0 | 0 | | | | | | | | | √ | | | | | | | | CH 9 |
| 1 | 0 | 0 | 1 | | | | | | | | | | √ | | | | | | | CH 10 |
| 1 | 0 | 1 | 0 | | | | | | | | | | | √ | | | | | | CH 11 |
| 1 | 0 | 1 | 1 | | | | | | | | | | | | √ | | | | | CH 12 |
| 1 | 1 | 0 | 0 | | | | | | | | | | | | | √ | | | | CH 13 |
| 1 | 1 | 0 | 1 | | | | | | | | | | | | | | √ | | | CH 14 |
| 1 | 1 | 1 | 0 | | | | | | | | | | | | | | | √ | | CH 15 |
| 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | √ | CH 16 |

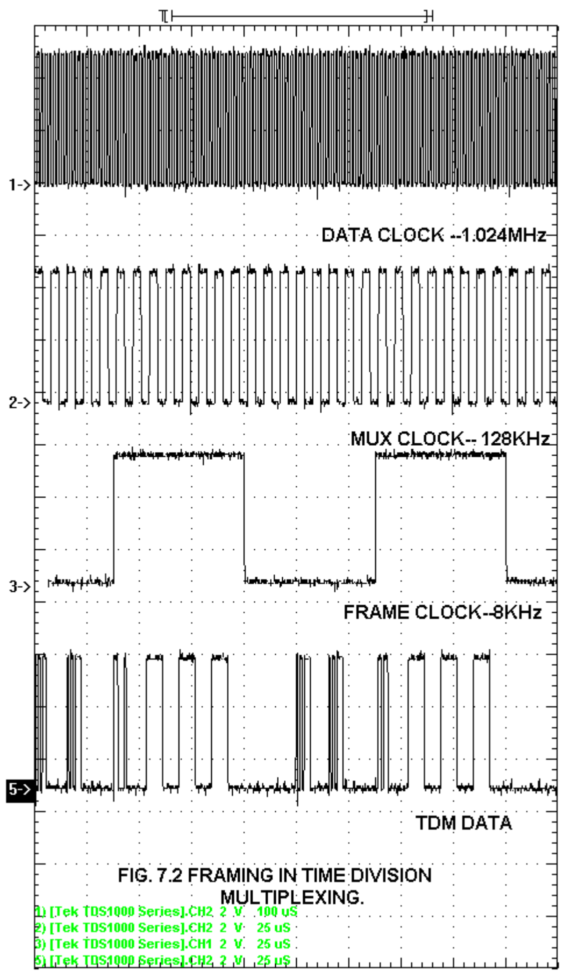



FIG. 7.2 FRAMING IN TIME DIVISION MULTIPLEXING.

Marker used in Time Division Multiplexing is a unique bit pattern placed at some fixed position in the frame & is used to determine the start of the frame at the receiver. Some times a different marker is used in alternate framers to counter the possibility of data bits containing the marker sequence generating a false marker.

**Double Marker**

The marker is supposed to be a unique bit pattern in the data stream, which is identified at the receiver to identify the beginning of frame. Question arises, what happens if the marker pattern is contained elsewhere as data?

To avoid this situation, the receiver circuit is usually designed to detect a marker repeating itself one in a frame. It is only the repetition after a certain number of bits, which allows the bit pattern to be accepted as the marker. It is highly impossible that the data will have this pattern repeating at the same position in every frame. However, to avoid even this situation, the marker is sometimes chosen to be different in alternate frames.

## EQUIPMENTS

- FO-A-P Kit with power supply
- Patch chords
- 20 MHz Dual Trace Oscilloscope
- 1 Meter fiber cable

## OBSERVATIONS

Observe & measure the frequencies for **DATA CLK, MUX CLK** &**FRM CLK**(frame clock). Ref fig: 7.2.

Observe the marker at test point **MRTX1** on the channel & frame clock on the CRO channel. Ref fig: 7.3.

Observe the second marker MRTX2 & frame clock similarly at the output of **OR** gate.

Now observe the test point of marker i.e.channel-1 with respect to MRTX1 &MRTX2 and plot the waveform.

Observe the channel-1 signal with respect to the frame clock. You will find the markers are present alternately in each frame.

## SWITCH FAULTS

Put switch **1 (SF1)** in Switch Fault section to **ON** position. This will open TX-marker signal at i/p of U9. This will open pin 8 of U9 (74LS150) i.e. channel one is removed at MUX section. No marker is transmitted and hence the receiver synchronization is disturbed.

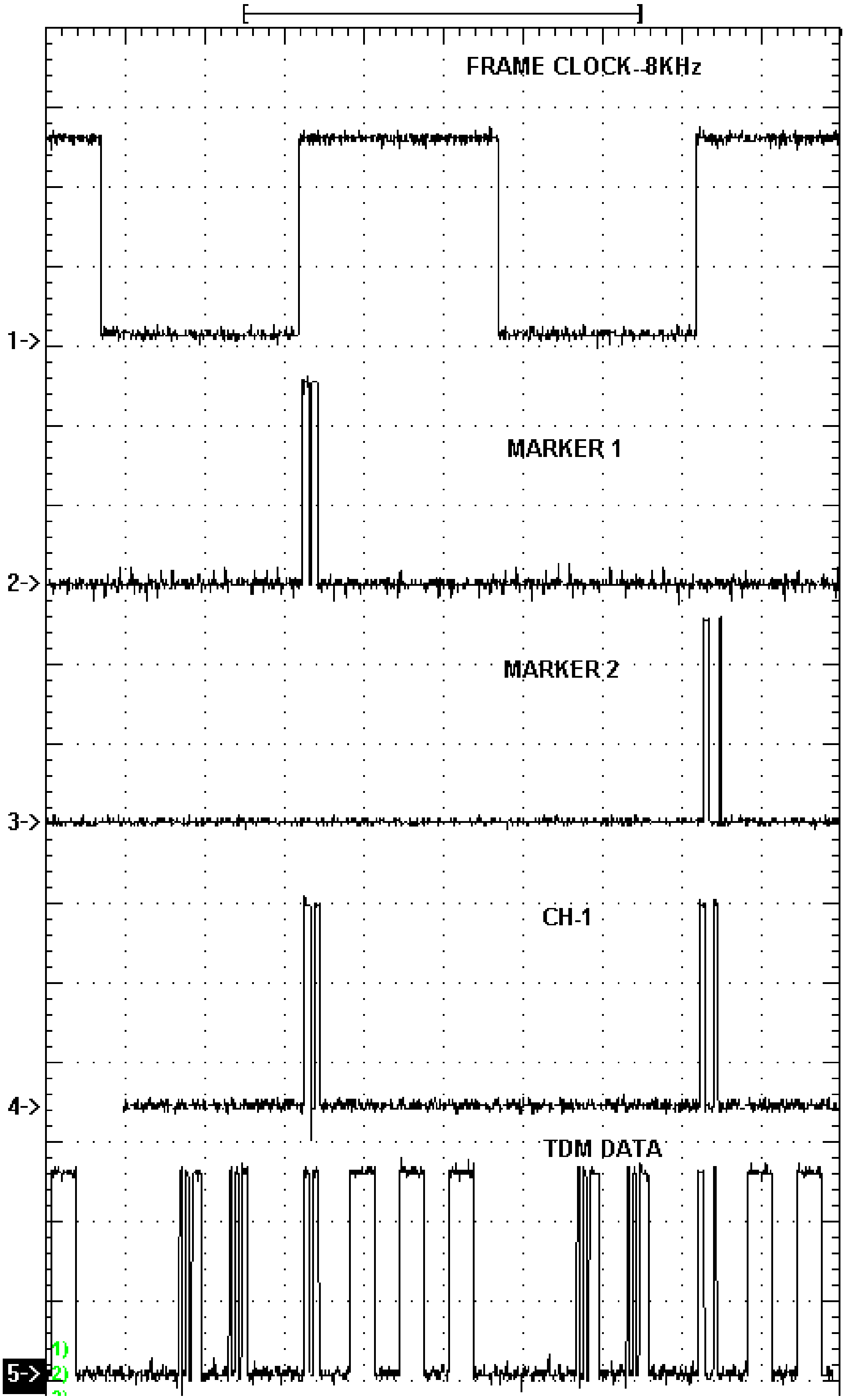



**FIG7.3 MARKER IN TDM**

**B. Study Of Channel Inputs & PCM Data In TDM.**

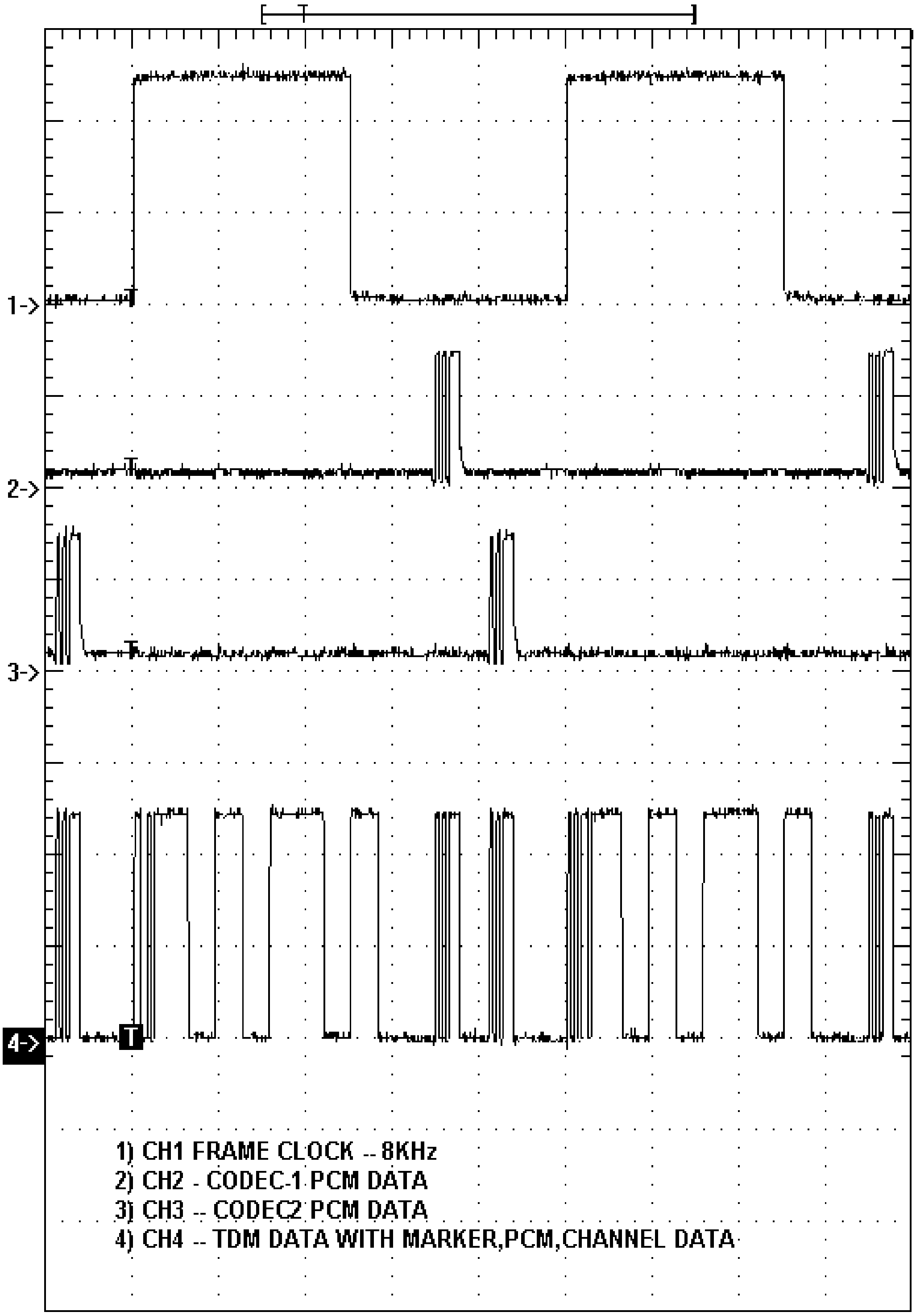
Once the frame is formed and marker is inserted for synchronization in the first channel, now remain the next 15 time slots free for data information. On the FO-A-P trainer kit we have used 8 ON/OFF switches as channel-2 to channel-9. Then next 2(10&11) slots are kept blank. In the 12th time slot the PCM data of codec-1 chip is inserted. Again 13th time slot is kept zero. In the 14th time slot the PCM data of codec-2 (IC14)is inserted. The 15th, 16th positions are again kept zero. The FIG.7.4 shows the complete Time Division Multiplexed Data.

## OBSERVATIONS

- Observe the test points of **TDM OUT** with respect to the frame clock on CRO.
- Put the channel switch **ON & OFF** & observe the position of each channel in a frame. Observe how the frame repeats itself at regular intervals.
- Observe the marker **MRTX1 & MRTX2** with respect to the frame data & see how the two markers are transmitted in alternate frame.
- Observe the **PCM** coded voice data at the **PCM OUT** test points of codec 1 & codec 2 with respect to the frame clock & also with respect to the **TDM OUT** test point. Observe the position of **PCM** data in the frame.

## SWITCH FAULTS

- Put switch **2 (SF1)** in Switch Fault section to **ON** position. This will open pin 4 of U9 (74LS150). The channel 6 signal goes in undefined state & o/p response is irrespective of the CH6 switch position.


1->
2->
3->
4->
1) CH1 FRAME CLOCK -- 8KHz
2) CH2 - CODEC-1 PCM DATA
3) CH3 -- CODEC2 PCM DATA
4) CH4 -- TDM DATA WITH MARKER,PCM,CHANNEL DATA

# EXPERIMENT
# NO. 8

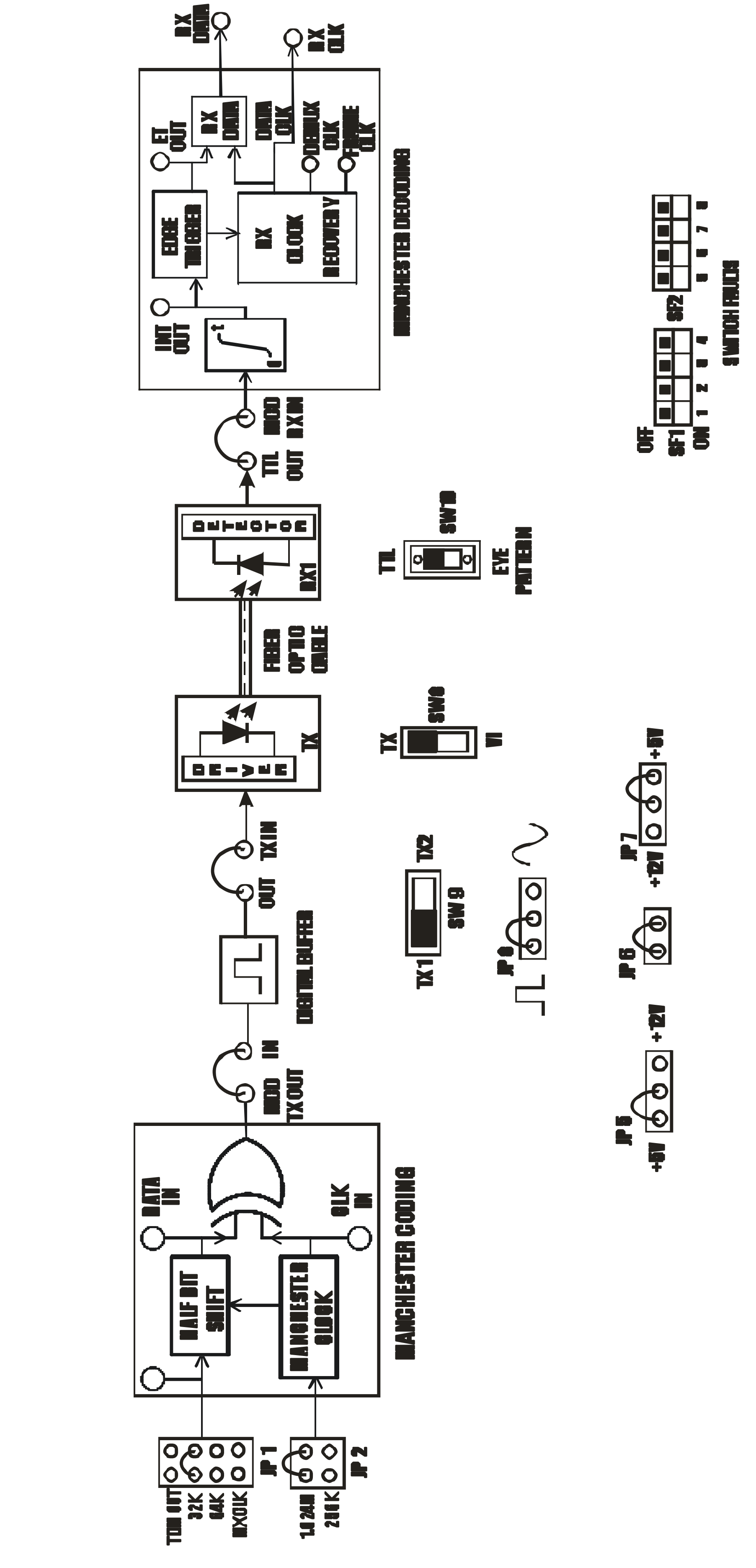



FIG. 8.1 STUDY OF MANCHESTER CODING AND DECODING.

# EXPERIMENT NO. 8

## NAME

Study of Manchester Coding & Decoding

## OBJECTIVE

The objective of this experiment is to study the techniques of Manchester coding & decoding in digital communication

## THEORY

A fiber optic digital communication system usually consists of the following:

- Optical transmitter including electrical to optical converter (E/O)
- Optical receiver including optical to electrical converter (O/E)
- Optical Fiber as data transmission medium
- Connectors, Couplers & Splicer
- Line Coder
- Line Decoder
- Timing recovery unit

In digital communication, the recovery of the clock used for transmission is essential. The Technique of timing recovery depends on the no. of "0" to "1" & "1" to "0" transitions. At times, one cannot ensure that the data to be transmitted has such transitions. To introduce the transitions in such a case, line coding is used. For low bit rate fiber optic transmission, the commonly used line coding technique is known as Manchester Coding, whereas for higher bit rates various coding schemes like mBnB coding schemes are used. Thus line coding is implemented in digital communication for the following reasons.

- For ease of timing recovery at the receiver
- To shape the spectrum of the signal as required by various elements in the communication system

The recovery of synchronous clock, synchronous both in terms of phase & frequency, at the receiver is a must for any synchronous digital link. A long string of zeros & ones, i.e. lack of transitions in data, could make the receiver clock lose synchronization. Line coding introduces sufficient transitions in the data for ease of clock recovery. Also a certain pattern of data, like a long string zeros or one, would result in a strong DC component in the transmitted signal. Such a DC component is often not desirable; line coding shifts the spectrum & avoids the DC component. Several methods of coding digital transmission for unique effects are possible each type of coding has inherent advantages. We will consider the simplest line coding technique as Manchester Coding. The data waveform is composed of a binary bit stream with each bit forming a cell, one clock period in length.
The Manchester coded waveform is generated in the following manner. If the Data bit is '1', the waveform will be positive for the first half of the bit cell. For a '0' data

bit, the first half of the bit cell will be a zero. A transition always occurs in the center of the bit cell, regardless of whether the bit is '1' or '0'.

## EQUIPMENTS

- FO-A-P Kit with power supply
- Patch chords
- 20 MHz Dual Trace Oscilloscope
- 1 Meter fiber cable

**NOTE:** Keep All Switch Faults In Off Position.

## PROCEDURE

- Make connections as shown in diagram.
- Keep jumper & switch settings as shown in fig.
- Observe the Test points of **DATA IN**&**MCDTX OUT**. The **MCDTXOUT** shows that when Data is “1” the first half bit of data is high whereas the second half is zero. While for Data as “0” the first half is a “0” level & second half is “1” as shown in the waveforms.
- Select the fiber optic transmitter TX1 SFH756 using jumper & switch settings as shown in fig.9.2.
- Slightly unscrew the cap of SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the one-meter fiber into the cap. Now tighten the cap by screwing it back.
- Slightly unscrew the cap of RX1 Photo Transistor with TTL logic output SFH551V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Observe the test points of **INT OUT, ET OUT**. The decoded output is obtained at **RX DATA** test point & the decoding clock at **DATA CLK** test point.FIG.8.2 shows Manchester coded & decoded signals for 32 KHz data input.(After inserting the fiber on both sides, align the fiber properly inside both SFH devices if required to get proper indication of o/p LEDs.)
- Observe the waveforms for other data by shifting **JP1** to **64 KHz & MX CLK.**
- Also observe the effect of changing Manchester clock by keeping **JP2** position at **256 KHz.** Now the Manchester coded output will be exact but the Manchester decoded output & clock may not match the data as the decoded clock circuit is tuned to 1.024MHz clock & not 256Khz.So for this observe only the Manchester coded data.

# SWITCH FAULTS

**NOTE:** Keep the connections as per the procedure. Now switch **ON** corresponding fault switch button to **ON** position & observe the different effects on the output. The faults are normally used one at a time.

- Put switch **5(SF2)** in Switch Fault section to **ON** position. This will open pin 2 MCDCLK of U19 (74HC86). This will affect the Manchester coding and the Manchester coded o/p will be in a wrong format.
- Put switch **6(SF2)** in Switch Fault section to **ON** position. This will disturb the Manchester decoding o/p and data will be disturbed. R71 comes in parallel to PR1 & hence the RC time constant changes & RX CLK gets disturbed.

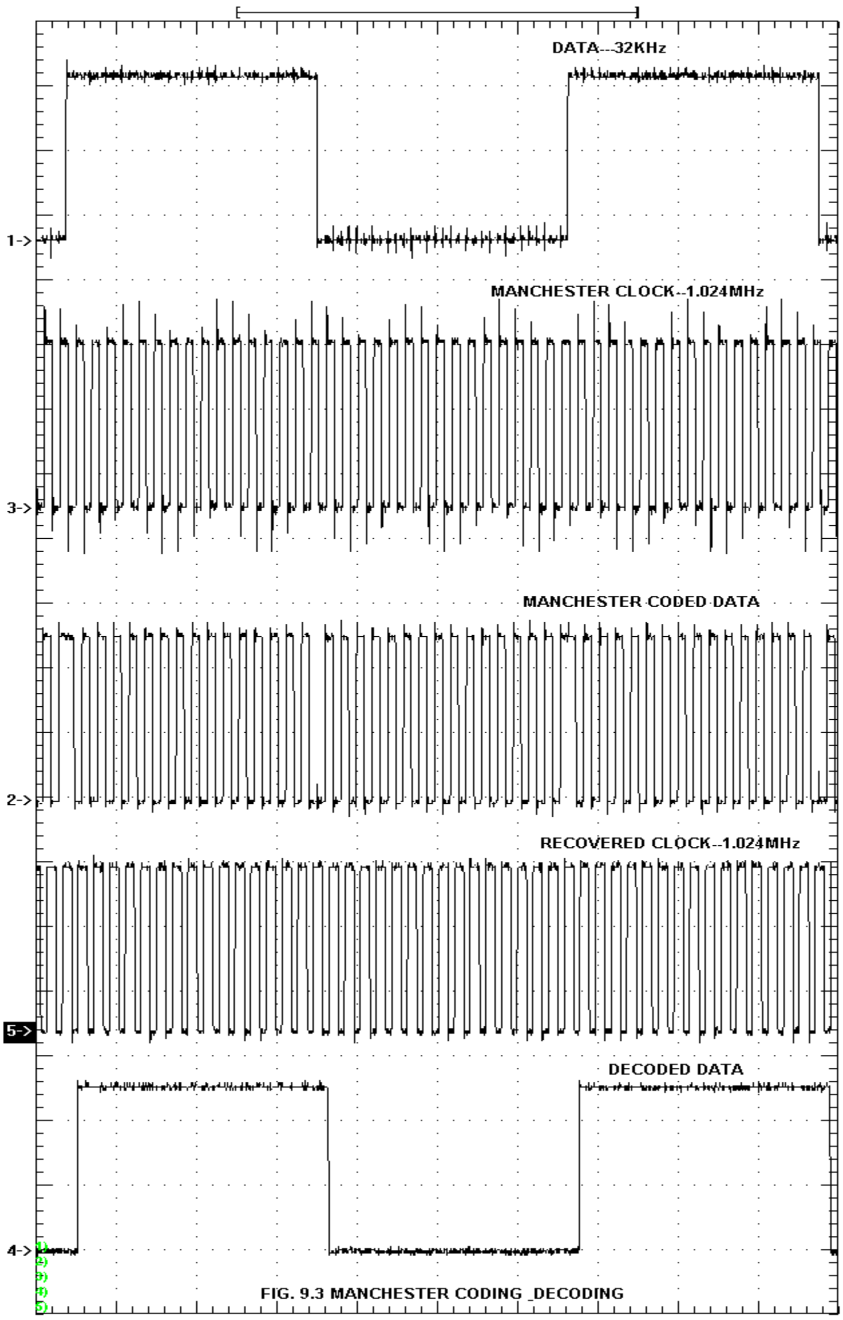



FIG. 9.3 MANCHESTER CODING _DECODING

# EXPERIMENT NO. 9

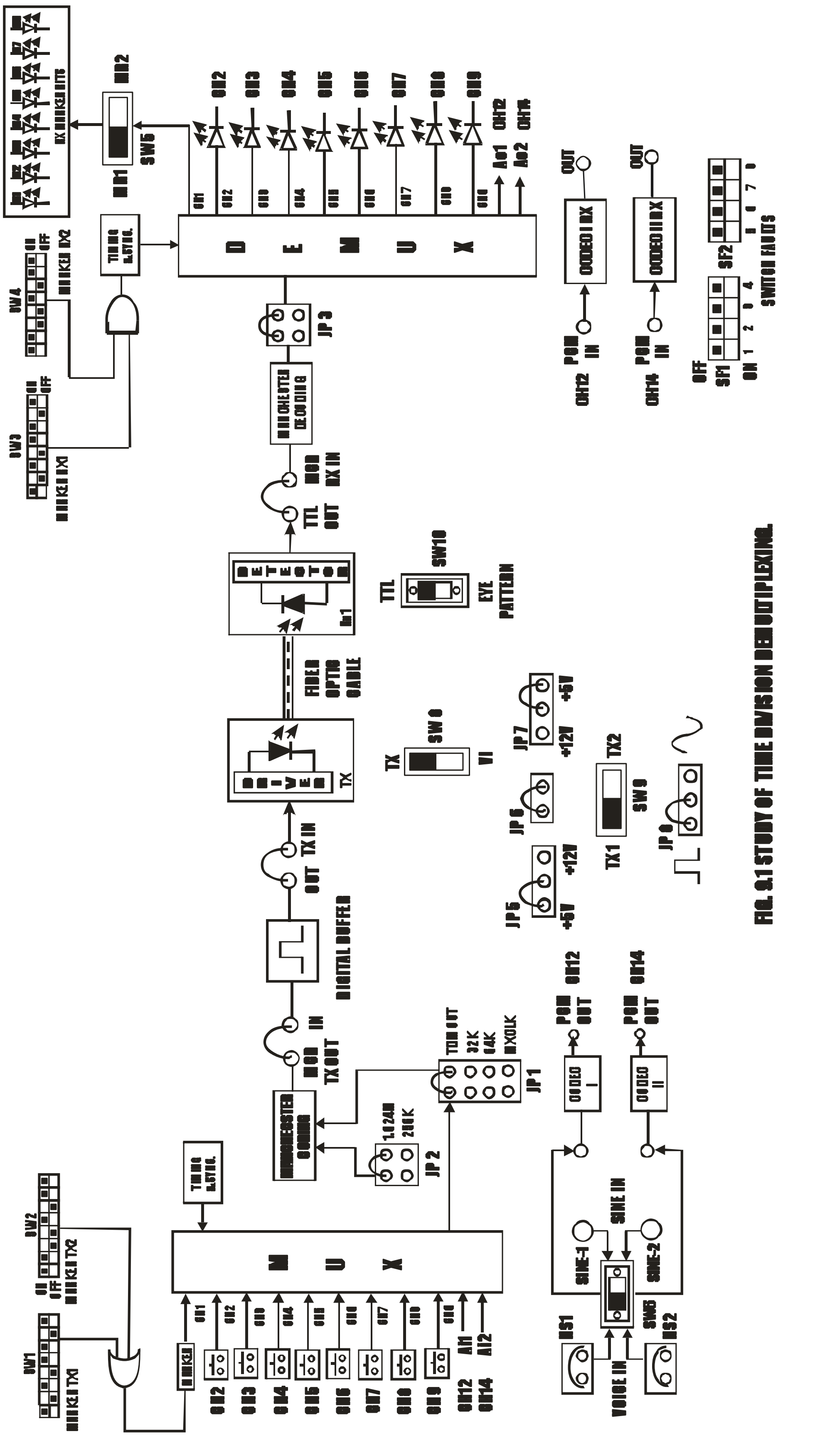



FIG. 9.1 STUDY OF TIME DIVISION DEMULTIPLEXING

# EXPERIMENT NO. 9

## NAME

Study of Time Division Demultiplexing

## OBJECTIVE

To objective of this experiment is to study the process of Time Division Demulplexing.

## THEORY

The process of Time Division Demultiplexing is the process of having one input & many outputs.
The TDM data on a single link is separated on time basis, depending on the Q0, Q1, Q2 & Q3 clocks at the receiver section.
The Manchester coded data is first decoded and the original TDM data is obtained. Also the clock & Timing signals are generated using the RX clock. The marker from the recovered TDM data is compared with the RXMP1 & RXMP2. When the markers are identified or when they are found similar by the RX marker comparator, the comparator generates a detection (sync) pulse. Thus two sync pulses are obtained at the receiver one for each marker.
These sync pulses are used to synchronize the TDM Data with the receiver clocks. The marker RX1 is kept same as marker TX1 & marker RX2 is kept similar to marker TX2 as shown in the block diagram.

## EQUIPMENTS

FO-A-P Kit with power supply
Patch chords
20 MHz Dual Trace Oscilloscope
1 Meter fiber cable

**NOTE:** Keep All Switch Faults In Off Position.

## PROCEDURE

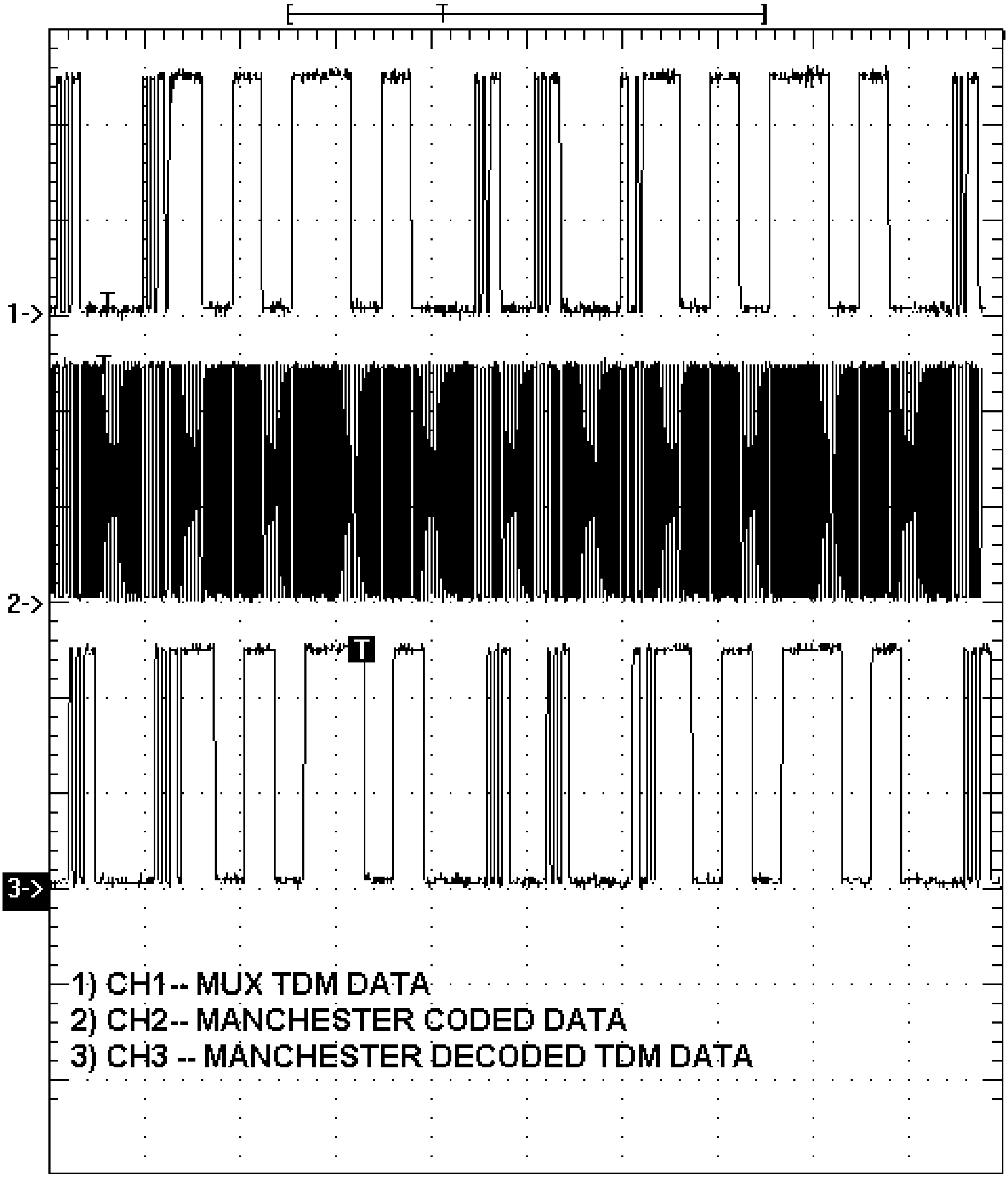
- Make the connections as shown in the diagram.
- Keep the jumper & switch settings as shown in fig.9.1.
- Select the fiber optic transmitter TX1 SFH756 using jumper & switch settings as shown in fig.9.1.
- Slightly unscrew the cap of SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the one-meter fiber into the cap. Now tighten the cap by screwing it back.
- Slightly unscrew the cap of RX1 Photo Transistor with TTL logic output SFH551V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Observe the detected signal at post **TTL OUT** on oscilloscope.

- Connect the received signal post **TTL OUT** to post **MCD RX IN**.
- Observe the integrated output at **INT OUT,** Observe edge triggered output at **ET OUT** as shown in fig.
- Observe the recovered clock as shown in FIG.9.3.at **RX CLK** post the duty cycles of the recovered clock is not the same as transmitter clock, but they are synchronized except for a slight transmission delay in the received clock.
- Observe decoded data at **RX DATA** post as shown in fig.9.4.
- Select marker 1 or 2 for display using switch SW5.
- Select **TX2 SFH450** using switch **SW9**& Repeat the above procedure. (After inserting the fiber on both sides, adjust the fiber alignment properly inside both SFH devices if required to get proper indication of o/p LEDs.)
- Repeat the above procedure for marker1 & 2 as given in the marker settings table below.

**Note**

The students can form hundreds of other combinations apart from the marker settings table. Few settings may not give proper TDM o/p. Avoid 4 consecutive 1's or 0's in single marker settings to get proper TDM data.

| TX MARKER -1 | TX MARKER -2 |
|---|---|
| 10010001 | 10110110 |
| 11011001 | 10110110 |
| 11011001 | 10001010 |
| 01011101 | 10001010 |
| 10101101 | 10001010 |
| 10101101 | 11001011 |
| 10111001 | 11001011 |
| 10110001 | 01101010 |
| 10011101 | 01100010 |
| 10010101 | 01011001 |


1->
2->
3->
1) CH1-- MUX TDM DATA
2) CH2-- MANCHESTER CODED DATA
3) CH3 -- MANCHESTER DECODED TDM DATA

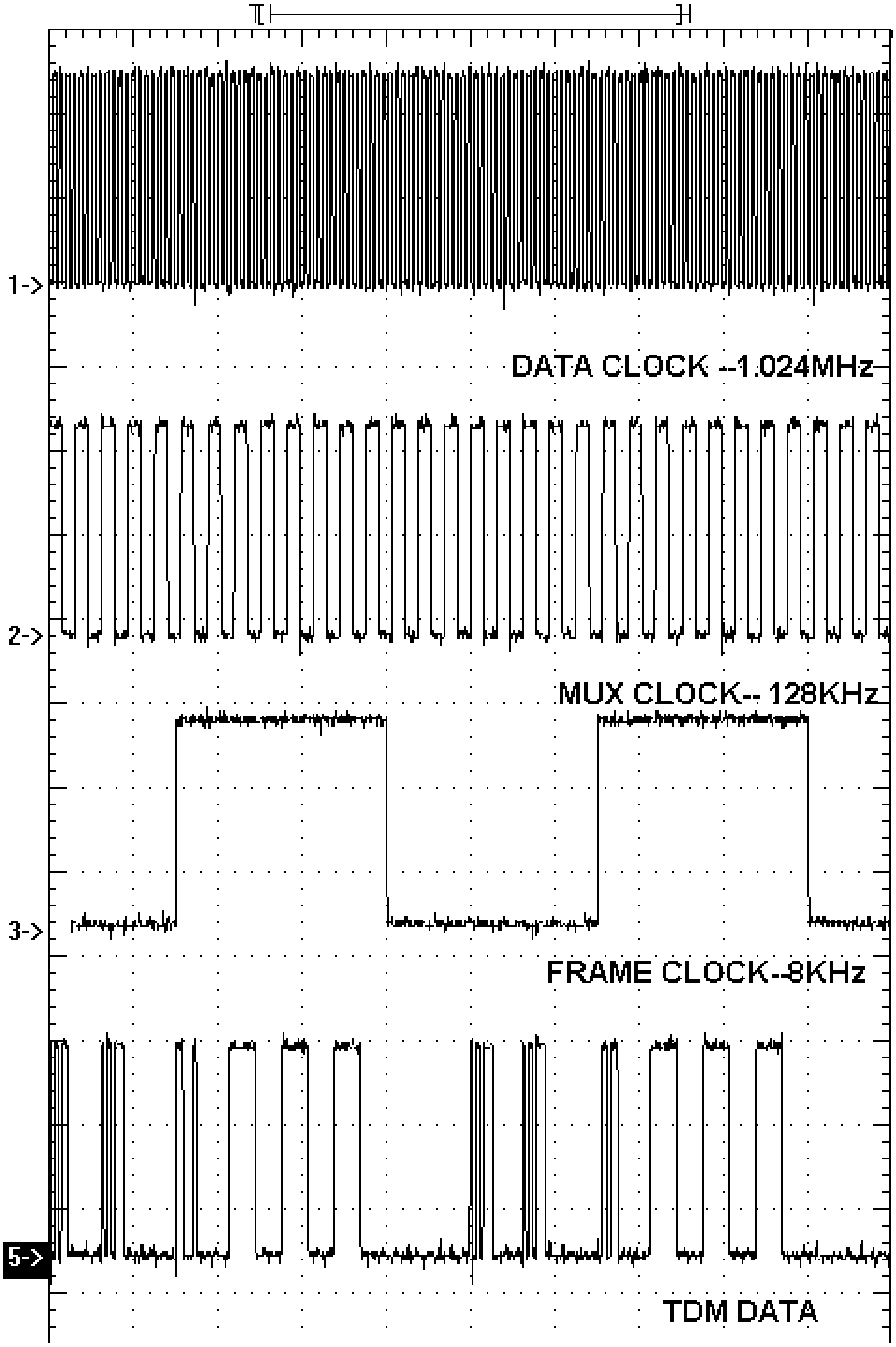



**FIG 9.3: FRAMING IN TIME DIVISION**

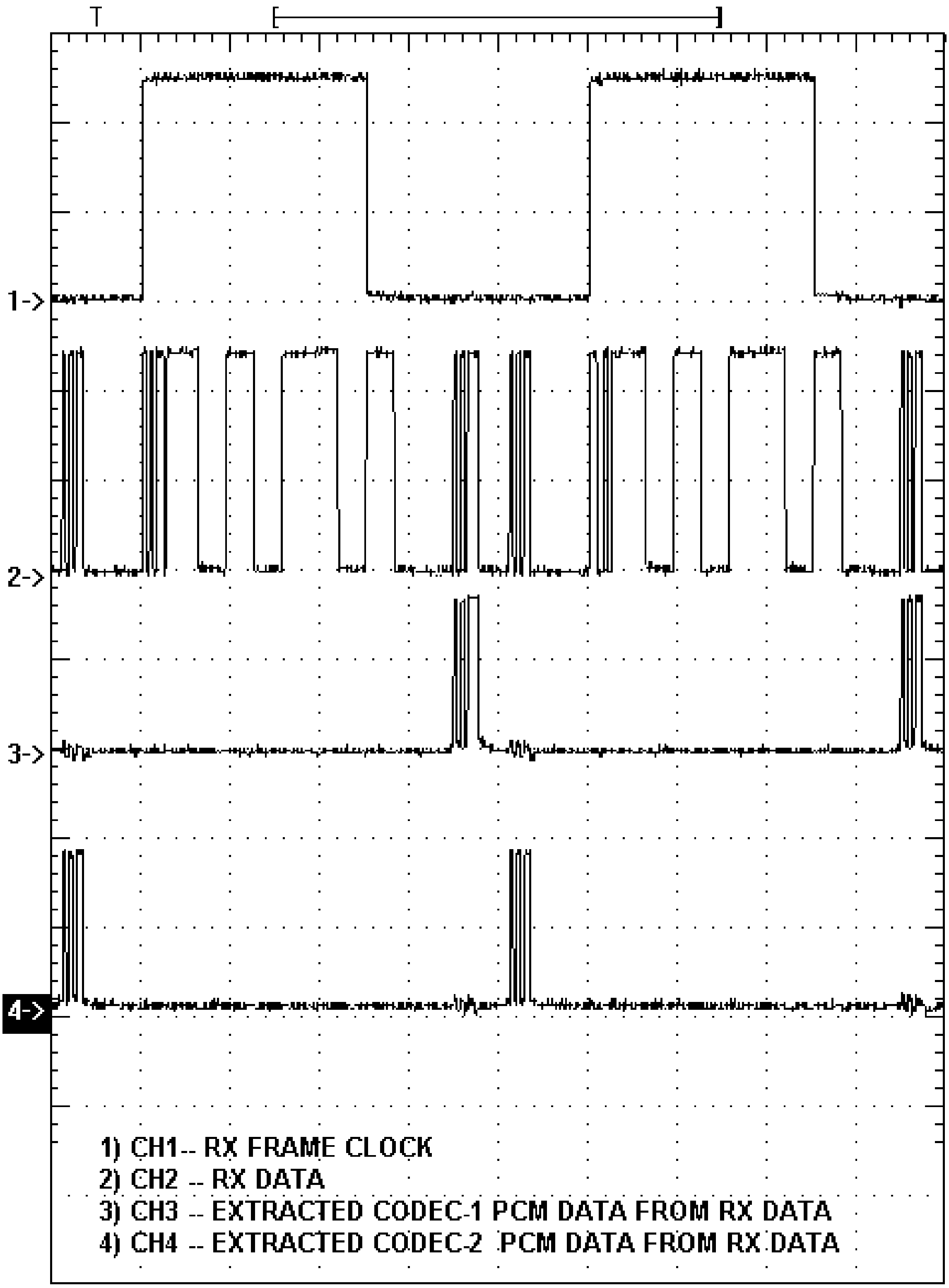



**FIG.9.4: EXTRACTED PCM DATA FROM RX DATA**

# SWITCH FAULTS

Switch fault section on is provided with 8 switches. These switches can be used to simulate fault conditions in various parts of the circuit. In order to keep FO-A-P kit fully operational, all switches should be set to OFF position before use. Switch faults are introduced one at a time.

- Put switch **1(SF1)** in Switch Fault section to **ON** position. This will open TX-marker signal at i/p of U9. This will open pin 8 of U9 (74LS150) i.e. channel one is removed at MUX section. No marker is transmitted and hence the receiver synchronization is disturbed.
- Put switch **2(SF1)** in Switch Fault section to **ON** position. This will open pin 4 of U9 (74LS150). The channel 6 I/P of Mux section is opened & hence the channel 6 state in the TDM DATA becomes independent of the switch position & hence the CH6 O/P LED indication is irrespective of the CH6 switch position.
- Put switch **3(SF1)** in Switch Fault section to **ON** position. This will open pin 13 of U9 (74LS150). This will remove the MUX clock QC. Due to which few channels at the MUX o/p will be lost. Since they will not be selected.
- Put switch **4(SF1)** in Switch Fault section to **ON** position. This will open pin 8 of U12 (145502). The MSIRX1 signal is removed. This will disable the codec selection during demux and codec response cannot be observed.
- Put switch **5(SF2)** in Switch Fault section to **ON** position. This will open pin 2 MCDCLK of U19 (74HC86). This will affect the Manchester coding and the Manchester coded o/p will be in a wrong format.
- Put switch **6(SF2)** in Switch Fault section to **ON** position. This will disturb the Manchester decoding o/p and data will be disturbed. R71 comes in parallel to PR1 & hence the RC time constant changes & RX CLK gets disturbed.
- Put switch **7(SF2)** in Switch Fault section to **ON** position. This will open pin 20 of U24 (74HCT154). Demux clock Q4 will be removed from Demux IC and hence few channels at i/p will not be selected.
- Put switch **8(SF2)** in Switch Fault section to **ON** position. This will open pin 11 of U35 (74LS374). This will open the latching signal of U35 and marker settings hence the U35 will not accept new marker settings & will display the previous settings till the latch signal is connected.

# EXPERIMENT NO. 10

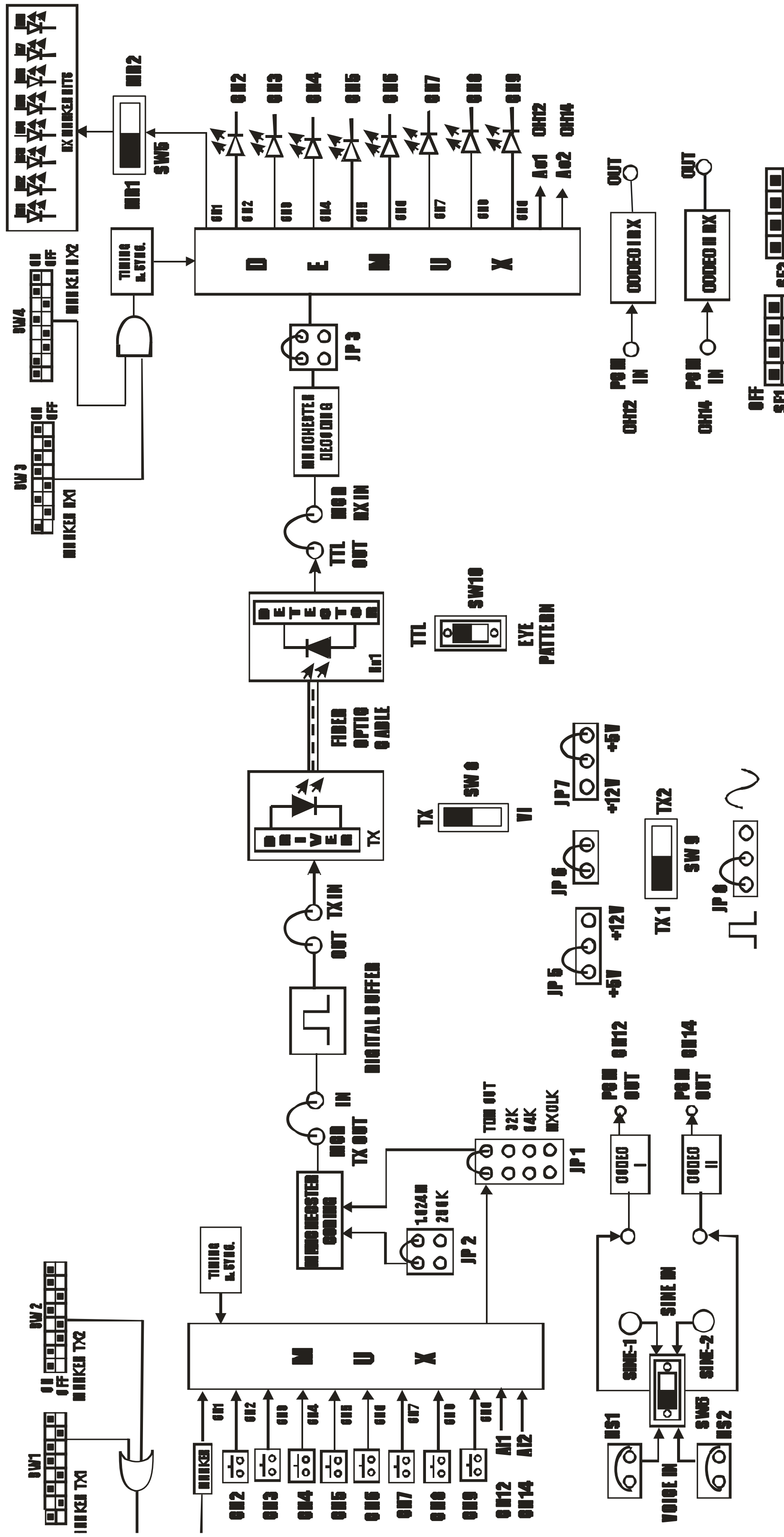



FIG.10.1 STUDY OF PCM VOICE CODING AND CODEC FREQUENCY RESPONSE

# EXPERIMENT NO. 10

## NAME

Study of PCM Voice Coding And Codec Frequency Response

## OBJECTIVE

The objective of this experiment is to study the linearized A-law PCM coding. The analog to digital conversion as well as the reverse process and the filtering characteristics of the CODEC chip 145502 used in this kit are also studied.

## THEORY

Present techniques of voice communication use standards such as A-law / μ - law companded PCM voice coding at 64 Kbits/sec. When the analog speech signal is converted to pulse code modulation, it is first filtered using a low pass filter with a cut-off at about 3.4 KHz. The analog signal is sampled at 8KHz as per the Nyquist Criteria. Each sample is quantized and coded into eight bits per sample.
Actually the voice signal features something undesirable and that is its amplitude range varies greatly from one conversion to another. If the quantization levels are uniformly spaced then it certainly creates problems. If the amplitude of the signal is small, quantization levels have to be closely spaced. This gives proper resolution. But if the signal amplitude is large then this fine resolution will result in increasing the number of code bits.

Hence, normally a technique of unequal spacing of quantization levels is used. In the Fiber FOL-A-P kit, the audio signal from the telephone handset is processed using telephone interface chip. It is then applied to voice coder. This chip actually performs analog to digital conversion and vice-versa.

The digital data output is in pulse code modulated form. The CODEC chip used exhibits both A-law and u-law companding techniques. Here, we have selected A-law of companding. The digital output of CODEC is connected as i/p to the multiplexer. The received digital data is again converter into analog form by the chip. The clock considerations and other features of the CODEC chip used in this kit can be studied in details from the datasheets provided with the manual.

## EQUIPMENTS

FO-A-P Kit with power supply
Patch chords
1 MHz Function Generator
20 MHz Dual Trace Oscilloscope

**NOTE:** Keep All Switch Faults In Off Position.

# PROCEDURE

- Make connections as shown in fig.10.1. Connect the power supply cables with proper polarity to FOL-A-P Kit. While connecting this, ensure that the power supply is OFF.
- Now voice communication can be done between two audio channels using telephone handset.
- Observe the effect on voice signal at various test points.
- Keep the switch **SW6** towards **SINE IN** position.
- Feed a sinusoidal signal of 1 KHz and of amplitude up to 2Vpp to **SINE 1 & SINE 2** input terminals. This gives an analog input to both the CODECs.
- Observe the reconstructed waveform at **OUT** post for CODEC I RX and at **OUT** post for CODEC II RX as shown in fig. Compare both the applied input and the reconstructed signal on oscilloscope as shown in **FIG. 10.2**.
  Draw waveforms. (The output signal may be delayed or phase shifted with respect to the input)
- Observe the signal changes at various test points.
- Vary the input frequency in steps and simultaneously observe the output signal. Measure the amplitude of the o/p signal for each input frequency.
- Find the frequency reading after which the response of the codec drops. This gives the bandwidth of the codec.
- Since it works for audio range we should get bandwidth at around 3.4 KHz.

# SWITCH FAULTS

**NOTE:** Keep the connections as per the procedure. Now switch **ON** corresponding fault switch button to **ON** position & observe the different effects on the output. The faults are normally used one at a time.

- Put switch **4(SF1)** in Switch Fault section to **ON** position. This will open pin 8 of U12 (145502). The MSIRX1 signal is removed. This will disable the codec selection during demux and codec1 response gets disturbed or may not generate any signal.

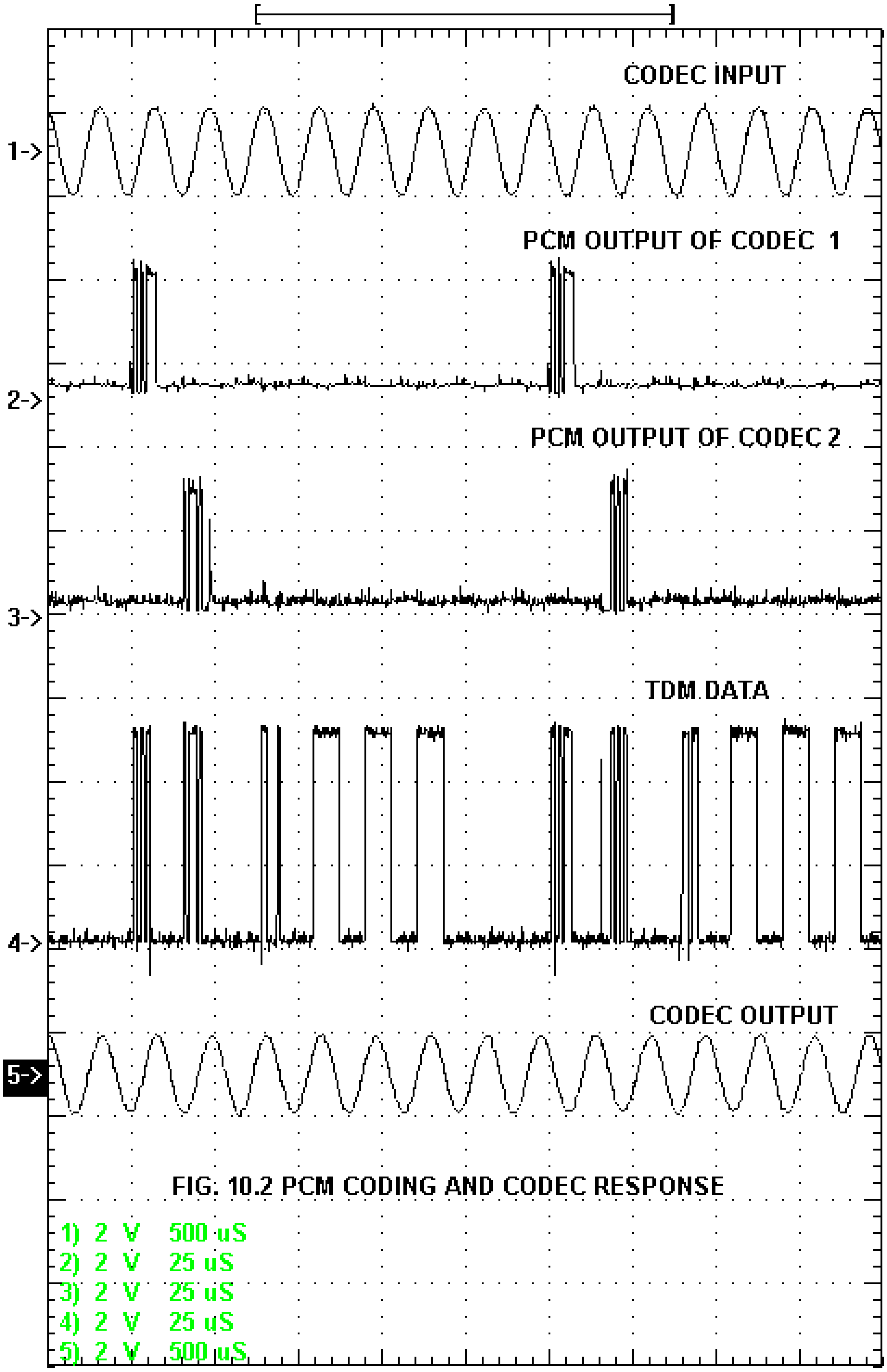



FIG. 10.2 PCM CODING AND CODEC RESPONSE

# EXPERIMENT NO. 11

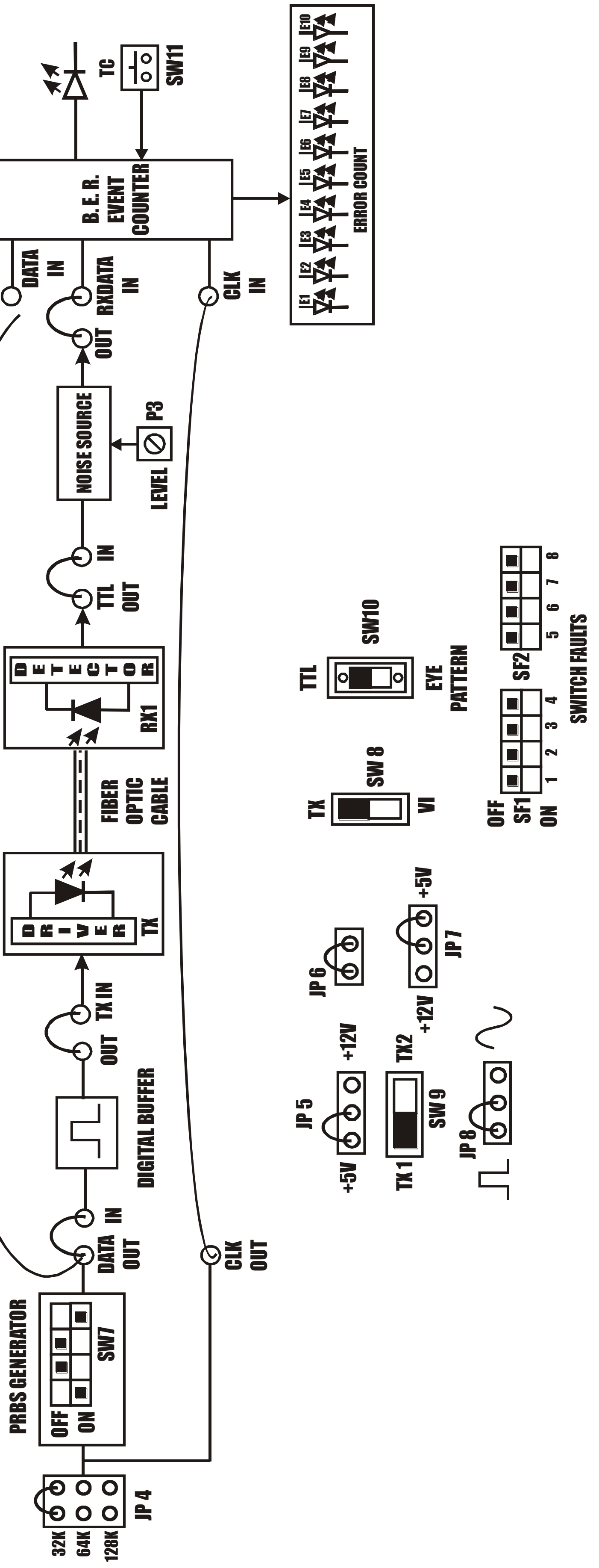



FIG.11.1 BLOCK DIAGRAM FOR BIT ERROR RATE MEASUREMENT.

# EXPERIMENT NO. 11

## NAME

Measurement of Bit Error Rate

## OBJECTIVE

To measure Bit error rate

## THEORY

Bit Error Rate

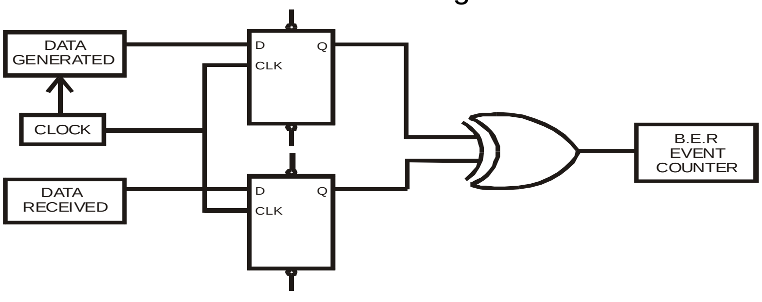
In telecommunication transmission, the bit error rate (BER) is a Ratio of bits that have errors relative to the total number of bits received in a transmission. The BER is an indication of how often a packet or other data unit has to be retransmitted because of an error. Too high BER may indicate that a slower data rate would actually improve overall transmission time for a given amount of transmitted data since the BER might be reduced, lowering the number of packets that had to be resent.

**Measuring Bit Error Rate**

A BERT (bit error rate tester) is a procedure or device that measures the BER for a given transmission. The BER or quality of the digital link is calculated from the number of bits received in error divided by the number of bits transmitted.

BER= (Bits in Error) / (Total bits transmitted)

Using a bench test setup, this is easily measured by means of a comparator in which the transmitted bits are matched in an XOR gate with the received bits. Fig shows the schematic of the device used for the following measurements.




**BIT ERROR RATE TESTER (B.E.R.T.)**

If the bits are alike at the XOR gate input, when clocked in from the D flip flop, the output is low. If they are different, the XOR output goes high, causing an event count. The event counter can be set for various time periods. In general, the longer the time period, the more accurate is the count.

A random character generator and white noise source should be used for these measurements.

The number of bit errors is dependent upon the amount of noise entering the system. White noise or background noise has an average or RMS value that is exceeded periodically by peaks that may get raised many times that level. These peaks exist only for a very short period of time. When the peak equals or exceeds the signal level, that is noise energy = bit energy, there is a 50/50 chance of error. The peak time periods can be calculated statistically from the error function.

In Link-B, PRBS sequence is generated by using a 4-bit right shift register whose feedback is completed by the EX-OR gate.

Let Initially 1001 be the 4-bit switch setting on the SW7.

| Clock States | D1 | D2 | D3 | D4 | |
|---|---|---|---|---|---|
| | | | A | B | C |
| 1 | 1 | 0 | 0 | 1 | 1 |
| 2 | 1 | 1 | 0 | 0 | 0 |
| 3 | 0 | 1 | 1 | 0 | 1 |
| 4 | 1 | 0 | 1 | 1 | 0 |
| 5 | 0 | 1 | 0 | 1 | 1 |
| 6 | 1 | 0 | 1 | 0 | 1 |
| 7 | 1 | 1 | 0 | 1 | 1 |
| 8 | 1 | 1 | 1 | 0 | 1 |
| 9 | 1 | 1 | 1 | 1 | 0 |
| 10 | 0 | 1 | 1 | 1 | 0 |
| 11 | 0 | 0 | 1 | 1 | 0 |
| 12 | 0 | 0 | 0 | 1 | 1 |
| 13 | 1 | 0 | 0 | 0 | 0 |
| 14 | 0 | 1 | 0 | 0 | 0 |
| 15 | 0 | 0 | 1 | 0 | 1 |
| 16 | 1 | 0 | 0 | 1 | 1 |

Thus the sequence repeats constantly with a period corresponding to 16 clock states.
Length of sequence = $2^4$ =16

Now the Pseudo Random Sequence pattern is C = 1010111100010011

## EQUIPMENTS

FO-A-P Kit with power supply
Patch chords
1 Meter Fiber cable
Patch chords
20 MHz Dual Channel Oscilloscope

**NOTE:** Keep All Switch Faults In Off Position.

## PROCEDURE

- Make connections as shown in fig.11.1. Connect the power supply cables with proper polarity to FO-A-P Kit. While connecting this, ensure that the power supply is OFF.
- Keep **PRBS** switch **SW7** as shown in fig.11.1 to generate PRBS signal.
- Keep switch **SW8** towards **TX** position.
- Keep switch **SW9** towards **TX1** position.
- Keep the switch **SW10** at fiber optic receiver output to **TTL** position.
- Select PRBS generator clock at 32 KHz by keeping jumper **JP4** at **32K** position.
- Keep Jumper **JP5** towards **+5V** position.

- Keep Jumpers **JP6** shorted.
- Keep Jumper **JP8** towards **pulse** position.
- Switch ON the power supply.
- Connect the post **DATA OUT** of PRBS Generator to the **IN** post of Digital Buffer and also to the **DATA IN** post of Bit Error Rate event counter.
- Connect the **OUT** post of Digital Buffer to **TX IN** post Transmitter.
- Slightly unscrew the cap of LED SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the one-meter fiber into the cap. Now tighten the cap by screwing it back.
- Slightly unscrew the cap of RX1 Photo Transistor with TTL logic output SFH551V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Connect detected signal **TTL OUT** to post **IN** of Noise Source.
- Connect post **OUT** of Noise Source to post **RXDATA IN** of Bit Error Rate event counter.
- Connect post **CLK OUT** of PRBS Generator to post **CLK IN** of Bit Error Rate event counter.
- Press Switch **SW11** to start counter.
- Vary pot **P3** for **Noise Level** to observe effect of noise level on the error count.
- Observe the Error Count LEDs for the error count in received signal in time 10 seconds as shown in **FIG. 11.2**.

**BER Measurement**

As per the definition the BER is a ratio of Error bits (Eb) to Total bits Transmitted (Tb) in a period of time t seconds.
i.e. BER = Eb / Tb
For eg. in this experiment if PRBS data is transmitted at 32Kbits per second ( i.e. jumper selection at 32KHz ) for a period of 10 seconds.
So total bits transmitted in 10 seconds (Tb) = 320Kbits.
The TTL OUT data & data with noise is fed to BER counter which compares the two data inputs at each clock input.
The counter displays the Error count (Eb) on LED in 10-bit binary form (e.g. 0000001010), which has to be converted in decimal form (it becomes 10) so the BER ratio then becomes
BER = 10 / (320 x 10 E 3)
= 0.00003125
i.e. the channel Bit Error Rate ratio is 3.1x10E-5 ( 3 / 100000 ) or in other words we can say that out of 100000 bits transmitted through the channel the channel gives 3 bits in error.

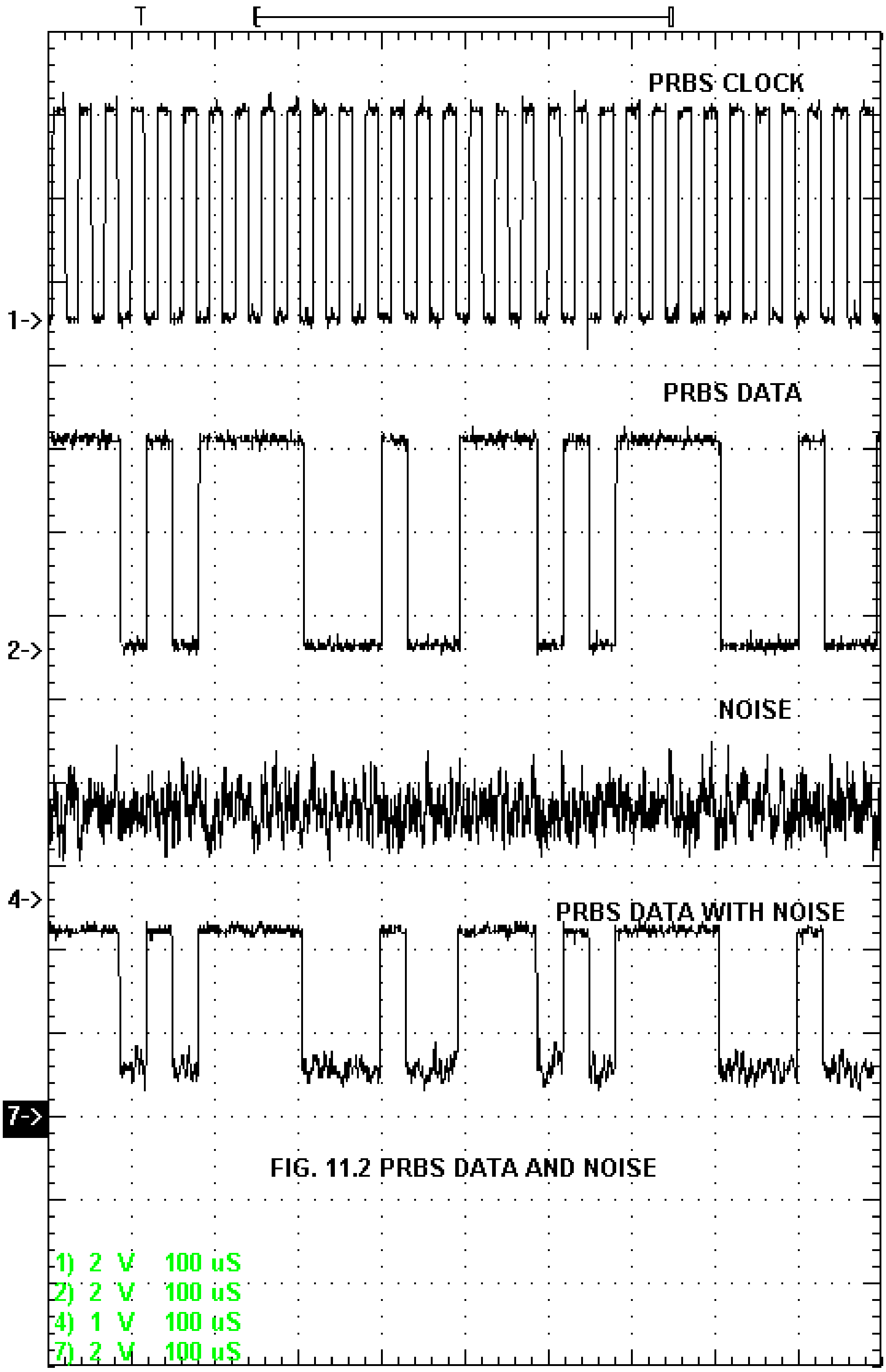



FIG. 11.2 PRBS DATA AND NOISE

# EXPERIMENT NO. 12

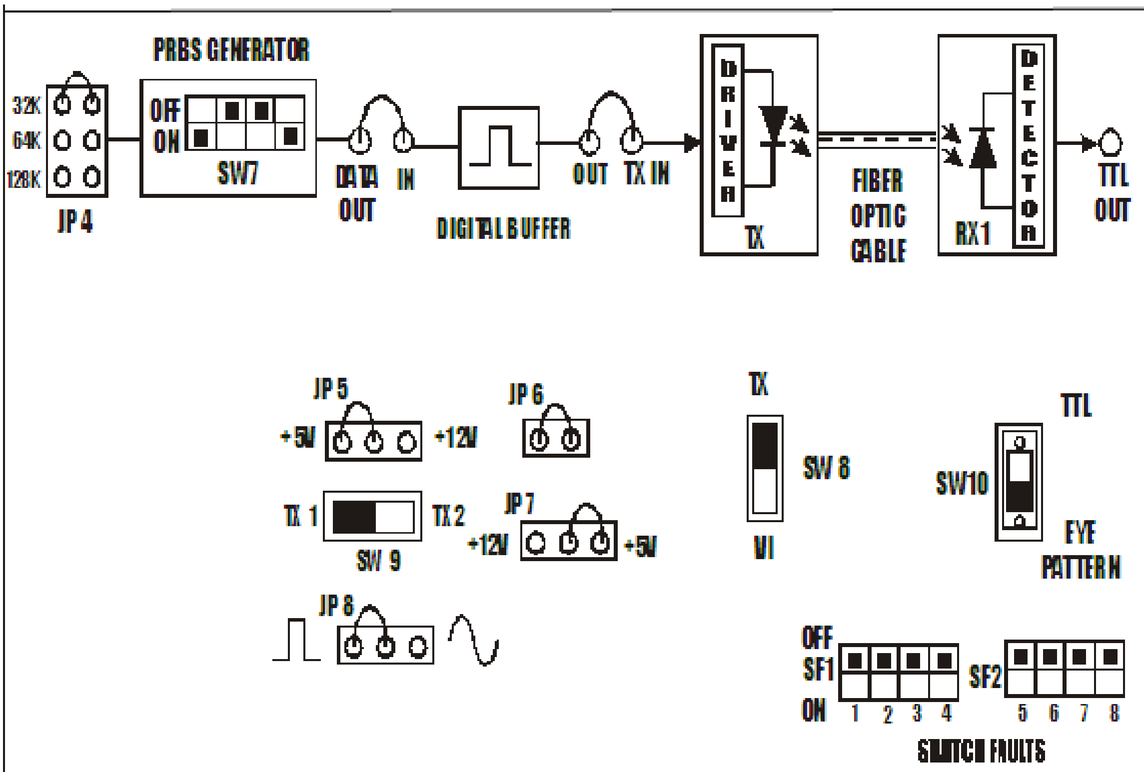



FIG. 12.1 BLOCK DIAGRAM FOR STUDY OF EYE PATTERN

# EXPERIMENT NO. 12

## NAME

Study of Eye Pattern

## OBJECTIVE

The objective of this experiment is to study eye pattern using fiber optic link.

## THEORY

The eye-pattern technique is a simple but powerful measurement method for assessing the data-handling ability of a digital transmission system. This method has been used extensively for evaluating the performance of wire systems and can also be applied to optical fiber data links. The eye-pattern measurements are made in the time domain and allow the effects of waveform distortion to be shown immediately on an oscilloscope.

An eye-pattern can be observed with the basic equipment shown in Fig. 12.1. The output from a pseudorandom data pattern generator is applied to the vertical input of an oscilloscope and the data rate is used to trigger the horizontal sweep. This results in the type of pattern shown in Fig. 12.2, which is called the **eye pattern** because the display shape resembles a human eye. To see how the display pattern is formed, consider the eight possible 4 -bit-long NRZ combinations. When these sixteen combinations are superimposed simultaneously, an eye pattern as shown in Fig. 12.2 is formed.

To measure system performance with the eye-pattern method, a variety of word patterns should be provided. A convenient approach is to generate a random data signal, because this is the characteristic of data streams found in practice. This type of signal generates ones and zeros at a uniform rate but in a random manner. A variety of pseudorandom pattern generators are available for this purpose. The word pseudorandom means that the generated combination or sequence of ones and zeros will eventually repeat but that it is sufficiently random for test purposes. A pseudorandom bit sequence comprises four different 2-bit-long combinations, eight different 3-bit- long combinations, sixteen different 4-bit-long combinations and so on (that is, sequences of different N-bit-long combinations) up to a limit set by the instrument. After this limit has been generated, the data sequence will repeat.

A great deal of system performance information can be deduced from the eye-pattern display. To interpret the eye pattern, follow the procedure ahead.

An eye pattern, also known as an eye diagram, is an oscilloscope display in which a digital signal from a receiver is repetitively sampled and applied to the vertical input, while the data rate is used to trigger the horizontal sweep.

If the signals are too long, too short, poorly synchronized with the system clock, too high, too low, too noisy, or too slow to change, or have too much undershoot or overshoot, this can be observed from the eye diagram. An open eye pattern corresponds to minimal signal distortion. Distortion of the signal waveform due to inter symbol interference and noise appears as closure of the eye pattern.

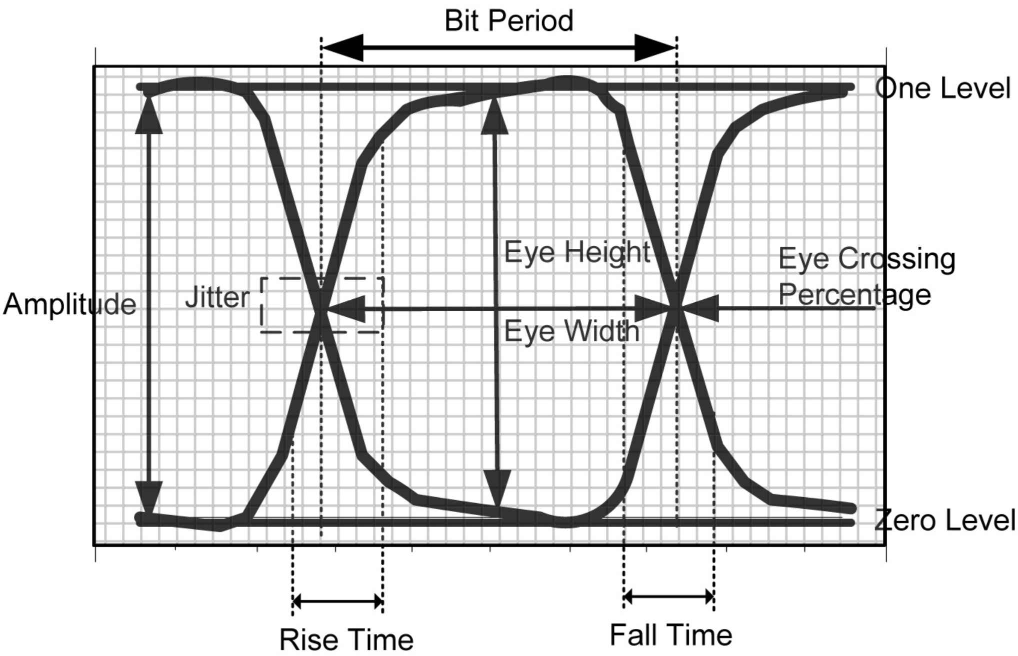



**FIG 12.3: Measurement i\of eye pattern**

## EQUIPMENTS

FO-A-P Kit with power supply
Patch chords
1 Meter Fiber cable
Patch chords
20 MHz Dual Channel Oscilloscope

**NOTE:** Keep All Switch Faults In Off Position.

## PROCEDURE

- Make connections as shown in fig.12.1. Connect the power supply cables with proper polarity to FO-A-P Kit. While connecting this, ensure that the power supply is OFF.
- Keep switch **SW7** as shown in fig.12.1. to generate PRBS signal.
- Keep switch **SW8** towards **TX** position.
- Keep switch **SW9** towards **TX1** position.
- Keep the switch **SW10** to **EYE PATTERN** position.
- Select PRBS generator clock at 32 KHz by keeping jumper **JP4** at **32K** position.
- Keep Jumper **JP5** towards **+5V** position.
- Keep Jumpers **JP6** shorted.
- Keep Jumper **JP8** towards **TTL** position.
- Switch ON the power supply.
- Connect the post **DATA OUT** of PRBS Generator to the **IN** post of digital buffer.
- Connect **OUT** post of digital buffer to **TX IN** post.
- Slightly unscrew the cap of LED SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the one-meter fiber into the cap. Now tighten the cap by screwing it back.

- Slightly unscrew the cap of RX1 Photo Transistor with TTL logic output SFH551V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Connect **CLK OUT** of PRBS Generator to **EXT. TRIG.** of oscilloscope.
- Connect detected signal **TTL OUT** to vertical channel **Y** input of oscilloscope. Then observe EYE PATTERN by selecting EXT. TRIG KNOB on oscilloscope as shown in **FIG 12.2**. Observe the Eye pattern for different clock frequencies. As clock frequency increases the EYE opening becomes smaller.
- As change in frequency (at JP4) observed effect on eye diagram.
- Connect TTL OUT post to Noise source IN post.
- Observed the eye diagram with noise.

| Frequency (Data Rate) | Amplitude (eye diagram) | Bit Period (eye diagram) |
|---|---|---|
| 32KHz | | |
| 64KHz | | |
| 128KHz | | |

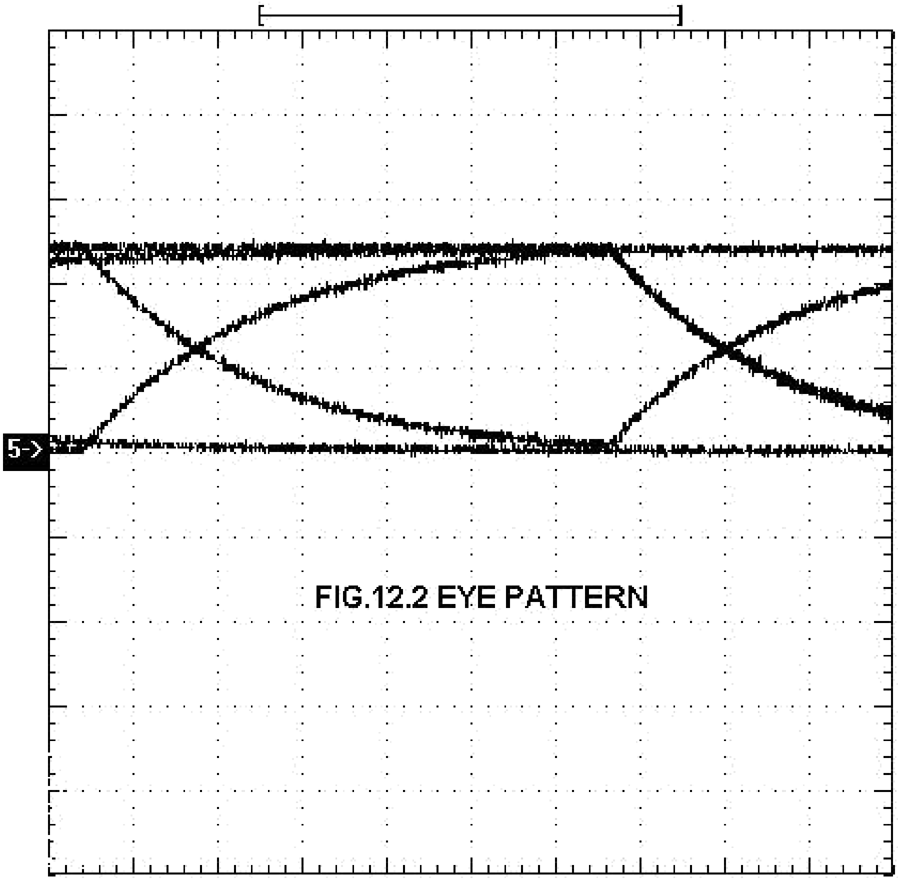



FIG.12.2 EYE PATTERN

# EXPERIMENT NO. 13

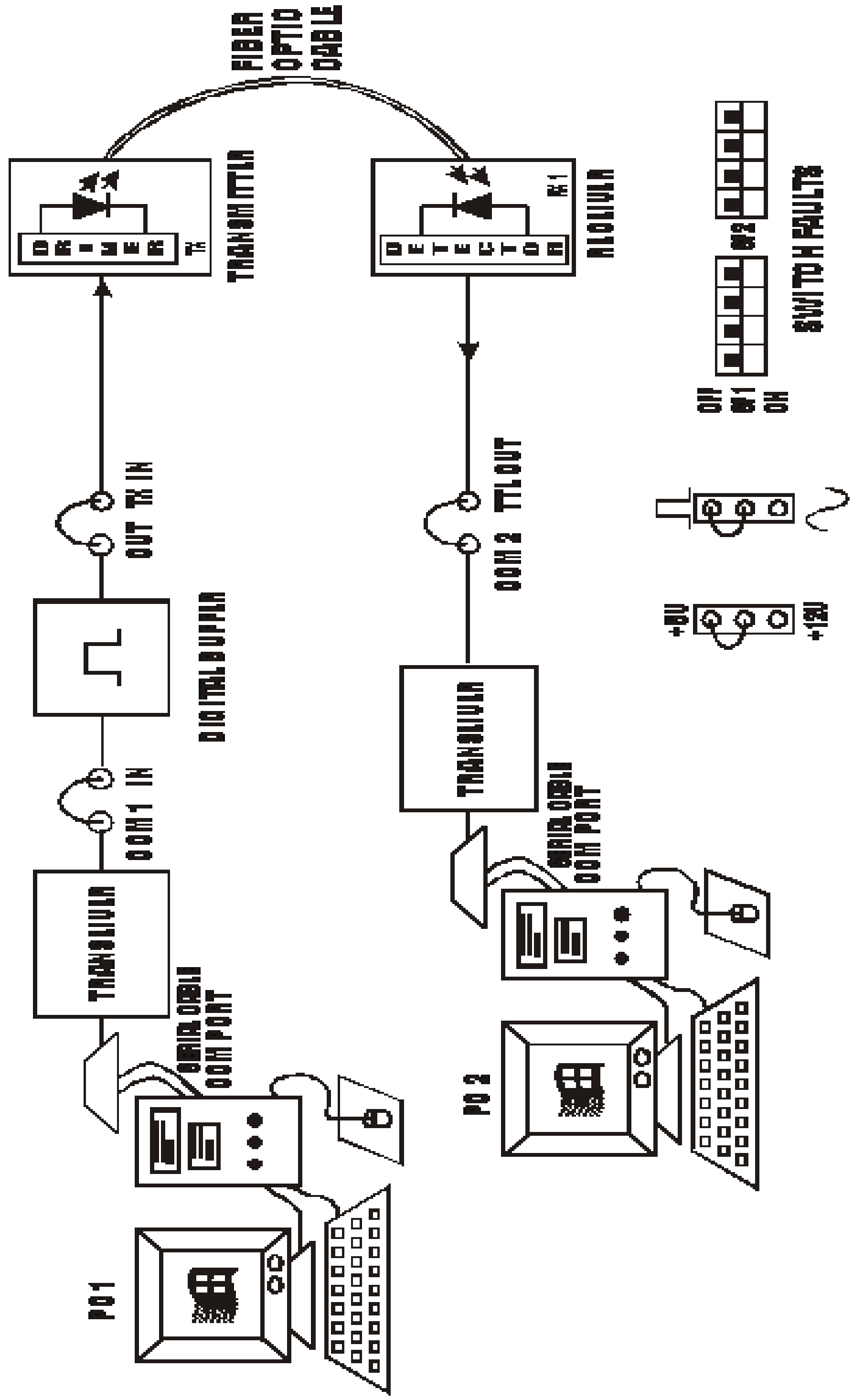



FIG. 6.1 BLOCK DIAGRAM FOR SETTING UP PC TO PC COMMUNICATION VIA FIBER OPTIC CABLE

# EXPERIMENT NO.13

## NAME

Study of Rs-232 Serial Communication between Two Computers Using Fiber Optic Digital Link

## OBJECTIVE

The objective of this experiment is to connect the RS-232 ports of two computers using Optical Fiber Digital Link, transmit data from one computer over this link and receive the same data on the other computer.

## THEORY

Microprocessor is a parallel device. It transfers the 8, 16 or 32 bit of data simultaneously over the data lines. The number of data lines depends upon the type of microprocessor used in the system. This is parallel I/O mode of the data transfer.
However in many situations the parallel data transfer is either impractical or impossible. This is very expensive and noisy especially when the distances are large. Also some devices such as CRT or CTD are not designed for parallel I/O. Moreover in many scientific and industrial process control applications, the devices under control are at the site or plant, which may be long enough from control room. In these situations, the serial I/O mode is used wherein only one bit at a time is transferred over a single cable. This cable may be a normal cable or an optical fiber.
Very important advantage of serial mode of data transfer is that it is inexpensive. Also the data is accurately transferred and received through the link. It is daily practice to put checks for the data and framing it. Uncorrupted data transfer is greatest advantage of serial mode, and exactly this is the reason behind the fact that serial mode is preferred in many applications. This plays vital role in many applications like PC-to-PC Data Communication, Industrial Process controls, Robotics, CNC and DNC (Distributed numerical control) and many more.
So it is necessary to have some system, which will perform serial I/O operation between PC and outside device using optical fiber link.

Function Of Max 232 (RS-232C Transceiver)
The computer communicates from serial COM port, which is at RS232C levels i.e. at 12V. Transceiver MAX232 performs the function of converting RS 232C signals to TTL levels & vice versa.

### Hardware Settings

Switch off the power supply of PC.
To perform this experiment, COM ports of PC are used. Onboard 9-pin D-type (female) connectors are provided for interfacing with the PC. Two Serial to USB converters are provided with this kit. Connect USB connector end of one cable to one of the com ports of Computer and the 9-pin D-type connector end to CN6 on FOL-A-P. Similarly connect other Serial cable to other Computer serial port & CN7 on FOL-A-P.

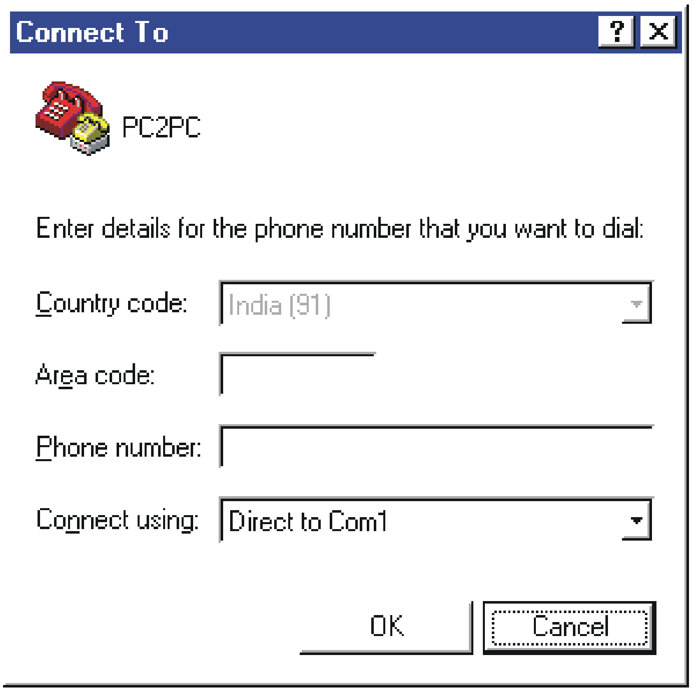
## EQUIPMENTS

- FOL-A-P
- 1mtr Fiber cable
- Connecting Patch chord
- Serial to USB converter_02 nos
- Power supply
- Computer with Hyper Terminal

**NOTE:** Keep All Switch Faults In Off Position.

## PROCEDURE

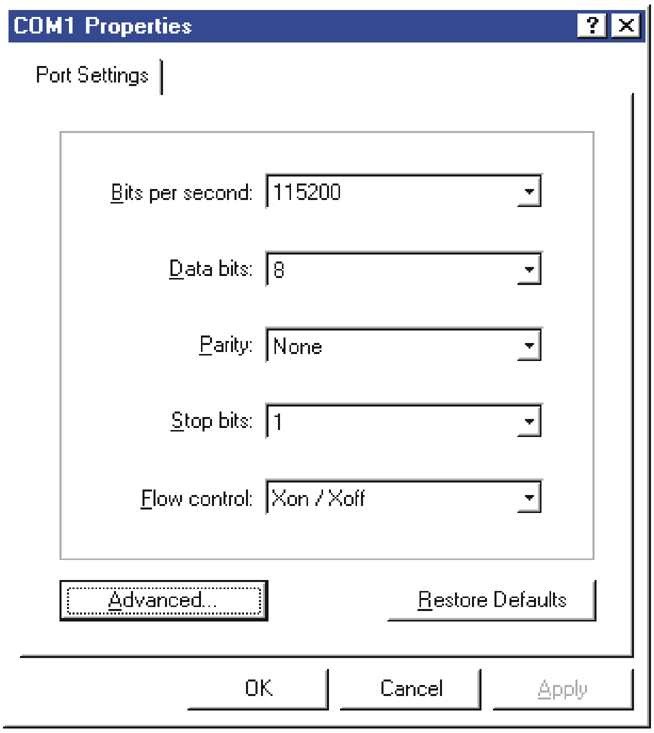
- Make connections as shown in fig. Connect the power supply cables with proper polarity to FOL-A-P Kit. While connecting this, ensure that the power supply is OFF.
- Keep the jumpers **JP5 on +5V** & **JP8 on Digital** side on FOL-A-P.
- Connect **COM1** post in RS-232 section to **IN** post of Digital Buffer Section.
- Connect the output of Digital Buffer post **OUT** to post **TX IN**.
- Slightly unscrew the cap of LED SFH756V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tighten the cap by screwing it back.
- Slightly unscrew the cap of **RX1** Photo Transistor with TTL logic output SFH551V. Do not remove the cap from the connector. Once the cap is loosened, insert the other end of fiber into the cap. Now tighten the cap by screwing it back.
- Connect the output of detector post **TTL OUT** to post **COM2** in RS-232 section.
- Refer to section – HARDWARE SETTINGS of this experiments and make the necessary connections or connect one end of Serial to USB converter cable to Computer COM1 port and other end to CN6 connector on FOL-A-P then connect second cable one end to second Computer COM1 port and other end to CN6 connector on FOL-A-P.
- Switch on the Computers.
- After putting ON one of the PC, go to START MENU, PROGRAMS, ACCESSORIES, COMMUNICATION and then Click on HYPER TERMINAL.
- A new Window will open, where in you Double Click on HYPERTRM, Two Widows will open, one at the background and another (small window) with title Connection Description which will be Active.
- Enter the **name** in the box by which you would like to store your connection, for e.g. (PC2PC), and Click **OK**. Also you could select the Icon provided below. The background window title will change to the name provided by you.
- Then specify connect using: by selecting **Direct to COM1** or port where your cable is connected and then click on **OK.** See Fig.




(**NOTE:** Please check the Port you have selected and the Ports you are connecting.)
Now Window with Title **COM 1 Properties** will appear, where Port Setting should be done as shown below and click on **OK**. See Fig.

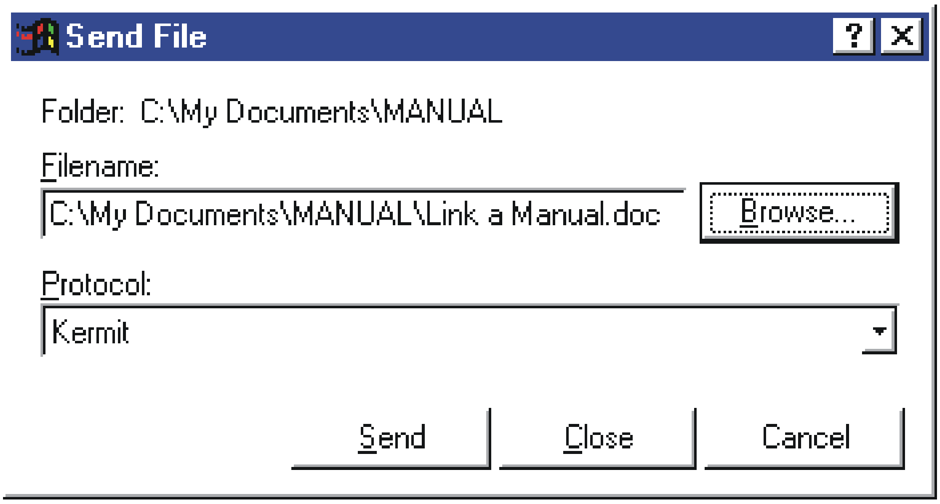
| | |
|---|---|
| Bits Per Second | Up to 115200 |

| | |
|---|---|
| Data Bits | 8 |
| Parity | None |
| Stop Bits | 1 |
| Flow Control | Xon / Xoff |

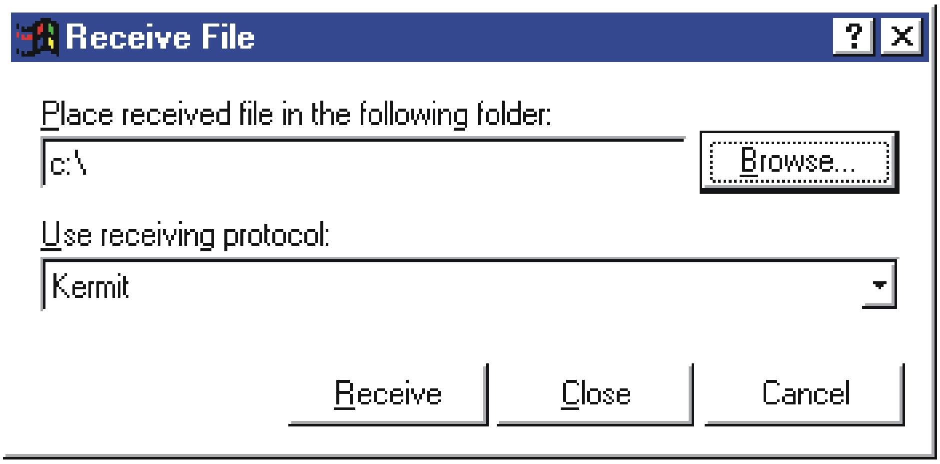


**NOTE:** For Bits Per Second setting you could select them for different speeds. Do not exceed it above 115200 bps.

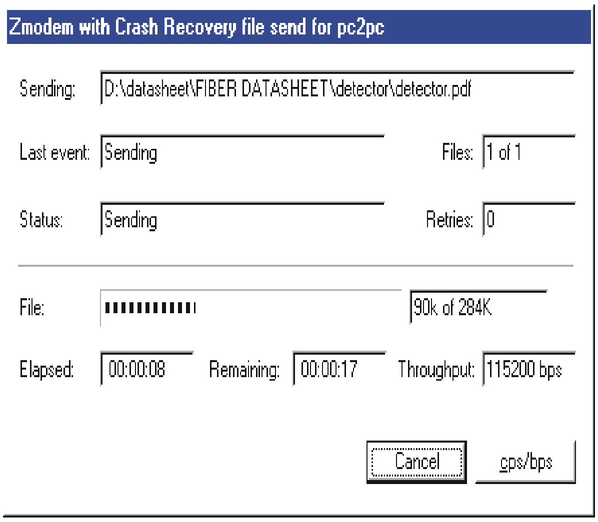
- After the above settings you click **OK**. The Background window will become Active
- Click on **File**, **Save As**, and save it in the Directory, which you want.
- Perform the same procedure (from 10. to 15.) on the computer to with whom you want to communicate.
- To start communicating between the two PCs Click on the **TRANSFER** Menu and again click on **Send File**. A window will be prompted having title Send File with **File Name** and **Protocol**. See Fig.

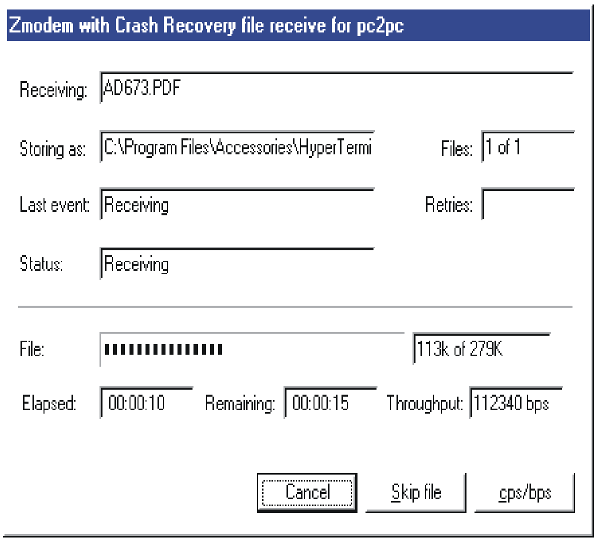



- Select **Browse** for the file, which you would like to send to the PC connected, select the file and Click on **Open**, the file name and address will be displayed in the small window. Then select the **Kermit** Protocol, (optional use protocols are X modem, Y modem and 1K X modem.)
- To receive the file on the PC Click on the **TRANSFER** Menu and again click on **Receive File**. A window will be Prompted having title Receive File with Location at which you want to store the Received file and Receiving Protocol. See Fig.




- Select **Browse** for the location where you would like to store the received file, select the folder and Click **OK**, the folder name and address will be displayed in the small window. Protocol to be selected should be **Kermit** and same as file transmitting PC.
- On the PC from which the selected file to be transmitted, click on **SEND**. A window will open showing file transfer status. Immediately at the Receiving PC Click **Receive** (otherwise Time Out Error will be displayed and communication will fail) .You will see a window showing file is being received in the form of packets. See Fig.




- After file is transferred both the windows in the (transmitting & receiving PCs) will close. Check for the received file in the folder where the file is stored.
- You can do this procedure vice-versa to transfer the file.
  **NOTE:** To change the bits per second rate (baud Rate) go to **properties** in the Hyper Terminal window, Click on **File Menu**, **Properties** and **configure**, in the bits per second select other rates from 110 to 115200 bps. You will observe that the rate at which the file is transferred will vary with the selected baud rate.

  Also you can observe and check that by removing fiber from detector packet increment and file transfer stops.

  Also ensure that at the left corner of the hyper terminal window connected time must be ON, if disconnected label is there go to call option and press call.

**NOTE:** If Hyper Terminal is not installed in your computer then, put on PC Go to My Computer, Control Panel, Add Remove Programs, Windows Setup, Communication, Check Hyper Terminal and Click OK. Also select detected COM port for PC to PC communication.